CHAPTER 2.1

Introduction to the Modem Layer

2.1.0.0.1. *Topics and Organization.*

- For the modem layer, we
  - study **general** and linear modulators, including
    - linear stream modulator (LSM), orthogonal frequency division multiplexing (OFDM) modulator, and orthogonal code division multiplexing (OCDM) modulator,
  - the design of **optimal** demodulators,
  - based on them, we **optimize** the modulators for spectral efficiency and low complexity decoding.
  - Then, we study the **demodulator** design for fading and MIMO channels
  - and build **equivalent** discrete time channel models for the design of the codec layer.

2.1.0.0.2. *Modem Layer.*

- The modem layer includes

$$\xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)} \boxed{\text{Equivalent Baseband Channel}} \xrightarrow{\boldsymbol{y}(t)} \boxed{\text{Demodulator}} \xrightarrow{\vec{\boldsymbol{y}}}$$

  - The modulator maps a vector of complex numbers $\vec{\boldsymbol{x}}$ to a complex valued continuous time baseband signal $\boldsymbol{x}(t)$.
  - The equivalent baseband channel replaces

$$\boxed{\text{I/Q Up Converter}} \xrightarrow{\boldsymbol{x}_{\mathrm{c}}(t)} \boxed{\text{Channel}} \xrightarrow{\boldsymbol{y}_{\mathrm{c}}(t)} \boxed{\text{I/Q Down Converter}}$$

    * The model is given in Section 1.4.11.
  - In the receiver, the demodulator converts continuous time baseband signals to discrete time signals $\vec{\boldsymbol{y}}$ without losing information on the messages.

2.1.0.0.3. *Layer Learning Objectives.*

- Understand and Implement modulators
- Design spectral efficient modulator with minimum inter-symbol-interference (ISI)
- Design optimal demodulators
- Derive equivalent discrete time channel models to replace the modem layer

CHAPTER 2.2

Modulator

2.2.1. Introduction

- This module helps you understand the design of linear modulators and sampling theorem
- The linear modulation includes
  - Linear stream modulation (LSM)
  - Orthogonal Frequency Division Multiplexing (OFDM)
  - Code Division Multiple Access (CDMA)

2.2.1.0.1. *Learning Objectives.*

- Understand the linear modulation
- Calculate the spectrum of the modulated signals

2.2.1.0.2. *Organization.*

- Linear stream modulation examples, sampling theorem
- Orthogonal Frequency Division Multiplexing (OFDM)
- Code Division Multiple Access (CDMA)

2.2.2. Linear Stream Modulation and Linear Modulation

- What are some examples of linear modulation?
- What are the basic terminologies about modulation?

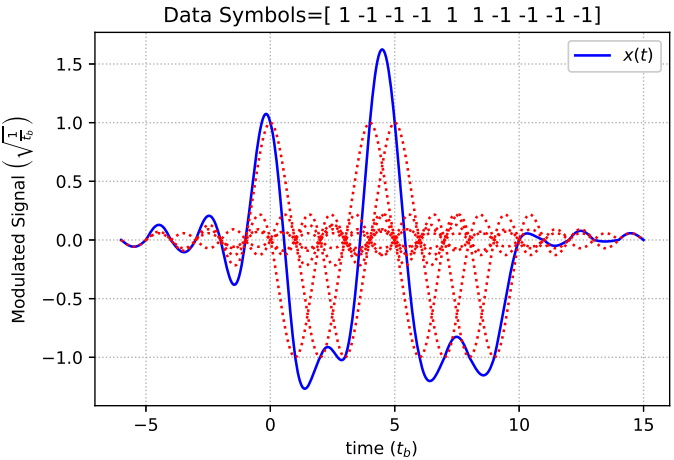
2.2.2.0.1. *A linear modulation example.*

EXAMPLE 2.2.1. (*Linear Modulation*) A sequence of complex valued SYMBOLS $[x_k]_k$ is modulated to a continuous waveform $x(t)$

- by a weighted summation of shifted pulses as

$$x(t) = \sum_k x_k u(t - kt_b),$$

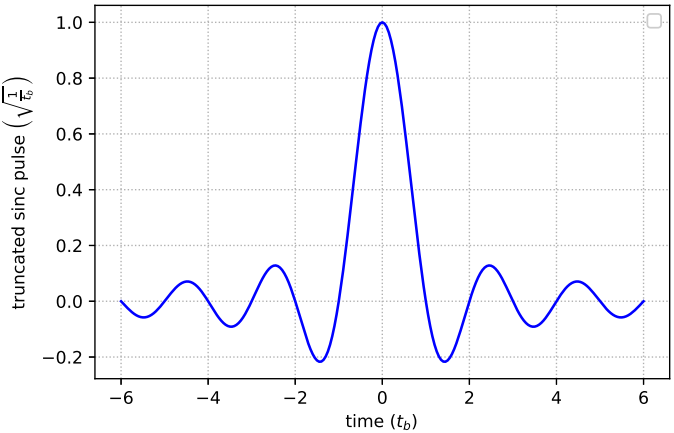
where t_b is called SYMBOL PERIOD; $\frac{1}{t_b}$ is SYMBOL RATE;


Data Symbols=[1 -1 -1 -1 1 1 -1 -1 -1 -1]
$x(t)$
Modulated Signal $\left(\sqrt{\frac{1}{t_b}}\right)$
time (t_b)
1.5
1.0
0.5
0.0
−0.5
−1.0
−5
0
5
10
15

- the pulse shape $u(t)$ looks similar to a normalized sinc function

$$\frac{1}{\sqrt{t_b}}\text{sinc}\left(t\frac{1}{t_b}\right);$$

$$\text{sinc}(x) \triangleq \frac{\sin(\pi x)}{\pi x}.$$


1.0
0.8
0.6
0.4
0.2
0.0
−0.2
truncated sinc pulse $\left(\sqrt{\frac{1}{t_b}}\right)$
−6
−4
−2
0
2
4
6
time (t_b)

- To facilitate low complexity receiver, the shifted PULSE SHAPES are desired to be orthogonal to each other, i.e., $\forall k_1 \neq k_2$,

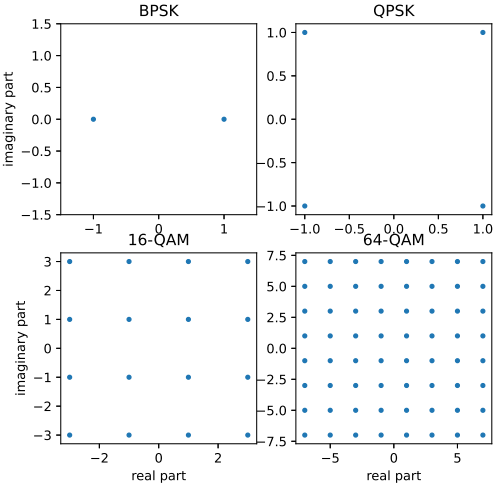
$$\begin{aligned} u(t-k_1 t_b) \quad &\perp \quad u(t-k_2 t_b) \Longleftrightarrow \\ \langle u(t-k_1 t_b), u(t-k_2 t_b)\rangle \quad &\triangleq \quad \int u(t-k_1 t_b) u^*(t-k_2 t_b) dt = 0, \end{aligned}$$

- We often normalize the pulse shape to have unit energy, i.e.,

$$\int |u(t)|^2 \, dt \quad = \quad 1.$$

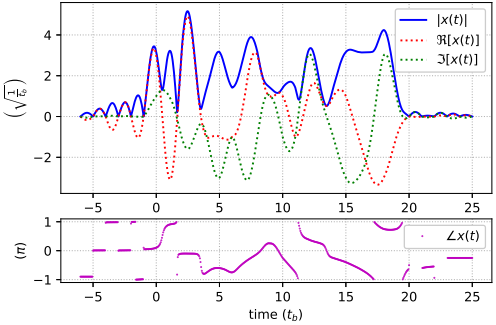
- There is a one-one mapping between the information message $\vec{i}$ and the symbols $\vec{x} = [x_k]_k$, which is also called a code word.
  - Note that the mapping of error control coded bits to symbols is also sometimes called modulation.
  - The conversion of baseband signals to passband signals is also sometimes called modulation.

- The symbols x_k's often belong to some finite size CONSTELLATION.


BPSK
QPSK
16-QAM
64-QAM
imaginary part
real part
imaginary part
real part

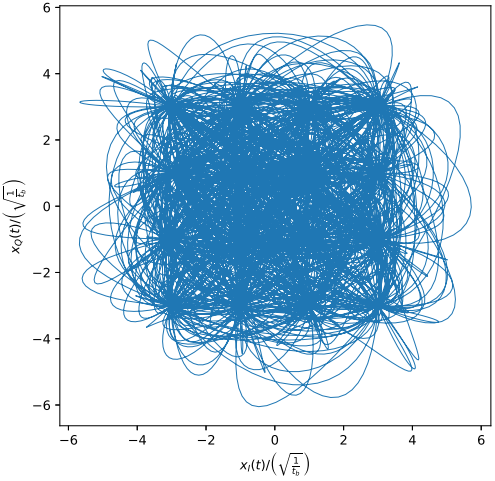
- For Binary Phase Shifted Keying (BPSK), $x_k \in \{1, -1\}$, a two point constellation.
- For Quadrature Phase Shifted Keying (QPSK), $x_k \in \{1+j, 1-j, -1+j, -1-j\}$, a four point constellation.
- For 16-Quadrature Amplitude Modulation (16-QAM), $x_k \in \{a + jb : \ a, b \in \{-3, -1, 1, 3\}\}$, a four point constellation.
- When signal to noise power ratio (SNR) is high, 4096-QAM could be used.

- Some examples of 16-QAM modulation are shown below.
  - A modulated signal of 20 symbols is



*

– The I/Q plot of a modulated signal of 1000 symbols is


6
4
2
0
−2
−4
−6
−6
−4
−2
0
2
4
6
$x_Q(t)/\left(\sqrt{\frac{1}{E_b}}\right)$
$x_I(t)/\left(\sqrt{\frac{1}{E_b}}\right)$

*

2.2.2.0.2. *Modulation.*

- We can formalize the above to obtain the following definitions.

DEFINITION 2.2.1. MODULATION is the process of mapping a discrete time signal $[x_k]_k$ to a continuous time signal $[x(t)]_{t\in\mathbb{R}}$.

2.2.2.0.3. *Linear stream modulation.*

Definition 2.2.2. The linear stream modulation (LSM) from $[x_k]_k$ to $[x(t)]_{t\in\mathbb{R}}$ is a weighted sum of *delayed* pulses, i.e.,

$$x(t) = \sum_k x_k u(t - kt_b).$$

Its Fourier transform is

$$\tilde{x}(f) = \sum_k x_k e^{-j2\pi f k t_b} \tilde{u}(f).$$

2.2.2.0.4. *Linear modulation*. Generalizing the linear stream modulation, we have the following.

DEFINITION 2.2.3. The LINEAR MODULATION from $[x_k]_k$ to $[x(t)]_{t\in\mathbb{R}}$ is a weighted sum of pulses, i.e.,

$$x(t) = \sum_k x_k u_k(t).$$

Its Fourier transform is

$$\tilde{x}(f) = \sum_k x_k \tilde{u}_k(f).$$

REMARK.

- Linear stream modulation is a special case of linear modulation with $u_k(t) = u(t - kt_b)$.
- It is often desirable to choose orthogonal pulse shapes such that $u_{k_1}(t) \perp u_{k_2}(t),\ k_1 \neq k_2$.

2.2.3. Sampling Theorem and the Generality of the Modulator

- Does the simple linear modulation lose generality, i.e., if we allow other (nonlinear) modulation, will we get a better transmitter?
  - To answer this question, we review the sampling theorem first.

2.2.3.0.1. *Sampling theorem.*

- Recall the sampling theorem below.

THEOREM 2.2.1. *If $x(t)$ is band limited to $(-\frac{1}{2t_s}, \frac{1}{2t_s})$, then for any chosen t, $x(t)$ can be reconstructed from its samples $\{x(kt_s) : k \in \mathbb{Z}\}$ by*

$$x(t) = \sum_k x(kt_s) sinc\left((t - kt_s)\frac{1}{t_s}\right). \tag{2.2.3.1}$$

2.2.3.0.2. *Generality of the linear modulator.*

PROBLEM 2.2.3.1. (*The Generality of the Linear Modulation*) Does there exist a transmitter mapping $\vec{i} \to x'(t)$, which maps a information sequence $\vec{i}$ to a band limited baseband signal $x'(t)$, that cannot be implemented by the linear modulator $\vec{i} \to \vec{x} = [x_k]_k \to x(t) = \sum_k x_k u_k(t - kt_b)$?

Solution:

- No.
  - Assume the $x'(t)$ is band limited to $[-\frac{b}{2}, \frac{b}{2}]$.
  - Let $t_b = \frac{1}{b}$, $x_k = \sqrt{t_b} x'(k t_b)$, and $u(t) = \frac{1}{\sqrt{t_b}} \text{sinc}\left(t \frac{1}{t_b}\right)$.
  - According to the sampling theorem, $x(t) = x'(t)$.
- Therefore, any transmitter mapping can be implemented by a linear modulator.

$\square$

2.2.3.0.3. *Degree of freedoms/s/Hz.*

PROBLEM 2.2.3.2. (*Degree of freedoms/s/Hz*) What is the maximum degree of freedoms/s/Hz, defined as the number of independent symbols that can be transmitted per second per Hz?

Solution:

- According to the sampling theorem, any signal with bandwidth b Hz, can be constructed with 1 symbol (complex number) per $\frac{1}{b}$ seconds.
  - Given n independent symbols, we can construct a signal of bandwidth of b Hz that needs at least $n\frac{1}{b}$ seconds to be transmitted.
  - Thus, the degree of freedoms/s/Hz is

$$n/(n\frac{1}{b})/b = 1 \text{ (symbols/s/Hz)}.$$

- The above is the ideal case. Using non-ideal pulse shape $u(t)$, which occupies larger band, the degree of freedom is less than 1.

- However, using multiple transmit and receive antennas, multiple spatial paths can be employed to transmit multiple symbols/s/Hz in the same spectrum.

□

2.2.3.0.4. *Exercises.*

EXERCISE 2.2.1. Assume $x(t)$ is band limited to $(-3, 7)$.

1. What is the minimum sampling frequency without aliasing?
2. How to reconstruct the signal from its samples?

Solution:

1. The minimum sampling frequency is $\frac{1}{t_s} = 7 - (-3) = 10$.
2. To reconstruct the signal from the samples, we could use a low pass filter centered at 2 with bandwidth 10. To reuse the fomula (2.2.3.1), which assume a filter centered at 0, we could move the signal band to be centered at 0 and move it back after the reconstruction:

$$\begin{aligned} x(t) &= e^{j2\pi 2t} \sum_k x(kt_s) e^{-j2\pi 2kt_s} \mathrm{sinc}\left((t - kt_s)\frac{1}{t_s}\right) \\ &= \sum_k x(kt_s) e^{j2\pi 2k(t-kt_s)} \mathrm{sinc}\left((t - kt_s)\frac{1}{t_s}\right). \end{aligned}$$

□

2.2.4. Power of Signals of Linear Modulation

- How to calculate the power of transmit signals of linear modulation?

2.2.4.0.1. *Signal power calculation.*

PROBLEM 2.2.4.1. (*Power of Linearly Modulated Signals*) In communications, we often want to minimize transmit signal power for a given date rate or maximize data rate for a given power. Therefore, it is important to calculate the transmit signal power.

- What is the power of linearly modulated signal?

Solution:

- Because the transmit signal depends on the information sequence, which is not deterministic, we should model the symbols $\boldsymbol{x}_k$'s as random variables.

- The **expected energy** of a linearly modulated signal is

$$
\begin{aligned}
e_{\boldsymbol{x}(t)} &= \mathrm{E}\left[\int |\boldsymbol{x}(t)|^2 dt\right] \\
&= \mathrm{E}\left[\int \left|\sum_k \boldsymbol{x}_k u_k(t)\right|^2 dt\right] \\
&= \mathrm{E}\left[\int \sum_{k_1} \boldsymbol{x}_{k_1} u_{k_1}(t) \sum_{k_2} \boldsymbol{x}^*_{k_2} u^*_{k_2}(t) dt\right] \\
&= \mathrm{E}\left[\sum_{k_1}\sum_{k_2} \boldsymbol{x}_{k_1} \left(\underbrace{\int u_{k_1}(t) u^*_{k_2}(t) dt}_{r_u(k_1,k_2)}\right) \boldsymbol{x}^*_{k_2}\right]
\end{aligned}
$$

$$
\begin{aligned}
&= \mathrm{E}\left[\underbrace{\left[\boldsymbol{x}^*_{k_2}\right]_{1,k_2}}_{\vec{\boldsymbol{x}}^\dagger}\underbrace{\left[r_u(k_1,k_2)\right]_{k_2,k_1}}_{R_u}\underbrace{\left[\boldsymbol{x}_{k_1}\right]_{k_1}}_{\vec{\boldsymbol{x}}}\right] \\
&= \mathrm{E}\left[\vec{\boldsymbol{x}}^\dagger R_u \vec{\boldsymbol{x}}\right] \\
&= \mathrm{E}\left[\left|R_u^{\frac{1}{2}}\vec{\boldsymbol{x}}\right|^2\right].
\end{aligned}
$$

– *Special case:* If the pulses are orthonormal, i.e., $r_u(k_1,k_2) = \delta[k_1 - k_2]$, where the discrete Dirac function is

$$
\delta[k] = \begin{cases} 1 & k = 0 \\ 0 & k \neq 0 \end{cases},
$$

then

$$e_{\boldsymbol{x}(t)} = \mathrm{E}\left[|\vec{\boldsymbol{x}}|^2\right] = \sum_k \mathrm{E}\left[|\boldsymbol{x}_k|^2\right].$$

- The **expected power** is

$$\frac{e_{\boldsymbol{x}(t)}}{\text{duration of } \boldsymbol{x}(t)}.$$

  - *Special case:*
    * For linear stream modulation with orthonormal pulses, where $u_k(t) = u(t - k_{t_b})$, the duration is approximately the number of symbols multiplies t_b;
    * We further assume a constant symbol energy $\mathrm{E}\left[|\boldsymbol{x}_k|^2\right] = e_s$.
    * Then, the expected power is $\frac{e_s}{t_b}$.

□

2.2.4.0.2. *Exercises.*

EXERCISE 2.2.2. Consider linearly modulated signal

$$\boldsymbol{x}(t) = \sum_k \boldsymbol{x}_k u_k(t).$$

If $u_k(t)$'s are not orthogonal but $\boldsymbol{x}_k$'s are zero mean, independent with variance e_s, calculate the energy of $\boldsymbol{x}(t)$.

Solution:

- The energy is

$$
\begin{aligned}
e_{\boldsymbol{x}(t)} &= \mathrm{E}\left[\int |\boldsymbol{x}(t)|^2 dt\right] \\
&= \mathrm{E}\left[\sum_{k_1}\sum_{k_2} \boldsymbol{x}_{k_1} r_u(k_1,k_2)\boldsymbol{x}^*_{k_2}\right] \\
&= \sum_{k_1}\sum_{k_2} r_u(k_1,k_2)\mathrm{E}\left[\boldsymbol{x}_{k_1}\boldsymbol{x}^*_{k_2}\right] \\
&= \sum_{k_1}\sum_{k_2} r_u(k_1,k_2) e_s \delta[k_1-k_2] \\
&= e_s \sum_{k_1} e_{u_k(t)}.
\end{aligned}
$$

□

2.2.5. Power Spectral Density of Signals of Linear Modulation

- How to calculate the power spectral density of transmit signals of linear modulation?
  - This will be used to determine the occupied bandwidth.

2.2.5.0.1. *Finding the power spectral density of linearly modulated signals.*

PROBLEM 2.2.5.1. (*Power Spectral Density of Linearly Modulated Signals*) In wireless communications, one can only transmit in an allocated band. Power spectral density of the transmit signal tells us what band the signal occupies. Therefore, it is important to calculate the transmit signal power spectral density.

- What is the power spectral density of linearly modulated signal?

Solution:

- Power spectral density is the expected power of a signal at frequency f, i.e.,

$$
\begin{aligned}
\mathrm{PSD}_{\boldsymbol{x}(t)}(f) &= \frac{1}{\text{duration of } \boldsymbol{x}(t)} \mathrm{E}\left[\left|\mathcal{F}\left(\boldsymbol{x}(t)\right)(f)\right|^2\right] \\
&= \frac{1}{\text{duration of } \boldsymbol{x}(t)} \mathrm{E}\left[\left|\mathcal{F}\left(\sum_l \boldsymbol{x}_l u_l(t)\right)(f)\right|^2\right] \\
&= \frac{1}{\text{duration of } \boldsymbol{x}(t)} \mathrm{E}\left[\left|\sum_l \boldsymbol{x}_l \tilde{u}_l(f)\right|^2\right] \\
&= \frac{\sum_{l_1}\sum_{l_2} \mathrm{E}\left[\boldsymbol{x}_{l_1}\boldsymbol{x}_{l_2}^*\right] \tilde{u}_{l_1}(f)\tilde{u}_{l_2}^*(f)}{\text{duration of } \boldsymbol{x}(t)}. \qquad (2.2.5.1)
\end{aligned}
$$

- We observe that both pulse shapes $u_l(t)$ and the correlation $\mathrm{E}\left[\boldsymbol{x}_{l_1}\boldsymbol{x}_{l_2}^*\right]$ can be designed to achieve certain desired power spectral density.
- *Special case:* If $\mathrm{E}\left[\boldsymbol{x}_{l_1}\boldsymbol{x}_{l_2}^*\right] = e_s\delta[l_1 - l_2]$, i.e., the symbols have zero correlation and constant energy e_s, then

$$\mathrm{PSD}_{\boldsymbol{x}(t)}(f) = \frac{e_s \sum_l |\tilde{u}_l(f)|^2}{\text{duration of } \boldsymbol{x}(t)}.$$

  - Furthermore, for *linear stream modulation*,
    * where $\tilde{u}_l(f) = \tilde{u}(f)e^{-j2\pi f l t_b}$ and the duration is approximately the number of symbols multiplies t_b, one has

$$\mathrm{PSD}_{\boldsymbol{x}(t)}(f) = \frac{e_s}{t_b}|\tilde{u}(f)|^2,$$

    which is determined by the pulse shape only.

* Thus, when *designing* the pulse shape for linear stream modulation, we want to minimize $|\tilde{u}(f)|^2$ for f outside the allocated band. Sinc function $u(t) = \frac{1}{\sqrt{t_b}}\text{sinc}\left(t\frac{1}{t_b}\right)$ is a perfect pulse shape in this regard but it has too large a tail in time domain. We will discuss the pulse shape design after the discussion of demodulator, which impose certain requirements on the pulse shape as well.

$\square$

2.2.5.0.2. *Exercises.*

EXERCISE 2.2.3.

1. Consider linear stream modulation

$$x(t) = \sum_{k=1}^{2} x_k u(t - kt_b),$$

where $u(t) = \mathrm{rect}_{t_b}(t)$, $x_1 = 2$, $x_2 = -3$.

1.1. Calculate the power of $x(t)$.

1.2. Calculate the power spectral density of $x(t)$. Is it band limited? Do you observe that the transmit symbols affect the spectrum as well?

2. Consider linear stream modulation with pulse

$$u(t) = \frac{1}{\sqrt{t_b}} \mathrm{sinc}\left(t \frac{1}{t_b}\right).$$

Calculate the PSD of the modulated signal. Is it band limited?

Solution:

1. 1.1.

$$
\begin{aligned}
\text{Power}_{x(t)} &= \frac{e_{\boldsymbol{x}(t)}}{\text{duration of } \boldsymbol{x}(t)} \\
&= \frac{\mathrm{E}\left[\int |\boldsymbol{x}(t)|^2 dt\right]}{\text{duration of } \boldsymbol{x}(t)} \\
&= \frac{2^2 t_b + (-3)^2 t_b}{2t_b} \\
&= 6.5
\end{aligned}
$$

1.2.

$$
\begin{aligned}
&\mathrm{PSD}_{\boldsymbol{x}(t)}(f) \\
= \quad & \frac{1}{\text{duration of } \boldsymbol{x}(t)} \mathrm{E}\left[\left|\mathcal{F}\left(\boldsymbol{x}(t)\right)(f)\right|^2\right] \\
= \quad & \frac{1}{\text{duration of } \boldsymbol{x}(t)} \mathrm{E}\left[\left|\mathcal{F}\left(\sum_l \boldsymbol{x}_l u_l(t)\right)(f)\right|^2\right] \\
= \quad & \frac{1}{\text{duration of } \boldsymbol{x}(t)} \mathrm{E}\left[\left|\sum_l \boldsymbol{x}_l \tilde{u}_l(f)\right|^2\right]
\end{aligned}
$$

$$
\begin{aligned}
= \quad & \frac{\left|2e^{-j2\pi f t_b} t_b \mathrm{sinc}\left(f t_b\right) - 3e^{-j2\pi f 2t_b} t_b \mathrm{sinc}\left(f t_b\right)\right|^2}{2t_b} \\
= \quad & \frac{t_b \left|2 - 3e^{-j2\pi f t_b}\right|^2 \left|\mathrm{sinc}\left(f t_b\right)\right|^2}{2}
\end{aligned}
$$

No. The PSD vanishes slowly as $|f|$ increases. Yes, the PSD is a function of the transmit symbols.

2.

$$\begin{aligned}
\mathrm{PSD}_{\boldsymbol{x}(t)}(f) &= \frac{e_s}{t_b}\left|\tilde{u}(f)\right|^2 \\
&= \frac{e_s}{t_b}\left|\sqrt{t_b}\mathrm{rect}_{\frac{1}{t_b}}(f)\right|^2 \\
&= e_s\mathrm{rect}_{\frac{1}{t_b}}(f).
\end{aligned}$$

Yes.

□

2.2.6. OFDM Modulation

- What is Orthogonal Frequency Division Multiplexing (OFDM) modulation and what is its power spectral density?
  - OFDM is often used to combat frequency selective fading channels and/or share the frequency spectrum among multiple users.

2.2.6.0.1. *The definition of OFDM modulation.*

DEFINITION 2.2.4. The ORTHOGONAL FREQUENCY DIVISION MULTIPLEXING (OFDM) MODULATION from $[x_k]_k$ to $[x(t)]_{t\in\mathbb{R}}$ is a linear *stream* modulation in frequency domain,

$$\tilde{x}(f) = \sum_l x_l \tilde{u}(f - lf_\Delta - f_0),$$

whose inverse Fourier transform is

$$x(t) = \left(\sum_l x_l e^{j2\pi(lf_\Delta + f_0)t}\right) u(t),$$

where $u(t)$ can be viewed as a window function. It is a linear modulation in time domain with wave forms

$$\begin{aligned} u_l(t) &= e^{j2\pi(lf_\Delta + f_0)t} u(t), \\ \tilde{u}_l(f) &= \tilde{u}(f - lf_\Delta - f_0). \end{aligned}$$

2.2.6.0.2. *An example pulse.*

EXAMPLE 2.2.2. We may let

$$\tilde{u}(f) = \frac{1}{\sqrt{f_\Delta}} \mathrm{sinc}\left(f \frac{1}{f_\Delta}\right)$$

so that the orthogonality

$$\tilde{u}_{l_1}(f) \perp \tilde{u}_{l_2}(f), \ \forall l_1 \neq l_2.$$

holds.

- With such kind of choices, symbol x_l is said to be multiplexed on l-th SUBCARRIER centered at $lf_\Delta + f_0$, where signals $x_l \tilde{u}(f - lf_\Delta - f_0)$ of different subcarriers, though overlapping, are orthogonal to each other. This is why the modulation is called orthogonal frequency division multiplexing.

- f_Δ is the spacing between adjacent subcarriers. The bandwidth is approximately the number of symbols multiplying f_Δ.
- Note that the corresponding $u(t) = \sqrt{f_\Delta}\mathrm{rect}_{\frac{1}{f_\Delta}}(t)$ is a square window.
- According to the sampling theorem in frequency domain, this modulation can duplicate any signal with finite duration. Thus, there is no loss of generality.

2.2.6.0.3. *Finding the power spectral density of OFDM signals*.

PROBLEM 2.2.6.1. (*Power Spectral Density of OFDM Signals*)

- What is the power spectral density of OFDM signals?

Solution:

- According to equation (2.2.5.1), the power spectral density of an OFDM modulated signal

$$\mathrm{PSD}_{\boldsymbol{x}(t)}(f) \quad = \quad \frac{\sum_{l_1}\sum_{l_2} \mathrm{E}\left[\boldsymbol{x}_{l_1}\boldsymbol{x}_{l_2}^*\right] \tilde{u}_{l_1}(f)\tilde{u}_{l_2}^*(f)}{\text{duration of } u(t)}.$$

- *Special case:* If $\mathrm{E}\left[\boldsymbol{x}_{l_1}\boldsymbol{x}_{l_2}^*\right] = e_s\delta[l_1 - l_2]$, i.e., the symbols have zero correlation and constant energy e_s, then

$$\mathrm{PSD}_{\boldsymbol{x}(t)}(f) \quad = \quad \frac{e_s \sum_l \left|\tilde{u}(f - lf_\Delta - f_0)\right|^2}{\text{duration of } u(t)}. \qquad (2.2.6.1)$$

  - When *designing* the pulse shape for OFDM modulation, frequency domain sinc function $\tilde{u}(f) = \frac{1}{\sqrt{f_\Delta}}\mathrm{sinc}\left(f\frac{1}{f_\Delta}\right)$ is a bad choice because it has too large a tail, leaking power outside the allocated band.

□

2.2.6.0.4. *A different view of OFDM using Dirac functions.*

PROBLEM 2.2.6.2. (*An Alternative View of OFDM Modulation and the Design of the Window Function*) We may use the Dirac pulse in frequency domain for OFDM modulation,

$$\tilde{x}(f) = \sum_{l} x_l \delta(f - l f_\Delta - f_0).$$

The advantage of this choice is that $\{\delta(f - l f_\Delta - f_0) : l \in \mathbb{Z}\}$ will remain orthogonal through the frequency selective fading channels, as we will discuss later.

1. What is the disadvantage of using Dirac function as frequency domain pulse?
2. How to mitigate it?
3. How to design the pulses for OFDM modulation so that the pulses are orthogonal and have tight spectrum?

Solution:

1. The disadvantage is that the time domain signal

$$x(t) = \sum_l x_l e^{j2\pi(lf_\Delta + f_0)t}$$

   has infinite support with non-diminishing power.

2. To mitigate it, we should multiply a window function $u(t)$ of finite support in time domain so that

$$x(t) = \left(\sum_l x_l e^{j2\pi(lf_\Delta + f_0)t}\right) u(t).$$

3. To design $u(t)$, note that $\sum_l x_l e^{j2\pi(lf_\Delta + f_0)t}$ is periodic with period $\frac{1}{f_\Delta}$. As long as $u(t)$ is constant for more than a period, all the information and properties of $\sum_l x_l e^{j2\pi(lf_\Delta + f_0)t}$ is preserved. For example, let $t_0 < t_1 < t_2 < t_3$ where

$t_2 = t_1 + \tau_h + \frac{1}{f_\Delta}$. We may choose

$$u(t) \quad = \quad \begin{cases} c & t_1 \leq t \leq t_2 \\ 0 & t < t_0 \text{ or } t > t_3 \end{cases},$$

- where the value of $u(t)$ for $t \in (t_0, t_1)$ or (t_2, t_3) is chosen to have a smooth transition to zero so that $|\tilde{u}(f)|^2$ has small tails.
- The part of $u(t)$, $t \in [t_1, t_1+\tau_h]$, is called a CYCLIC PREFIX as it has the same value as that of $t \in [t_2 - \tau_h, t_2]$. The prefix length τ_h is chosen to be larger than the frequency selective fading channel's impulse response's support length. The reason is discussed in the demodulation of OFDM signals in frequency selective fading channels.

- The signal $x(t)$, $t \in [t_0, t_3]$, is called an OFDM SYMBOL. The OFDM symbol period is $t_b = t_3 - t_0$.
- Such $u(t)$ may result in non-orthogonal $\tilde{u}_l(f) = \tilde{u}(f - lf_\Delta - f_0)$, $\forall l$. A suboptimal receiver get around of this by discarding received signals outside time interval $[t_1 + \tau_h, t_2]$, effectively obtaining a square window and sinc pulses in frequency domain, which are orthogonal.

$\square$

2.2.6.0.5. *Exercises.*

EXERCISE 2.2.4. Assume $x_{-1} = 1$, $x_0 = -1$, $x_1 = 1$, $u(t) = \sqrt{f_\Delta}\mathrm{rect}_{\frac{1}{f_\Delta}}(t)$, $f_0 = 0$, $f_\Delta = 400$ kHz. Calculate the OFDM modulated signal $x(t)$.

Solution:

$$
\begin{aligned}
x(t) &= \left(\sum_l x_l e^{j2\pi(lf_\Delta + f_0)t}\right) u(t) \\
&= \left(e^{-j2\pi 4\times 10^5 t} - 1 + e^{j2\pi 4\times 10^5 t}\right) \\
&\quad \times 200\sqrt{10}\mathrm{rect}_{2.5\times 10^{-6}}(t) \\
&= \left(\cos\left(2\pi 4 \times 10^5 t\right) - 1\right) \\
&\quad \times 200\sqrt{10}\mathrm{rect}_{2.5\times 10^{-6}}(t).
\end{aligned}
$$

$\square$

2.2.7. OFDM+LSM Modulation

- How to combine OFDM and LSM to utilize both frequency and time resources?

2.2.7.0.1. *OFDM+LSM Modulation.*

PROBLEM 2.2.7.1. (*OFDM+LSM Modulation*)

1. Combine OFDM and linear stream modulation to produce a modulation method.
2. Analyze its power spectral density.

Solution:

1. Recall that the signal of one OFDM symbol period is $\left(\sum_l x_l e^{j2\pi(lf_\Delta+j}\right.$ Let t_b be the OFDM symbol period. We can delay the an OFDM symbol by kt_b and transmit different symbols. The OFDM+LSM signal is

$$x(t) = \sum_k \sum_l x_{k,l} \underbrace{u(t-kt_b)e^{j2\pi(lf_\Delta+f_0)(t-kt_b)}}_{u_{k,l}(t)}, \qquad (2.2.7.1)$$

whose Fourier transform is

$$\tilde{x}(f) = \sum_k \sum_l x_{k,l} \tilde{u}(f-lf_\Delta-f_0)e^{-j2\pi fkt_b}.$$

2. Observe that equation (2.2.7.1) is a linear stream modulation indexed by k, l. The PSD result in equation (2.2.5.1) applies. Alternatively, assume $l = 1, 2, ..., n$. Since the OFDM symbols have no overlap, their expected energy in frequency

domain adds up to be n times that of one OFDM symbol, while the duration is a nt_b. If the symbols of an OFDM symbol have zero correlation and constant energy e_s, then the power spectral density

$$\mathrm{PSD}_{\boldsymbol{x}(t)}(f) \quad = \quad \frac{e_s \sum_l |\tilde{u}(f - lf_\Delta - f_0)|^2}{t_b}$$

is the same as equation (2.2.6.1).

□

2.2.8. OCDM Modulation

- For linear modulation, can we create pulses $u_k(t)$'s that occupy the same part of time and frequency but are orthogonal?

2.2.8.0.1. *Definition of orthogonal code division multiplexing (OCDM) modulation.*

Definition 2.2.5. The Orthogonal Code Division Multiplexing (OCDM) modulation from $[x_k]_k$ to $[x(t)]_{t\in\mathbb{R}}$ is a linear modulation

$$\boldsymbol{x}(t) \quad = \quad \sum_l \boldsymbol{x}_l u_l(t),$$

with orthogonal pulses, i.e., $u_l(t) \perp u_k(t),\ l \neq k$, occupying the same frequency spectrum and time duration.

2.2.8.0.2. *Example.*

EXAMPLE 2.2.3. The two wave pulses in figure 2.2.8.1 are possible OCDM pulse shapes.

- Observe that they have sharp edges in time domain, implying large bandwidth occupation in frequency domain. Therefore, they need to be improved to be used in practice.

2.2.8.0.3. *Example.*

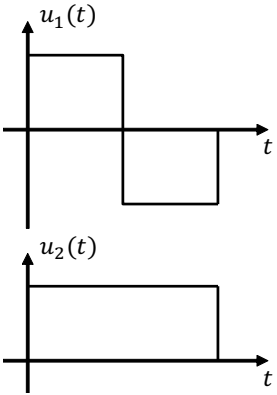


FIGURE 2.2.8.1. Two orthogonal pulses

EXAMPLE 2.2.4.

2.2.8.0.4. *How to construct orthogonal pulses?*

PROBLEM 2.2.8.1. (*Construction of OCDM Pulses*) Can you use orthonormal sequences $\{\vec{c}_k \in \mathbb{C}^{m_s \times 1} : \langle \vec{c}_{k_1}, \vec{c}_{k_2} \rangle = \delta[k_1 - k_2]\}$ and orthonormal pulses $\{u(t - mt_c) : \langle u(t - m_1 t_c), u(t - m_2 t_c) \rangle = \delta[m_1 - m_2]\}$ to construct orthogonal pulses with support length roughly $m_s t_c$?

Solution:

- We may use the linear stream modulation to modulate each sequence $\vec{c}_k = [c_{k,m}]_m$ as

$$u_k(t) = \sum_{m=1}^{m_s} c_{k,m} u(t - mt_c).$$

- To verify the orthogonality, we calculate

$$
\begin{aligned}
& \langle u_{k_1}(t), u_{k_2}(t) \rangle \\
= & \left\langle \sum_{m_1=1}^{m_s} c_{k_1,m_1} u(t - m_1 t_c), \sum_{m_2=1}^{m_s} c_{k_2,m_2} u(t - m_2 t_c) \right\rangle \\
= & \sum_{m_2=1}^{m_s} \sum_{m_1=1}^{m_s} c_{k_1,m_1} c^*_{k_2,m_2} \langle u(t - m_1 t_c), u(t - m_2 t_c) \rangle \\
= & \sum_{m_2=1}^{m_s} \sum_{m_1=1}^{m_s} c_{k_1,m_1} c^*_{k_2,m_2} \delta[m_1 - m_2] \\
= & \sum_{m_1} c_{k_1,m_1} c^*_{k_2,m_1} \\
= & \delta[k_1 - k_2].
\end{aligned}
$$

- Incidentally, since Fourier transform doesn't change inner products,

$$\tilde{u}_k(f) = \sum_m c_{k,m} e^{-j2\pi f m t_c} \tilde{u}(f)$$

are orthonormal pulses and so are

$$u'_k(t) = \sum_m c_{k,m} e^{-j2\pi t m t_c} \tilde{u}(t),$$

whose duration is approximately $\frac{1}{t_c}$.

$\square$

2.2.8.0.5. *Example: pulses from Hadamard matrices.*

EXAMPLE 2.2.5. • If $m_s = 2^n$,the matrix $[\vec{c}_k]_{1,k=1:m_s}$ of the orthogonal sequences could be real valued and equal to

$$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, n = 1 \quad , \quad \frac{1}{\sqrt{4}}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}, n = 2, ...,$$

where the Hadamard matrices can be constructed iteratively as

$$M_{n+1} = \begin{bmatrix} M_n & M_n \\ M_n & -M_n \end{bmatrix}.$$

- If $m_s \neq 2^n$,the matrix $[\vec{c}_k]_{1,k=1:m_s}$ of the orthogonal sequences could be the discrete Fourier transform matrix

$$[\vec{c}_k]_{1,k=1:m_s} = \frac{1}{\sqrt{m_s}} \left[e^{j2\pi \frac{km}{m_s}} \right]_{m=0:m_s-1,k=0:m_s-1} . \quad (2.2.8.1)$$

- The pulse could be

$$u(t) = \frac{1}{\sqrt{t_c}} \text{sinc}\left(t \frac{1}{t_c}\right),$$

which has a bandwidth of $\frac{1}{t_c}$.

2.2.8.0.6. *Exercises.*

EXERCISE 2.2.5. For $m_s = 3$, use (2.2.8.1) to find $[\vec{c}_k]_{1,k=1:m_s}$.

Solution:

$$
\begin{aligned}
\left[\vec{c}_k\right]_{1,k=1:m_s} &= \frac{1}{\sqrt{m_s}}\left[e^{j2\pi\frac{km}{m_s}}\right]_{m=0:m_s-1,k=0:m_s-1} \\
&= \frac{1}{\sqrt{3}}\left[e^{j2\pi\frac{km}{3}}\right]_{m=0:2,k=0:2} \\
&= \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & e^{j2\pi\frac{1}{3}} & e^{j2\pi\frac{2}{3}} \\ 1 & e^{j2\pi\frac{2}{3}} & e^{j2\pi\frac{4}{3}} \end{bmatrix}.
\end{aligned}
$$

$\square$

2.2.9. OCDM+LSM Modulation

- How to combine the Orthogonal Code Division Multiplexing (OCDM) modulation with the linear stream modulation (LSM)?
- How does the power spectral density look like? What is spectrum spreading?
  - OCDM is often used to combat frequency selective fading in time domain and/or share the frequency spectrum among multiple users.

2.2.9.0.1. *OCDM+LSM Modulation.*

PROBLEM 2.2.9.1. (*OCDM+LSM Modulation*)

1. *Modulation:* Combine OCDM and linear stream modulation to produce a modulation method.
2. What orthogonality of the pulses do you desire?
3. Give an example of OCDM+LSM signal using pulses

$$u_k(t) = \sum_{m=1}^{m_s} c_{k,m} u(t - mt_c),$$

created from linear stream modulation of orthonormal codes $\vec{c}_k$'s.

4. *PSD:* Analyze the power spectral density of OCDM+LSM signals when the pulses are created from linear stream modulation of orthonormal codes $\vec{c}_k$'s.
5. *Bandwidth:* Compare the power spectral density of OCDM+LSM with that of LSM with the same symbol period t_b. Which

one occupies larger bandwidth? Does OCDM waste spectrum?

Solution:

1. *OCDM+LSM:* Recall that the OCDM modulated signal is $\tilde{x}(f) = \sum_k x_k u_k(t)$, where $u_l(t)$'s are orthonormal. Let t_b be the OCDM symbol period. We can delay the an OCDM symbol by lt_b and transmit different symbols. As a result, the OCDM+LSM signal is

$$x(t) = \sum_k \sum_l x_{k,l} \underbrace{u_k(t - lt_b)}_{u_{k,l}(t)}, \tag{2.2.9.1}$$

   which is a linear modulation with pulses indexed by k, l.
   - If $\boldsymbol{x}(t)$ is transmitted from a base station, $\boldsymbol{x}_{k,l}$ may be information intended for the k-th user. Thus, the signals for different users are orthogonal and separable at a user.
2. To reduce complexity as discussed later, we want all the delayed pulses to be orthogonal, i.e., $u_{k_1}(t - l_1 t_b) \perp u_{k_2}(t -$

$l_2 t_b)$, $l_1 \neq l_2$ or $k_1 \neq k_2$. If we further normalize each pulse energy to be one, we have

$$\langle u_{k_1}(t - l_1 t_b), u_{k_2}(t - l_2 t_b)\rangle \quad = \quad \delta[l_1 - l_2]\delta[k_1 - k_2].$$

3. If $\langle u(t - m_1 t_c), u(t - m_2 t_c)\rangle = \delta[m_1 - m_2]$ and $t_b = m_s t_c$, then the orthogonality $\langle u_{k_1}(t - l_1 t_b), u_{k_2}(t - l_2 t_b)\rangle = \delta[l_1 - l_2]\delta[k_1 - k_2]$ holds. An example of the base pulse is

$$u(t) \quad = \quad \frac{1}{\sqrt{t_c}} \mathrm{sinc}\left(t \frac{1}{t_c}\right).$$

4. *PSD:* Observe that equation (2.2.9.1) is a linear stream modulation indexed by k, l. The PSD result in equation (2.2.5.1) applies and we obtain

$$\mathrm{PSD}_{\boldsymbol{x}(t)}(f) \quad = \quad \frac{\sum_{l_1,l_2,k_1,k_2} \mathrm{E}\left[\boldsymbol{x}_{k_1,l_1} \boldsymbol{x}^*_{k_2,l_2}\right] \tilde{u}_{k_1}(f) e^{-j2\pi f l_1 t_b} \tilde{u}^*_{k_2}(f) e^{j2\pi f l_2 t_b}}{\text{duration of } \boldsymbol{x}(t)}.$$

- *Special Cases:*

– If the symbols have zero correlation, i.e., $\mathrm{E}\left[\boldsymbol{x}_{k_1,l_1}\boldsymbol{x}^*_{k_2,l_2}\right] = e_s\delta[l_1-l_2]\delta[k_1-k_2]$,

$$\begin{aligned}\mathrm{PSD}_{\boldsymbol{x}(t)}(f) &= \frac{\sum_{k_1} ne_s\left|\tilde{u}_{k_1}(f)\right|^2}{nt_b} \\ &= \frac{e_s\sum_k\left|\tilde{u}_k(f)\right|^2}{t_b},\end{aligned}$$

where n is the number of OCDM symbol indexed by l.

* Further, if the orthonormal pulses are built from orthonormal codes, i.e., $u_k(t) = \sum_{m=1}^{m_s} c_{k,m}u(t-mt_c)$, and if $k = 1, ..., m_2$,

$$\tilde{u}_k(f) = \sum_m c_{k,m}e^{-j2\pi fmt_c}\tilde{u}(f),$$

$$|\tilde{u}_k(f)|^2 = \sum_{m_1}\sum_{m_2} c_{k,m_1} c^*_{k,m_2} e^{-j2\pi f(m_1-m_2)t_c} |\tilde{u}(f)|^2,$$

$$\begin{aligned}
\sum_{k=1}^{m_s} |\tilde{u}_k(f)|^2 &= \sum_{m_1=1}^{m_s}\sum_{m_2=1}^{m_s}\sum_{k=1}^{m_s} c_{k,m_1} c^*_{k,m_2} e^{-j2\pi f(m_1-m_2)t_c} |\tilde{u}(f)|^2 \\
&= \sum_{m_1=1}^{m_s}\sum_{m_2=1}^{m_s} \delta[m_1-m_2] e^{-j2\pi f(m_1-m_2)t_c} |\tilde{u}(f)|^2 \qquad (2.2.9.2) \\
&= m_s |\tilde{u}(f)|^2
\end{aligned}$$

where equation (2.2.9.2) follows from that the columns of matrix $[\vec{c}_k]_{1,k=1:m_s}$ are orthonormal. Therefore,

$$\mathrm{PSD}_{\boldsymbol{x}(t)}(f) = \frac{e_s m_s \, |\tilde{u}(f)|^2}{t_b},$$

which makes sense because m_s symbols of e_s energy is transmitted within t_b and $\int \mathrm{PSD}_{\boldsymbol{x}(t)}(f)df = \frac{e_s m_s}{t_b}$, where $\int |\tilde{u}(f)|^2 \, df = 1$ due to unit energy $u(t)$.

5. *Bandwidth:*
   - Recall the PSD of LSM,

$$\mathrm{PSD}_{\boldsymbol{x}(t)}(f) = \frac{e_s}{t_b} |\tilde{u}'(f)|^2$$

For sinc pulse,

$$u'(t) = \frac{1}{\sqrt{t_b}} \mathrm{sinc}\left(t \frac{1}{t_b}\right),$$

we have

$$\tilde{u}'(f) = \sqrt{t_b} \mathrm{rect}_{\frac{1}{t_b}}(f).$$

Thus, the bandwidth is $\frac{1}{t_b}$.

- For OCDM+LSM with sinc base pulse,

$$u(t) = \frac{1}{\sqrt{t_c}}\text{sinc}\left(t\frac{1}{t_c}\right),$$

we have

$$\tilde{u}(f) = \sqrt{t_c}\text{rect}_{\frac{1}{t_c}}(f).$$

Thus, the bandwidth is $\frac{1}{t_c} = m_s\frac{1}{t_b}$, which is m_s times the bandwidth of LSM. This is why

  - $\vec{c}_k$'s are called SPREADING CODES,
  - OCDM is called SPREAD SPECTRUM COMMUNICATIONS,
  - and m_s is called the SPREADING FACTOR. It is also called SPREADING GAIN, for reasons that will be clear in the discussion of demodulation.

- Note that OCDM also transmit m_s times symbols of LSM within t_b. Thus, we are not wasting the spectrum.

$\square$

2.2.9.0.2. *Exercises.*

EXERCISE 2.2.6.

1. In UMTS mobile communication standard, a possible spreading factor is $m_s = 256$ and the symbol period is $t_b = \frac{1}{15\text{k}}$, calculate the chip rate $\frac{1}{t_c}$, which will be the signal bandwidth if the sinc pulse is used.
2. Does the pulse $u_k(t - lt_b)$ and $u_k(t - (l+1)t_b)$ have overlap?

Solution:

- The chip rate is

$$\begin{aligned} \frac{1}{t_c} &= \frac{1}{\frac{t_b}{m_s}} \\ &= 256 * 15\text{k} = 3.84\text{M}. \end{aligned}$$

- Yes. In practice, a pulse's support is more than one symbol time.

□

2.2.10. Summary

2.2.10.0.1. *Linear Modulation.*

- $\xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)=\sum_l \boldsymbol{x}_l u_l(t)}$

$$\begin{aligned} \boldsymbol{x}(t) &= \sum_l \boldsymbol{x}_l u_l(t) \\ \tilde{\boldsymbol{x}}(f) &= \sum_l \boldsymbol{x}_l \tilde{u}_l(f) \end{aligned}$$

- If the symbols have zero correlation and constant energy e_s, then the power spectral density is

$$\mathrm{PSD}_{\mathrm{LM}}(f) = \frac{e_s \sum_l |\tilde{u}_l(f)|^2}{\text{duration of } \boldsymbol{x}(t)}.$$

2.2.10.0.2. *Linear Stream Modulation (LSM)* .

- $\xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)=\sum_l \boldsymbol{x}_l u(t-lt_b)}$

$$\boldsymbol{x}(t) = \sum_l \boldsymbol{x}_l u(t-lt_b)$$

$$\tilde{\boldsymbol{x}}(f) = \sum_l \boldsymbol{x}_l e^{-j2\pi f l t_b} \tilde{u}(f)$$

- $$\mathrm{PSD}_{\mathrm{LSM}}(f) = \frac{e_s}{t_b} \left|\tilde{u}(f)\right|^2$$

- Example

  –

$$u(t) = \frac{1}{\sqrt{t_b}} \mathrm{sinc}\left(t \frac{1}{t_b}\right)$$

2.2.10.0.3. *Orthogonal Frequency Division Multiplexing (OFDM).*

- $\xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)=\left(\sum_l \boldsymbol{x}_l e^{j2\pi(lf_\Delta+f_0)t}\right)u(t)}$

$$\tilde{\boldsymbol{x}}(f) = \sum_l \boldsymbol{x}_l \tilde{u}(f - lf_\Delta - f_0)$$

$$\boldsymbol{x}(t) = \left(\sum_l \boldsymbol{x}_l e^{j2\pi(lf_\Delta+f_0)t}\right) u(t),$$

- $$\mathrm{PSD}_{\mathrm{OFDM}}(f) = \frac{e_s \sum_l |\tilde{u}(f - lf_\Delta - f_0)|^2}{\text{duration of } u(t)}$$

- Examples

–

$$\tilde{u}(f) = \frac{1}{\sqrt{f_\Delta}} \mathrm{sinc}\left(f \frac{1}{f_\Delta}\right)$$

$$u(t) = \sqrt{f_\Delta} \mathrm{rect}_{\frac{1}{f_\Delta}}(t)$$

– or

$$u(t) = \begin{cases} c & t_1 \leq t \leq t_2 \\ 0 & t < t_0 \text{ or } t > t_3 \end{cases},$$

where $t_0 < t_1 < t_2 = t_1 + \tau_h + \frac{1}{f_\Delta} < t_3$; the value of $u(t)$ for $t \in (t_0, t_1)$ or (t_2, t_3) is chosen to have a smooth transition to zero so that $|\tilde{u}(f)|^2$ has small tails. The part of $u(t)$, $t \in [t_1, t_1 + \tau_h]$, is called a CYCLIC PREFIX.

2.2.10.0.4. *Orthogonal Code Division Multiplexing (OCDM)*

.

- **OCDM**

$$\boldsymbol{x}(t) \quad = \quad \sum_{l} \boldsymbol{x}_l u_l(t),$$

where $u_l(t) \perp u_k(t)$, $l \neq k$, occupying the same frequency spectrum and time duration.

- **OCDM+LSM**

$$- \xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)=\sum_k \sum_l \boldsymbol{x}_{k,l} u_k(t-lt_b)}$$

$$\boldsymbol{x}(t) \quad = \quad \sum_{k} \sum_{l} \boldsymbol{x}_{k,l} \underbrace{u_k(t - lt_b)}_{u_{k,l}(t)}$$

- Examples

–

$$u_k(t) = \sum_{m=1}^{m_s} c_{k,m} u(t - mt_c),$$

where vectors $\vec{c}_k = [c_{k,m}]_m$'s are orthonormal,

–

$$u(t) = \frac{1}{\sqrt{t_c}} \mathrm{sinc}\left(t \frac{1}{t_c}\right),$$

–

$$\mathrm{PSD}_{\mathrm{OCDM}}(f) = \frac{e_s m_s \left|\tilde{u}(f)\right|^2}{t_b},$$

which has a bandwidth of $\frac{1}{t_c}$.

2.2.10.0.5. *Optimization of Modulator.*

- You may have noticed that not much except for the pulse $u(t)$ can be optimized for linear modulation.
- After the study of demodulator, we will study how to design $u(t)$ to have low complexity receiver and high spectral efficiency.

Appendices

2.2.A. Homework Problems

PROBLEM 2.2.A.1. *

(*Implementation of a Linear Modulator*) Plot realizations of signals of linear stream modulation,

$$\boldsymbol{x}(t) \quad = \quad \sum_{l=1}^{n} \boldsymbol{x}_l u(t - lt_b),$$

for various cases so that you gain pictorial impression of them. If $x(t)$ is complex valued, you can plot $|x(t)|$, $\angle x(t)$, $x_{\mathrm{I}}(t)$, $x_{\mathrm{Q}}(t)$. You may also plot the Fourier transform $\tilde{x}(f)$.

- You may select $n = 7$,
- x_l from BPSK, QPSK, 8-PSK, 16-QAM constellations,
- $u(t) = \mathrm{sinc}(t\frac{1}{t_b})$ or square waves or any pulse you find online.

PROBLEM 2.2.A.2.

(*Energy of Constellations*) Assume symbol $\boldsymbol{x} \in S$ is uniformly distributed over constellation S. For each of the following constellations, calculate the expected symbol energy $\mathrm{E}\left[|\boldsymbol{x}|^2\right]$ and choose a such that the expected energy of $a\boldsymbol{x}$ is 1.

1. BPSK: $S = \{m : m \in \{-1, 1\}\}$:
2. QPSK: $S = \{m_{\mathrm{I}} + jm_{\mathrm{Q}} : m_{\mathrm{I}}, m_{\mathrm{Q}} \in \{-1, 1\}\}$
3. 8-PSK: $S = \left\{e^{jm\frac{2\pi}{8}} : m \in \{0, 1, ..., 7\}\right\}$
4. 16-QAM: $S = \{m_{\mathrm{I}} + jm_{\mathrm{Q}} : m_{\mathrm{I}}, m_{\mathrm{Q}} \in \{-3, -1, 1, 3\}\}$

PROBLEM 2.2.A.3. *

(*Plotting the Power Spectral Density* of LSM) Find and plot the power spectral density of signals of linear stream modulation,

$$x(t) = \sum_{l=1}^{n} x_l u(t - lt_b).$$

Assume the symbol energy $e_s = 1$, symbol rate $\frac{1}{t_b} = 5$ MHz, and

$$u(t) = \frac{1}{\sqrt{t_b}} \mathrm{sinc}\left(t \frac{1}{t_b}\right).$$

What is the bandwidth?

PROBLEM 2.2.A.4. *

(*Estimate the Power Spectral Density* of LSM) We know the power spectral density of signals of linear stream modulation

$$\boldsymbol{x}(t) = \sum_{l=1}^{n} \boldsymbol{x}_l u(t - lt_b)$$

is

$$\mathrm{PSD}(f) = \frac{e_s}{t_b} \left|\tilde{u}(f)\right|^2,$$

when the symbols $\boldsymbol{x}_l$'s are independently drawn from a finite size constellation, and have zero mean and constant energy $\mathrm{E}\left[\left|\boldsymbol{x}_l\right|^2\right] =$

e_s. One of your colleagues claims that he can build a digital PSD analyzer by simply plotting

$$\frac{\left|\mathcal{F}\left(x(t)\right)(f)\right|^2}{nt_b}$$

for a realization of $\boldsymbol{x}(t)$ because the empirical power spectral density

$$\mathbf{PSD}(f) \triangleq \frac{\left|\mathcal{F}\left(\boldsymbol{x}(t)\right)(f)\right|^2}{nt_b} \overset{n\to\infty}{\longrightarrow} \mathrm{PSD}(f).$$

Prove or disprove his claim. Hint: Do the following.

1. Plot multiple realizations of the empirical power spectral density $\mathbf{PSD}(f)$, their average, and $\mathrm{PSD}(f)$ in a graph. Repeat it for increasing n's. Repeat it for real valued and complex valued constellations. Form your own conjecture.

2. Observe

$$\frac{\left|\mathcal{F}\left(\boldsymbol{x}(t)\right)(f)\right|^2}{nt_b}$$

$$= \underbrace{\frac{\left|\sum_{l=1}^{n} \boldsymbol{x}_l e^{-j2\pi f l t_b}\right|^2}{n}}_{\boldsymbol{z}_n(f)} \frac{|\tilde{u}(f)|^2}{t_b}.$$

Examine whether $\mathrm{Var}\left(\boldsymbol{z}_n(f)\right) \stackrel{n\to\infty}{\longrightarrow} 0$. [Hint: There will be four fold summations. Check the cases when some of the indexes are equal.]

PROBLEM 2.2.A.5. *

(*Correlation Functions of Signals* of LSM) Define the correlation function

$$r_{\boldsymbol{x}}(t_1, t_2) \quad = \quad \mathrm{E}\left[\boldsymbol{x}(t_1)\boldsymbol{x}^*(t_2)\right]$$

of a linear stream modulation signal

$$\boldsymbol{x}(t) \quad = \quad \sum_{l=1}^{n} \boldsymbol{x}_l u(t - lt_b),$$

where $\boldsymbol{x}_l$'s are i.i.d. random variables with zero mean and variance e_s.

1. (6/10 pts) Calculate $r_{\boldsymbol{x}}(t_1, t_2)$.
2. (2/10 pts) Calculate $r_{\boldsymbol{x}}(t_1, t_1)$.
3. (2/10 pts) Calculate $r_{\boldsymbol{x}}(t_1+t_b, t_2+t_b)$. What do you observe?

PROBLEM 2.2.A.6. *

(*Implementation of OFDM Modulator*) Plot realizations of signals of OFDM modulation,

$$\tilde{\boldsymbol{x}}(f) \quad = \quad \sum_{l} \boldsymbol{x}_l \tilde{u}(f - lf_\Delta - f_0)$$

$$\boldsymbol{x}(t) \quad = \quad \left(\sum_l \boldsymbol{x}_l e^{j2\pi(lf_\Delta + f_0)t} \right) u(t),$$

for both frequency and time domains and for various cases so that you gain pictorial impression of them. If $x(t)$ is complex valued, you can plot $|x(t)|$, $\angle x(t)$, $x_{\mathrm{I}}(t)$, $x_{\mathrm{Q}}(t)$.

- You may select $\boldsymbol{x}_l$ from QPSK and/or 16-QAM constellations,
- $u(t) = \mathrm{sinc}(t\frac{1}{t_b})$ or square waves or any pulse you find interesting.

PROBLEM 2.2.A.7. *

(*Plot the Power Spectral Density* of OFDM) Plot the power spectral density of signals of OFDM modulation,

$$x(t) \quad = \quad \left(\sum_{l=1}^{l_c} x_l e^{j2\pi(lf_\Delta + f_0)t} \right) u(t),$$

Assume the symbol energy $e_s = 1$, the subcarrier spacing $f_\Delta = 100$ kHz, number of subcarriers $l_c = 50$, $f_0 = f_\Delta(-\frac{l_c+1}{2})$, and

$$u(t) = \sqrt{f_\Delta}\mathrm{rect}_{\frac{1}{f_\Delta}}(t).$$

PROBLEM 2.2.A.8. *

(*Implementation of OCDM+LSM Modulator*) Assume

$$u_k(t) = \sum_{m=1}^{m_s} c_{k,m} u(t - mt_c);$$

where $m_s = 4$, $k = 1, ..., m_s$,

$$[c_{k,m}]_{m,k} = \frac{1}{\sqrt{4}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

$$u(t) = \frac{1}{\sqrt{t_c}} \text{sinc}\left(t\frac{1}{t_c}\right).$$

1. Plot $u_k(t)$, $k = 1, ..., m_s$.
2. Plot realizations of signals of OCDM+LSM modulation,

$$\boldsymbol{x}(t) = \sum_k \sum_l \boldsymbol{x}_{k,l} \underbrace{u_k(t - lt_b)}_{u_{k,l}(t)}.$$

and $\tilde{\boldsymbol{x}}(f)$ for various cases so that you gain pictorial impression of them. If $x(t)$ is complex valued, you can plot $|x(t)|$, $\angle x(t)$, $x_{\mathrm{I}}(t)$, $x_{\mathrm{Q}}(t)$. You may select $\boldsymbol{x}_l$ from QPSK and/or 16-QAM constellations.

PROBLEM 2.2.A.9. *

(*Plotting the Power Spectral Density* of OCDM+LSM) Find and plot the power spectral density of signals of OCDM modulation,

$$\boldsymbol{x}(t) = \sum_{k=1}^{m_s} \sum_{l} \boldsymbol{x}_{k,l} \underbrace{u_k(t - lt_b)}_{u_{k,l}(t)},$$

where

$$u_k(t) = \sum_{m=1}^{m_s} c_{k,m} u(t - mt_c);$$

vectors $\vec{c}_k = [c_{k,m}]_m$'s are orthonormal;

$$u(t) = \frac{1}{\sqrt{t_c}} \mathrm{sinc}\left(t \frac{1}{t_c}\right).$$

Assume the symbol energy $e_s = 1$.

CHAPTER 2.3

Demodulator for AWGN Channels

2.3.1. Introduction

$$\xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)} \boxed{\text{Equivalent Baseband Channel}} \xrightarrow{\boldsymbol{y}(t)} \boxed{\text{Demodulator}} \xrightarrow{\vec{\boldsymbol{y}}}$$

- This module helps you design the optimal demodulator of a receiver.
  - A demodulator coverts continuous time received signal to discrete time signal (near) optimally.
  - The optimality is in the sense that the output of the demodulator $\vec{\boldsymbol{y}}$ is a sufficient statistic of the information sequence $\vec{\boldsymbol{i}}$. That is, information about $\vec{\boldsymbol{i}}$ is not lost during the conversion.
- We begin with a discussion of sufficient statistics and then show how to design optimal demodulators using two approaches.

- Then, we derive the equivalent discrete time channel model of the demodulator output $\vec{\boldsymbol{y}}$ as a function of the modulator input $\vec{\boldsymbol{x}}$.
- We consider an additive white Gaussian noise (AWGN) channel for the equivalent baseband channel. Fading channels are considered in another chapter.

2.3.1.0.1. *Learning Objectives.*

- Can design optimal demodulator
- Understand matched filtering
- Can derive equivalent discrete time channel model

2.3.1.0.2. *Subjects.*

- Sufficient statistics and optimal detector
- Signal space projections
- Optimal demodulators for linear modulations.
- Matched filtering.
- Equivalent discrete time baseband channel model.

2.3.1.0.3. *Questions to Keep in Mind.*

- For every step of signal processing, ask yourself whether you are losing information.

2.3.1.0.4. *Organization.*

- Sufficient statistics
- Sampling and likelihood
- Likelihood and matched filter
- Signal space method
- Discrete time implementation of matched filter

2.3.2. Sufficient Statistics

$$\xrightarrow{\vec{\boldsymbol{i}}} \boxed{\text{Encoder}} \xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)} \boxed{\text{AWGN Channel}} \xrightarrow{\boldsymbol{y}(t)=\boldsymbol{x}(t)+\boldsymbol{n}(t)} \boxed{\text{Demodulator}} \xrightarrow{\vec{\boldsymbol{y}}}$$

- At a receiver, we want to convert continuous time signal $\boldsymbol{y}(t)$ to discrete time signal $\vec{\boldsymbol{y}}$ without losing information about $\vec{\boldsymbol{i}}$.
- What do we mean by without losing information?
  - This has been formalized in statistics by the concept of sufficient statistic.

2.3.2.0.1. *Definition of sufficient statistic and its relation to mutual information.*

DEFINITION 2.3.1. Let $\boldsymbol{x}$ and $\boldsymbol{y}$ be two random variables. A function $T(\boldsymbol{y})$ is a SUFFICIENT STATISTIC for $\boldsymbol{x}$ if $\boldsymbol{x}$ and $\boldsymbol{y}$ are independent conditioned on $T(\boldsymbol{y})$.

LEMMA 2.3.1. *For the Markov chain $\boldsymbol{x} \to \boldsymbol{y} \to T(\boldsymbol{y})$, $T(\boldsymbol{y})$ is a sufficient statistic for $\boldsymbol{x}$ if and only if $I(\boldsymbol{x};\boldsymbol{y}|T(\boldsymbol{y})) = 0$ or $I(\boldsymbol{x};\boldsymbol{y}) = I(\boldsymbol{x};T(\boldsymbol{y}))$.*

PROOF. See [4]. □

2.3.2.0.2. *Exercise.*

EXERCISE 2.3.1. (*Sufficient Statistic Example*) The transmitted signal is $\boldsymbol{x} \in \{1, -1\}$. The received signal is

$$\boldsymbol{y} \quad = \quad \boldsymbol{x} + \boldsymbol{n},$$

where $\boldsymbol{n} \sim \mathcal{CN}(0, 1)$ is a circularly symmetric complex Gaussian noise independent of the transmit signal. Is the multiple to one mapping $\boldsymbol{z} = \Re[\boldsymbol{y}]$ a sufficient statistic for $\boldsymbol{x}$?

Solution:

- Yes, because

$$f_{\boldsymbol{y}|\boldsymbol{x},\boldsymbol{z}}(y|x,z) \quad = \quad f_{\boldsymbol{y}|\boldsymbol{z}}(y|z)$$

  where conditioned on $\boldsymbol{z} = z$, we know $\Re[\boldsymbol{y}] = z$ and $\Im[\boldsymbol{y}]$ only depends on the imaginary part of the noise, which is independent of the transmit signal $\boldsymbol{x}$.

$\square$

2.3.3. Dense Samples is a Sufficient Statistic

$$\xrightarrow{\vec{\boldsymbol{i}}} \boxed{\text{Encoder}} \xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)} \boxed{\text{AWGN Channel}} \xrightarrow{\boldsymbol{y}(t)=\boldsymbol{x}(t)+\boldsymbol{n}(t)} \boxed{\text{Demodulator}} \xrightarrow{\vec{\boldsymbol{y}}}$$

- Can we use the sampling approach to obtain a discrete time sufficient statistic in AWGN channels?
- What sampling frequency is sufficient?

2.3.3.0.1. *Optimal sampling frequency of band limited channels.*

PROBLEM 2.3.3.1. We consider AWGN channel

- where the received signal is

$$\boldsymbol{y}(t) = \boldsymbol{x}(t) + \boldsymbol{n}(t);$$

- the transmit signal is $\boldsymbol{x}(t)$ whose spectrum is within $(-\frac{b}{2}, \frac{b}{2})$;
- the circularly symmetric complex Gaussian noise $\boldsymbol{n}(t)$'s power spectral density is

$$\tilde{r}_{\boldsymbol{n}}(f) = N_0 \mathrm{rect}_b(f),$$

  i.e., the noise is white within the signal bandwidth.

1. Is the infinite length vector $[\boldsymbol{y}(kt_s)]_k$, where $\frac{1}{t_s} \geq b$, a sufficient statistic for detecting the information message?

2. Find statistics of the noise in

$$\boldsymbol{y}(kt_s) \quad = \quad \boldsymbol{x}(kt_s) + \boldsymbol{n}(kt_s). \tag{2.3.3.1}$$

What happens if $\frac{1}{t_s} = b$?

Solution:

1. Yes, because if we know $[\boldsymbol{y}(kt_s)]_k$, we can reconstruct

$$\boldsymbol{y}(t) \quad = \quad \left(\sum_k \boldsymbol{y}(kt_s)\text{sinc}\left((t-kt_s)\frac{1}{t_s}\right)\right)$$

   in mean square sense. Therefore, there is no loss of information.

2. We know $\boldsymbol{n}(kt_s)$ is a circularly symmetric complex Gaussian random variable.
   - The correlation/covariance function of $\boldsymbol{n}(t)$ is

$$r_{\boldsymbol{n}}(\tau) \quad = \quad N_0 b\text{sinc}(\tau b).$$

     Therefore, the covariance matrix of $[\boldsymbol{n}(kt_s)]_k$ is $\Sigma = [\text{E}\left[\boldsymbol{n}(k_1t_s)\boldsymbol{n}^*(k_2t_s)\right]]_{k_1,k_2}$, where

$$\text{E}\left[\boldsymbol{n}(k_1t_s)\boldsymbol{n}^*(k_2t_s)\right] \quad = \quad r_{\boldsymbol{n}}\left((k_1-k_2)t_s\right).$$

- In summary,

$$[\boldsymbol{n}(kt_s)]_k \quad \sim \quad \mathcal{CN}\left(\vec{0}, [N_0 b \text{sinc}\left((k_1 - k_2)t_s b\right)]_{k_1,k_2}\right).$$

  The noise variance is

$$\text{Var}\left(\boldsymbol{n}(kt_s)\right) \quad = \quad N_0 b.$$

- If $\frac{1}{t_s} = b$,

$$[\boldsymbol{n}(kt_s)]_k \quad \sim \quad \mathcal{CN}\left(\vec{0}, N_0 b I\right), \tag{2.3.3.2}$$

  i.e., the covariance matrix is a scaled identity matrix and the noise samples are i.i.d..

□

2.3.3.0.2. *Conclusion.*

- Any sequence of samples with higher than Nyquist-Shannon sampling rate of band limited signal will not lose information and is thus a sufficient statistic. In practice, we often over sample at four times the Nyquist-Shannon sampling rate.
- The sampled relation in (2.3.3.1) together with the statistics of $\boldsymbol{n}(kt_s)$ is an equivalent discrete time channel model.

2.3.3.0.3. *Exercise.*

EXERCISE 2.3.2. (*Sufficient Statistic Example*) The transmitted signal is $\boldsymbol{x} \in \{1, -1\}$. The received signal is

$$\boldsymbol{y} \quad = \quad \boldsymbol{x} + \boldsymbol{n},$$

where $\boldsymbol{n} \sim \mathcal{CN}(0, 1)$ is a circularly symmetric complex Gaussian noise independent of the transmit signal. Is the multiple to one mapping $\boldsymbol{z} = \Re[\boldsymbol{y}]$ a sufficient statistic for $\boldsymbol{x}$?

Solution:

- Yes, because

$$f_{\boldsymbol{y}|\boldsymbol{x},\boldsymbol{z}}(y|x,z) \quad = \quad f_{\boldsymbol{y}|\boldsymbol{z}}(y|z)$$

  where conditioned on $\boldsymbol{z} = z$, we know $\Re[\boldsymbol{y}] = z$ and $\Im[\boldsymbol{y}]$ only depends on the imaginary part of the noise, which is independent of the transmit signal $\boldsymbol{x}$.

$\square$

2.3.4. Likelihood and Matched Filter Output are Sufficient Statistics

$$\xrightarrow{\vec{\boldsymbol{i}}} \boxed{\text{Encoder}} \xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)} \boxed{\text{AWGN Channel}} \xrightarrow{\boldsymbol{y}(t)=\boldsymbol{x}(t)+\boldsymbol{n}(t)} \boxed{\text{Demodulator}} \xrightarrow{\vec{\boldsymbol{y}}}$$

- What is likelihood? How is it related to sufficient statistics?
  - We define likelihood and show that the likelihood is a sufficient statistic. See [4] for a proof.
- We use the samples to calculate the likelihood and derive another sufficient statistic, matched filter output, for linear modulations.

2.3.4.0.1. *Definition of likelihood.*

DEFINITION 2.3.2. (*Likelihood*) Let the relation between two random variables $\boldsymbol{x} \in \mathcal{X}$ and $\boldsymbol{y} \in \mathcal{Y}$ be $p_{\boldsymbol{y}|\boldsymbol{x}}$, which can be viewed as channel transition probability with $\boldsymbol{x}$ as the channel input and $\boldsymbol{y}$ as the channel output. The receiver uses $\boldsymbol{y}$ to calculate the LIKELIHOOD

$$\begin{aligned} \vec{\boldsymbol{z}} &= \vec{f}(\boldsymbol{y}) \\ &\triangleq \left[p_{\boldsymbol{y}|\boldsymbol{x}}(\boldsymbol{y}|x)\right]_{x\in\mathcal{X}}, \end{aligned} \tag{2.3.4.1}$$

which is the probability that the received signal is $\boldsymbol{y}$ conditioned on the the transmitted signal is x, $\forall x \in \mathcal{X}$.

REMARK 2.3.1. The $\boldsymbol{y}$ in (2.3.4.1) is a random variable. Thus, $\vec{\boldsymbol{z}}$ is a random vector.

2.3.4.0.2. *Example.*

EXAMPLE 2.3.1. If $\mathcal{X} = \{1, -1\}$, the likelihood is $\vec{\boldsymbol{z}} = \begin{bmatrix} p_{\boldsymbol{y}|\boldsymbol{x}}(\boldsymbol{y}|1) \\ p_{\boldsymbol{y}|\boldsymbol{x}}(\boldsymbol{y}|-1) \end{bmatrix}$.

2.3.4.0.3. *Likelihood is a sufficient statistic.*

THEOREM 2.3.1. *(**Likelihood is a Sufficient Statistic**) For two related random variables $\boldsymbol{x}$ and $\boldsymbol{y}$ of finite alphabets, $(\mathcal{X}, \mathcal{Y}, p_{\boldsymbol{x}}, p_{\boldsymbol{y}|\boldsymbol{x}})$, the likelihood $\vec{\boldsymbol{z}}$ calculated in*

$$\begin{aligned} \vec{\boldsymbol{z}} &= \vec{f}(\boldsymbol{y}) \\ &\triangleq \left[p_{\boldsymbol{y}|\boldsymbol{x}}(\boldsymbol{y}|x)\right]_{x\in\mathcal{X}}, \end{aligned}$$

is a sufficient statistic with respect to $\boldsymbol{x}$, i.e.,

$$I(\boldsymbol{x};\boldsymbol{y}) = I(\boldsymbol{x};\vec{\boldsymbol{z}}),$$

which implies

$$\boldsymbol{x} \to \vec{\boldsymbol{z}} \to \boldsymbol{y}$$

is a Markov chain, i.e.,

$$p_{\boldsymbol{x},\boldsymbol{y}|\vec{z}} = p_{\boldsymbol{x}|\vec{z}}\, p_{\boldsymbol{y}|\vec{z}}. \tag{2.3.4.2}$$

2.3.4.0.4. *Proportional likelihood is a sufficient statistic.*

- We can further refine the above theorem by throwing away some irrelevant part of the likelihood. It is summarized in the following theorem.

THEOREM 2.3.2. *(**Proportional Likelihood is a Sufficient Statistic**, **Fisher–Neyman Factorization Theorem**) Consider two related random variables $\boldsymbol{x}$ and $\boldsymbol{y}$ of finite alphabets, $(\mathcal{X}, \mathcal{Y}, p_{\boldsymbol{x}}, p_{\boldsymbol{y}|\boldsymbol{x}})$. Define proportional likelihood $\vec{\boldsymbol{z}}$ as*

$$\vec{\boldsymbol{z}} \triangleq \vec{g}(\boldsymbol{y})$$

such that

$$p_{\boldsymbol{y}|\boldsymbol{x}}(\boldsymbol{y}|x) = g_1(\boldsymbol{y}) g_2(\vec{g}(\boldsymbol{y}), x), \ \forall x \in \mathcal{X}$$

Then $\vec{\boldsymbol{z}}$ is a sufficient statistic with respect to $\boldsymbol{x}$, i.e.,

$$I(\boldsymbol{x}; \boldsymbol{y}) = I(\boldsymbol{x}; \vec{\boldsymbol{z}}),$$

which implies

$$\boldsymbol{x} \rightarrow \quad \vec{\boldsymbol{z}} \quad \rightarrow \boldsymbol{y}$$

is a Markov chain, i.e.,

$$p_{\boldsymbol{x},\boldsymbol{y}|\vec{\boldsymbol{z}}} \quad = \quad p_{\boldsymbol{x}|\vec{\boldsymbol{z}}} p_{\boldsymbol{y}|\vec{\boldsymbol{z}}}.$$

2.3.4.0.5. *The matched filter output is a sufficient statistic.*

- We find another sufficient statistic from the calculation of the likelihood below.

PROBLEM 2.3.4.1. (**Likelihood, Sufficient Statistics, and Matched Filter**) Recall the communication system

$$\xrightarrow{\vec{\boldsymbol{i}}} \boxed{\text{Encoder}} \xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)} \boxed{\text{AWGN Channel}} \xrightarrow{\boldsymbol{y}(t)=\boldsymbol{x}(t)+\boldsymbol{n}(t)} \boxed{\text{Demod}}$$

- where $\vec{\boldsymbol{i}} \in \{0,1\}^n$ is the message to be sent or the information vector;
- $\vec{\boldsymbol{x}}$ is the code word;
- $\boldsymbol{x}(t) = x_{\vec{\boldsymbol{i}}}(t)$ is the baseband signal and is band limited within $\left(-\frac{b}{2}, \frac{b}{2}\right)$;
- $\boldsymbol{y}(t)$ is the output of the equivalent baseband AWGN channel;

- $\boldsymbol{n}(t)$ is the circularly symmetric complex Gaussian noise process with power spectral density $\tilde{r}_{\boldsymbol{n}}(f) = N_0 \mathrm{rect}_b(f)$.

1. Calculate the likelihood of the received signal conditioned on the information $\vec{\boldsymbol{i}}$ in order to find a sufficient statistic $\vec{\boldsymbol{y}}$ for detecting $\vec{\boldsymbol{i}}$ for arbitrary modulation.
2. Based on the likelihood, derive a sufficient statistic for linear modulation for complex and real valued $\vec{\boldsymbol{x}}$.
3. Based on the sufficient statistic for linear modulation, derive a sufficient statistic for linear stream modulation.
4. Based on the sufficient statistic for linear modulation, derive a sufficient statistic for OFDM modulation.
5. Based on the sufficient statistic for linear modulation, derive a sufficient statistic for OCDM+LSM modulation.

Solution:

1. The received signal is continuous in time and its distribution is not defined. But $[\boldsymbol{y}(kt_s)]_k$, where $\frac{1}{t_s} = b$, does not lose information about $\boldsymbol{y}(t)$ according to the sampling theorem. Therefore, we start from the likelihood of length m samples $\vec{\boldsymbol{y}}_s = [\boldsymbol{y}(kt_s)]_{k=\lfloor -\frac{m}{2} \rfloor + 1 : \lfloor \frac{m}{2} \rfloor}$ and let $m \to \infty$.

- The likelihood of $\vec{\boldsymbol{y}}_s$ conditioned on the message is

$$
\begin{aligned}
& f_{\vec{\boldsymbol{y}}_s|\vec{\boldsymbol{i}}}(\vec{y}_s|\vec{i}) \\
= \quad & \frac{1}{(\pi N_0 b)^m} e^{-\frac{\sum_{k=\left\lfloor -\frac{m}{2} \right\rfloor+1}^{\left\lfloor \frac{m}{2} \right\rfloor} |y(kt_s)-x_{\vec{i}}(kt_s)|^2}{N_0 b}} && (2.3.4.3) \\
= \quad & \frac{1}{(\pi N_0 b)^m} e^{-\frac{d_{\mathrm{E}}^2\left(\frac{1}{\sqrt{b}}\vec{y}_s, \frac{1}{\sqrt{b}}\vec{x}_{\vec{i}}\right)}{N_0}} \\
\overset{m\to\infty}{\propto} \quad & e^{-\frac{\int_{-\infty}^{\infty} |y(t)-x_{\vec{i}}(t)|^2 dt}{N_0}} && (2.3.4.4) \\
= \quad & e^{-\frac{d_{\mathrm{E}}^2\left(y(t), x_{\vec{i}}(t)\right)}{N_0}} \\
= \quad & e^{-\frac{\int_{-\infty}^{\infty} |y(t)|^2 dt - 2\Re\left[\int_{-\infty}^{\infty} y(t) x_{\vec{i}}^*(t) dt\right] + \int_{-\infty}^{\infty} |x_{\vec{i}}(t)|^2 dt}{N_0}} \\
= \quad & \underbrace{e^{-\frac{\int_{-\infty}^{\infty} |y(t)|^2 dt}{N_0}}}_{g_1([y(t)]_t)} \underbrace{e^{-\frac{-2\Re\left[\left\langle [y(t)]_t, \left[x_{\vec{i}}(t)\right]_t \right\rangle\right] + \int_{-\infty}^{\infty} |x_{\vec{i}}(t)|^2 dt}{N_0}}}_{g_2(\vec{g}([y(t)]_t), \vec{i})} && (2.3.4.5)
\end{aligned}
$$

- where (2.3.4.3) follows from $\boldsymbol{y}(kt_s) - x_{\vec{i}}(kt_s) = \boldsymbol{n}(kt_s)$ and

$$[\boldsymbol{n}(kt_s)]_k \quad \sim \quad \mathcal{CN}\left(\vec{0}, N_0 bI\right)$$

  according to equation (2.3.3.2);
- (2.3.4.4) follows from that $\frac{1}{(\pi N_0 b)^m}$ does not depend on $\vec{i}$;
  * $\vec{x}_{\vec{i}} \triangleq [x_{\vec{i}}(kt_s)]_{k=\lfloor -\frac{m}{2} \rfloor + 1 : \lfloor \frac{m}{2} \rfloor}$;
  * e^x is a continuous function for the limit to move to the exponent;
  * $x_{\vec{i}}(t) = \sum_k x_{\vec{i}}(kt_s) \frac{1}{\sqrt{b}} \sqrt{b} \mathrm{sinc}\left(b\left(t - kt_s\right)\right)$;
  * $y(t) = \sum_k y(kt_s) \frac{1}{\sqrt{b}} \sqrt{b} \mathrm{sinc}\left(b\left(t - kt_s\right)\right)$;

* $\left\{\sqrt{b}\mathrm{sinc}\left(b\left(t-kt_s\right)\right) : k \in \mathbb{Z}\right\}$ is a orthonormal set, resulting in the distance square

$$
\begin{aligned}
d_{\mathrm{E}}^2\left(\frac{1}{\sqrt{b}}\vec{y}, \frac{1}{\sqrt{b}}\vec{x_{\vec{i}}}\right) &\triangleq \sum_{k=\left\lfloor -\frac{m}{2}\right\rfloor+1}^{\left\lfloor \frac{m}{2}\right\rfloor} \left|\frac{1}{\sqrt{b}}y(kt_s) - \frac{1}{\sqrt{b}}x_{\vec{i}}(kt_s)\right|^2 \\
&\xrightarrow{m\to\infty} \int_{-\infty}^{\infty} \left|y(t) - x_{\vec{i}}(t)\right|^2 dt \\
&= d_{\mathrm{E}}^2\left(y(t), x_{\vec{i}}(t)\right).
\end{aligned}
$$

- According to Theorem theorem 2.3.2, any one of the following $\vec{\boldsymbol{y}} = \vec{g}([\boldsymbol{y}(t)]_t)$ is a sufficient statistic:

$$
\begin{aligned}
\text{distance: } \vec{\boldsymbol{y}} = \left[d_{\mathrm{E}}^2\left(\boldsymbol{y}(t), x_{\vec{i}}(t)\right)\right]_{\vec{i}\in\{0,1\}^n} &= \left[\int_{-\infty}^{\infty} |\boldsymbol{y}(t) - x_{\vec{i}}(t)|^2 dt\right]_{\vec{i}\in\{0,1} \\
\text{correlation: } \vec{\boldsymbol{y}} = \left[\langle \boldsymbol{y}(t), x_{\vec{i}}(t)\rangle\right]_{\vec{i}\in\{0,1\}^n} &= \left[\int_{-\infty}^{\infty} \boldsymbol{y}(t) x_{\vec{i}}^*(t) dt\right]_{\vec{i}\in\{0,1\}^n}, \\
\Re[\text{correlation}]: \vec{\boldsymbol{y}} = \left[\Re\left[\langle \boldsymbol{y}(t), x_{\vec{i}}(t)\rangle\right]\right]_{\vec{i}\in\{0,1\}^n} &= \left[\Re\left[\int_{-\infty}^{\infty} \boldsymbol{y}(t) x_{\vec{i}}^*(t) dt\right]\right]_{\vec{i}\in\{0,1}
\end{aligned}
$$

- The above sufficient statistics have a length that is exponential in n. For linear modulation, it will simply below.

2. Consider **linear modulation**,

$$
x_{\vec{i}}(t) = \sum_l x_{\vec{i},l} u_l(t).
$$

- If $\vec{\boldsymbol{x}}$ is *complex or real valued*,

$$\begin{aligned}\langle \boldsymbol{y}(t), x_{\vec{i}}(t)\rangle &= \left\langle \boldsymbol{y}(t), \sum_l x_{\vec{i},l} u_l(t)\right\rangle \\ &= \sum_l x^*_{\vec{i},l} \langle \boldsymbol{y}(t), u_l(t)\rangle .\end{aligned}$$

  Thus, the matched filter outputs,

$$\vec{\boldsymbol{y}} = [\langle \boldsymbol{y}(t), u_l(t)\rangle]_l ,$$

  is a sufficient statistic according to theorem 2.3.2.
- If $\vec{\boldsymbol{x}}$ is *real valued*,

$$\begin{aligned}\Re\left[\langle \boldsymbol{y}(t), x_{\vec{i}}(t)\rangle\right] &= \Re\left[\left\langle \boldsymbol{y}(t), \sum_l x_{\vec{i},l} u_l(t)\right\rangle\right] \\ &= \sum_l x^*_{\vec{i},l} \Re\left[\langle \boldsymbol{y}(t), u_l(t)\rangle\right] .\end{aligned}$$

Thus, the real part of the matched filter outputs,

$$\Re\left[\vec{\boldsymbol{y}}\right] = \left[\Re\left[\langle \boldsymbol{y}(t), u_l(t)\rangle\right]\right]_l,$$

is a sufficient statistic according to theorem 2.3.2.

- The above are called matched filter outputs because the inner product/projection can be obtained from a filter output sampled at t_0, i.e.,

$$\begin{aligned}\langle \boldsymbol{y}(t), u_l(t)\rangle &= \int \boldsymbol{y}(t) u_l^*(t) dt \\ &= y(t) * u_l^*(t_0 - t)\big|_{t=t_0}.\end{aligned}$$

- Since the filter impulse response $u_l^*(t_0 - t)$ matches the pulses used for modulation, $\vec{\boldsymbol{y}}$ is also called MATCHED FILTER OUTPUTS.

3. For **linear stream modulation**,

$$u_l(t) = u(t - lt_b).$$

The sufficient statistic is

$$\begin{aligned}\vec{\boldsymbol{y}} &= [\langle \boldsymbol{y}(t), u_l(t)\rangle]_l \\ &= [\langle \boldsymbol{y}(t), u(t-lt_b)\rangle]_l \\ &= \left[\boldsymbol{y}(t) * u^*(t_0-t)|_{t=t_0+lt_b}\right]_l,\end{aligned}$$

which can be obtained from the output of **one** filter with impulse response $u^*(t_0-t)$, sampled at $t = t_0 + lt_b$, $\forall l$.

4. For **OFDM** modulation,

$$u_l(t) = e^{j2\pi(lf_\Delta+f_0)t}u(t).$$

The sufficient statistic is

$$\begin{aligned}\vec{\boldsymbol{y}} &= [\langle \boldsymbol{y}(t), u_l(t)\rangle]_l \\ &= \left[\left\langle \boldsymbol{y}(t), e^{j2\pi(lf_\Delta+f_0)t}u(t)\right\rangle\right]_l \\ &= \left[\int \boldsymbol{y}(t)e^{-j2\pi(lf_\Delta+f_0)t}u^*(t)dt\right]_l.\end{aligned}$$

If

$$u(t) = \begin{cases} c & t_1 \leq t \leq t_2 \\ 0 & t < t_0 \text{ or } t > t_3 \end{cases},$$

where $t_0 < t_1 < t_2 = t_1 + \tau_h + \frac{1}{f_\Delta} < t_3$, a near sufficient and simple statistic is

$$\vec{\boldsymbol{y}} = \left[\int_{t_1+\tau_h}^{t_1+\tau_h+\frac{1}{f_\Delta}} \boldsymbol{y}(t) e^{-j2\pi(lf_\Delta+f_0)t} c^* dt \right]_l,$$

by throwing away part of $\boldsymbol{y}(t)$ and only employ a $\frac{1}{f_\Delta}$ wide time segment.

- As we will see later, the near sufficient statistic leads to a low complexity equivalent channel model in time invariant frequency selective fading channels.

5. For **OCDM+LSM** modulation, the sufficient statistic is

$$\vec{\boldsymbol{y}} = \left[\left[\underbrace{\langle \boldsymbol{y}(t), u_k(t - lt_b)\rangle}_{\boldsymbol{y}_{k,l}}\right]_k\right]_l .$$

If the pulses are

$$u_k(t - lt_b) = \sum_{m=1}^{m_s} c_{k,m} u(t - lm_s t_c - mt_c),$$

where m_s is the spreading code length, and the symbol period $t_b = m_s t_c$. Then,

$$\begin{aligned}
\boldsymbol{y}_{k,l} = \langle \boldsymbol{y}(t), u_{k,l}(t) \rangle &= \left\langle \boldsymbol{y}(t), \sum_{m=1}^{m_s} c_{k,m} u(t - l m_s t_c - m t_c) \right\rangle \\
&= \sum_{m=1}^{m_s} c_{k,m}^* \langle \boldsymbol{y}(t), u(t - l m_s t_c - m t_c) \rangle \\
&= \left\langle \left[\langle \boldsymbol{y}(t), u(t - l m_s t_c - m t_c) \rangle \right]_{m=1:m_s}, \vec{c}_k \right\rangle,
\end{aligned}$$

which can also be obtained from the output of a single matched filter.

□

2.3.5. A Signal Space Approach

- The sufficient statistics discussed so far can be alternatively proved using projections of the received signal to the space of the signal.

2.3.5.0.1. *Projections to the signal space is a sufficient statistic.*

THEOREM 2.3.3. *(Signal Space Approach to Sufficient Statistics) Consider communicating n bits $\vec{\boldsymbol{i}} \in \{0,1\}^n$ by signals $x_{\vec{\boldsymbol{i}}}(t)$. The channel output is*

$$\boldsymbol{y}(t) \quad = \quad x_{\vec{\boldsymbol{i}}}(t) + \boldsymbol{n}(t), \qquad -\frac{t_w}{2} < t < \frac{t_w}{2},$$

where $\boldsymbol{n}(t)$ is a circularly symmetric complex Gaussian noise process.

- *The signal space has a $k_b \leq 2^n$ dimensional orthonormal basis $\{b_k(t) : k = 1, ..., k_b\}$, i.e.,*

$$\begin{aligned} x_{\vec{i}}(t) &= \sum_{k=1}^{k_b} d_{\vec{i},k} b_k(t), \\ \langle b_{k_1}(t), b_{k_2}(t) \rangle &= \delta[k_1 - k_2]. \end{aligned}$$

- *If*

1. *$b_k(t), k = 1, ..., k_b$, are eigen-functions of the noise covariance function;*
2. *or the noise is white over a space containing the signal space as a sub-space,*

- *then*
  1. *the projection $[\langle \boldsymbol{y}(t), b_k(t)\rangle]_{k=1,...,k_b}$ of $\boldsymbol{y}(t)$ onto the signal space is a sufficient statistic;*
  2. *the projection $[\langle \boldsymbol{y}(t), x_{\vec{i}}(t)\rangle]_{\vec{i}\in\{0,1\}^n}$ of $\boldsymbol{y}(t)$ onto the signals is a sufficient statistic;*
  3. *the projection $[\langle \boldsymbol{y}(t), u_k(t)\rangle]_{k=1,...,m}$ of $\boldsymbol{y}(t)$ onto the signal pulses is a sufficient statistic, where*

$$x_{\vec{i}}(t) = \sum_{k=1}^{m} x_{\vec{i},k} u_k(t),$$

and $\{u_k(t) : k = 1, ..., m\}$ may not be an orthogonal set.

REMARK.
- This theorem is proved in [4].

- The theorem also proves that the samples $[\boldsymbol{y}(kt_s)]_k$, $\frac{1}{t_s} = b$, is a sufficient statistic
  - because $\left\{\sqrt{b}\text{sinc}\left(b\left(t - k\frac{1}{b}\right)\right) : k \in \mathbb{Z}\right\}$ is a orthonormal basis of the signal space
  - and $\boldsymbol{y}(k\frac{1}{b}) = \sqrt{b}\left\langle \boldsymbol{y}(t), \sqrt{b}\text{sinc}\left(b\left(t - k\frac{1}{b}\right)\right)\right\rangle$.

2.3.6. Equivalent Discrete Time AWGN Channel Models

$$\xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)} \boxed{\text{AWGN Channel}} \xrightarrow{\boldsymbol{y}(t)=\boldsymbol{x}(t)+\boldsymbol{n}(t)} \boxed{\text{Demodulator}} \xrightarrow{\vec{\boldsymbol{y}}}$$

$$\xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Equivalent Discrete Time Channel}} \xrightarrow{\vec{\boldsymbol{y}}}$$

- Based on the sufficient statistics derived earlier, how do we find equivalent channel models that describe the relation between $\vec{\boldsymbol{x}}$ and $\vec{\boldsymbol{y}}$?

2.3.6.0.1. *Equivalent Discrete Time AWGN Channel Models.*

PROBLEM 2.3.6.1. (**Equivalent Discrete Time AWGN Channel Models**)

- Consider AWGN channel

$$\xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)} \boxed{\text{AWGN Channel}} \xrightarrow{\boldsymbol{y}(t)=\boldsymbol{x}(t)+\boldsymbol{n}(t)} \boxed{\text{Demodulator}} \xrightarrow{\vec{\boldsymbol{y}}}$$

  - where $\vec{\boldsymbol{x}}$ is the code word;
  - $\boldsymbol{x}(t)$ is the baseband signal and is band limited within $\left(-\frac{b}{2}, \frac{b}{2}\right)$;
  - $\boldsymbol{y}(t)$ is the output of the equivalent baseband AWGN channel;
  - $\boldsymbol{n}(t)$ is the circularly symmetric complex Gaussian noise process with power spectral density $\tilde{r}_{\boldsymbol{n}}(f) = N_0 \mathrm{rect}(f)$.

- For sufficient statistics $\vec{\boldsymbol{y}}$, derive the equivalent discrete time channel models, $\vec{\boldsymbol{y}} = [\boldsymbol{y}_l]_l = \text{func}(\vec{\boldsymbol{x}})$ for the following modulations.

1. Linear modulation.
2. Linear stream modulation.
3. OFDM modulation.
4. OCDM+LSM modulation.

Solution:

1. For *linear modulation*,

$$\boldsymbol{x}(t) = \sum_l \boldsymbol{x}_l u_l(t),$$

the sufficient statistic is

$$\vec{\boldsymbol{y}} = \left[\langle \boldsymbol{y}(t), u_l(t)\rangle\right]_l .$$

- Examine

$$
\begin{aligned}
\boldsymbol{y}_l &= \langle \boldsymbol{y}(t), u_l(t) \rangle \\
&= \left\langle \sum_{l_1} \boldsymbol{x}_{l_1} u_{l_1}(t) + \boldsymbol{n}(t), u_l(t) \right\rangle \\
&= \sum_{l_1} \boldsymbol{x}_{l_1} \underbrace{\langle u_{l_1}(t), u_l(t) \rangle}_{r_u(l_1,l)} + \underbrace{\langle \boldsymbol{n}(t), u_l(t) \rangle}_{\boldsymbol{n}_l} \\
&= \sum_{l_1} \boldsymbol{x}_{l_1} r_u(l_1, l) + \boldsymbol{n}_l \\
&= \boldsymbol{x}_l r_u(l, l) + \underbrace{\sum_{l_1 : l_1 \neq l} \boldsymbol{x}_{l_1} r_u(l_1, l)}_{\text{Inter-symbol Interference (ISI)}} + \boldsymbol{n}_l,
\end{aligned}
$$

where

$$
r_u(l_1, l) \triangleq \langle u_{l_1}(t), u_l(t) \rangle .
$$

- *Noise Covariance:* The noise $\boldsymbol{n}_l$ is the output of a filter with impulse response $u_l^*(-t)$. The input is a circularly symmetric complex Gaussian (CSCG) process $\boldsymbol{n}(t)$. Thus, $\boldsymbol{n}_l$ is a CSCG random variable with zero mean. The covariance is

$$\begin{aligned}
r_{\boldsymbol{n}}(l,k) &= \mathrm{E}\left[\boldsymbol{n}_l \boldsymbol{n}_k^*\right] \\
&= \mathrm{E}\left[\langle \boldsymbol{n}(t), u_l(t)\rangle \langle \boldsymbol{n}(t), u_k(t)\rangle^*\right] \\
&= \mathrm{E}\left[\langle \boldsymbol{w}(t), u_l(t)\rangle \langle \boldsymbol{w}(t), u_k(t)\rangle^*\right] \qquad (2.3.6.1) \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathrm{E}\left[\boldsymbol{w}(t)\boldsymbol{w}^*(t')\right] u_k(t') u_l^*(t) dt dt' \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} N_0 \delta(t-t') u_k(t') u_l^*(t) dt dt' \\
&= N_0 r_u(k,l),
\end{aligned}$$

where we in equation (2.3.6.1), we have replaced $\boldsymbol{n}(t)$ with a CSCG white noise process $\boldsymbol{w}(t)$, whose power

spectral density $\tilde{r}_{\boldsymbol{w}}(f) = N_0$, because $\boldsymbol{x}(t)$ and thus $u_l(t)$ is band limited to $\left(-\frac{b}{2}, \frac{b}{2}\right)$ and $\langle \boldsymbol{w}(t), u_l(t) \rangle = \langle \boldsymbol{n}(t), u_l(t) \rangle$ in distribution.

- *Channel Model:* Define $R_u \triangleq [r_u(k,l)]_{l,k}$. We can write the discrete time channel model in matrix form

$$\vec{\boldsymbol{y}} = R_u \vec{\boldsymbol{x}} + \vec{\boldsymbol{n}},$$

where $\vec{\boldsymbol{n}} = [\boldsymbol{n}_l]_l \sim \mathcal{CN}(\vec{0}, N_0 R_u)$.

  - *Whitening:* It is easier to work with white noise. Therefore, we whiten the noise by multiplying $\vec{\boldsymbol{y}}$ by the *whitening matrix* $R_u^{-\frac{1}{2}}$

$$\begin{aligned} \vec{\boldsymbol{y}}_1 = R_u^{-\frac{1}{2}} \vec{\boldsymbol{y}} &= R_u^{-\frac{1}{2}} R_u \vec{\boldsymbol{x}} + \underbrace{R_u^{-\frac{1}{2}} \vec{\boldsymbol{n}}}_{\vec{\boldsymbol{n}}_1} \\ &= R_u^{\frac{1}{2}} \vec{\boldsymbol{x}} + \vec{\boldsymbol{n}}_1, \end{aligned}$$

* where $\vec{\boldsymbol{n}}_1 \sim \mathcal{CN}(\vec{0}, N_0 I)$ because

$$\begin{aligned}
\mathrm{E}\left[\vec{\boldsymbol{n}}_1 \vec{\boldsymbol{n}}_1^\dagger\right] &= \mathrm{E}\left[\left(R_u^{-\frac{1}{2}} \vec{\boldsymbol{n}}\right)\left(R_u^{-\frac{1}{2}} \vec{\boldsymbol{n}}\right)^\dagger\right] \\
&= R_u^{-\frac{1}{2}} \mathrm{E}\left[\vec{\boldsymbol{n}} \vec{\boldsymbol{n}}^\dagger\right] R_u^{-\frac{1}{2}} \\
&= R_u^{-\frac{1}{2}} N_0 R_u R_u^{-\frac{1}{2}} = N_0 I.
\end{aligned}$$

* Note that $\vec{\boldsymbol{y}}_1$ is also a sufficient statistic if $R_u^{-\frac{1}{2}}$ is full rank because $\vec{\boldsymbol{y}}$ can be reconstructed from $\vec{\boldsymbol{y}}_1$.
* From the eigen-decomposition $R_u = V \Lambda V^\dagger$, we obtain $R_u^{\frac{1}{2}} = V \Lambda^{\frac{1}{2}} V^\dagger$ and $R_u^{-\frac{1}{2}} = V \Lambda^{-\frac{1}{2}} V^\dagger$.

– *For Orthonormal Pulses:* If the pulses are orthonormal, $R_u = I$, the channel model becomes

$$\vec{\boldsymbol{y}} = \vec{\boldsymbol{x}} + \vec{\boldsymbol{n}},$$

where $\vec{\boldsymbol{n}} = [\boldsymbol{n}_l]_l \sim \mathcal{CN}(\vec{0}, N_0 I)$ and

$$\boldsymbol{y}_l = \boldsymbol{x}_l + \boldsymbol{n}_l.$$

– If $\vec{\boldsymbol{x}}$ is *real valued*,

$$\Re\left[\vec{\boldsymbol{y}}\right] = \left[\Re\left[\langle \boldsymbol{y}(t), u_l(t)\rangle\right]\right]_l.$$

is also a sufficient statistic. Thus, the equivalent channel model is

$$\Re\left[\vec{\boldsymbol{y}}\right] = \Re\left[R_u\right]\vec{\boldsymbol{x}} + \vec{\boldsymbol{n}}_{\mathrm{I}},$$

where $\vec{\boldsymbol{n}}_{\mathrm{I}} = \Re\left[\vec{\boldsymbol{n}}\right] \sim \mathcal{N}(\vec{0}, \frac{N_0}{2}\Re\left[R_u\right])$ due to the CSCG property.

* *Whitening:* We can again whiten the noise by

$$\vec{\boldsymbol{y}}_2 = \Re\left[R_u\right]^{-\frac{1}{2}} \Re\left[\vec{\boldsymbol{y}}\right] = \Re\left[R_u\right]^{-\frac{1}{2}} \Re\left[R_u\right] \vec{\boldsymbol{x}} + \underbrace{\Re\left[R_u\right]^{-\frac{1}{2}} \vec{\boldsymbol{n}}}_{\vec{\boldsymbol{n}}_2}$$

$$= \Re\left[R_u\right]^{\frac{1}{2}} \vec{\boldsymbol{x}} + \vec{\boldsymbol{n}}_2,$$

· where $\vec{\boldsymbol{n}}_2 \sim \mathcal{N}(\vec{0}, \frac{N_0}{2} I)$.

2. For *linear stream modulation*,

$$u_l(t) = u(t - lt_b).$$

- The discrete time channel model is the same as that of linear modulation where the correlation of the pulses is

$$\begin{aligned} r_u(k,l) &= \langle u(t - kt_b), u(t - lt_b) \rangle \\ &= r_u\left((l-k)t_b\right), \end{aligned}$$

$$r_u(\tau) = \int u(t+\tau)u^*(t)dt.$$

– Orthonormal pulses require

$$r_u\left((l-k)t_b\right) = \delta[l-k].$$

3. For *OFDM modulation*,

$$u_l(t) = e^{j2\pi(lf_\Delta+f_0)t}u(t).$$

- The discrete time channel model is the same as that of linear modulation where the correlation of the pulses is

$$\begin{aligned} r_u(k,l) &= \left\langle e^{j2\pi(kf_\Delta+f_0)t}u(t), e^{j2\pi(lf_\Delta+f_0)t}u(t)\right\rangle \\ &= \left\langle e^{j2\pi(k-l)f_\Delta t}u(t), u(t)\right\rangle \\ &= \int |u(t)|^2 e^{-j2\pi(l-k)f_\Delta t}dt. \end{aligned}$$

– If

$$u(t) = \begin{cases} \sqrt{f_\Delta} & t_1 + \tau_h \leq t \leq t_1 + \tau_h + \frac{1}{f_\Delta} \\ 0 & \text{else} \end{cases},$$

then the pulses are orthonormal and thus $r_u(k,l) = \delta[k-l]$.

4. For OCDM+LSM modulation, the pulses are

$$u_{k,l}(t) = u_k(t - lt_b).$$

- Then,

$$\begin{aligned} \boldsymbol{y}_{k,l} &= \langle \boldsymbol{y}(t), u_{k,l}(t) \rangle \\ &= \sum_{k_1,l_1} \boldsymbol{x}_{k_1,l_1} \underbrace{\langle u_{k_1}(t - l_1 t_b), u_k(t - lt_b) \rangle}_{r_u(k_1,k,l_1,l)} + \underbrace{\langle \boldsymbol{n}(t), u_{k,l}(t) \rangle}_{\boldsymbol{n}_{k,l}}, \end{aligned}$$

where

$$\begin{aligned} & \mathrm{E}\left[\boldsymbol{n}_{k,l}\boldsymbol{n}^*_{k_1,l_1}\right] \\ = \quad & N_0 \left\langle u_{k_1,l_1}(t), u_{k,l}(t)\right\rangle . \end{aligned}$$

– If the pulses are orthonormal, i.e.,

$$r_u(k_1,k,l_1,l) \quad = \quad \delta[k_1-k]\delta[l_1-l],$$

then

$$\boldsymbol{y}_{k,l} \quad = \quad \boldsymbol{x}_{k,l} + \boldsymbol{n}_{k,l},$$

where $\boldsymbol{n}_{k,l} \sim \mathcal{CN}\left(0, N_0\right)$, $\forall k, l$, are i.i.d..

□

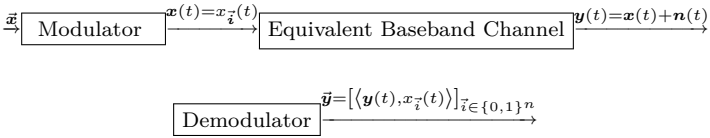
2.3.7. Summary

2.3.7.0.1. *Arbitrary Modulation.*

- Consider *AWGN* channel

$$\boldsymbol{y}(t) \quad = \quad \boldsymbol{x}(t) + \boldsymbol{n}(t),$$

where $\boldsymbol{n}(t)$ is a circularly symmetric complex Gaussian noise process with power spectral density $\tilde{r}_{\boldsymbol{n}}(f) = N_0 \mathrm{rect}_b(f)$; $\boldsymbol{x}(t) = x_{\vec{\boldsymbol{i}}}(t)$ is the baseband signal and is band limited within $\left(-\frac{b}{2}, \frac{b}{2}\right)$.



- Any of the following is a *sufficient statistic* for message $\vec{\boldsymbol{i}}$:

$$
\begin{aligned}
\text{Samples: } \vec{\boldsymbol{y}} &= \left[\boldsymbol{y}(kt_s)\right]_k, \ \frac{1}{t_s} \geq b, \\
\text{distance: } \vec{\boldsymbol{y}} &= \left[d_{\mathrm{E}}^2\left(\boldsymbol{y}(t), x_{\vec{i}}(t)\right)\right]_{\vec{i}\in\{0,1\}^n} \\
&= \left[\int_{-\infty}^{\infty} |\boldsymbol{y}(t) - x_{\vec{i}}(t)|^2 dt\right]_{\vec{i}\in\{0,1\}^n}, \\
\text{correlation: } \vec{\boldsymbol{y}} &= \left[\langle \boldsymbol{y}(t), x_{\vec{i}}(t)\rangle\right]_{\vec{i}\in\{0,1\}^n} \qquad (2.3.7.1) \\
&= \left[\int_{-\infty}^{\infty} \boldsymbol{y}(t) x_{\vec{i}}^*(t) dt\right]_{\vec{i}\in\{0,1\}^n}, \\
\Re[\text{correlation}]: \vec{\boldsymbol{y}} &= \left[\Re\left[\langle \boldsymbol{y}(t), x_{\vec{i}}(t)\rangle\right]\right]_{\vec{i}\in\{0,1\}^n} \\
&= \left[\Re\left[\int_{-\infty}^{\infty} \boldsymbol{y}(t) x_{\vec{i}}^*(t) dt\right]\right]_{\vec{i}\in\{0,1\}^n}.
\end{aligned}
$$

2.3.7.0.2. *Linear Modulation.*

- If $\vec{\boldsymbol{x}}$ is *complex or real* valued,

$$\xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)=\sum_l \boldsymbol{x}_l u_l(t)} \boxed{\text{Equivalent Baseband Channel}} \xrightarrow{\boldsymbol{y}(t)=\boldsymbol{x}(t)+\boldsymbol{n}(t)}$$

$$\boxed{\text{Demodulator}} \xrightarrow[=R_u\vec{\boldsymbol{x}}+\vec{\boldsymbol{n}}]{\vec{\boldsymbol{y}}=[\langle \boldsymbol{y}(t),u_l(t)\rangle]_l} \boxed{R_u^{-\frac{1}{2}} \times \cdot} \xrightarrow[=R_u^{\frac{1}{2}}\vec{\boldsymbol{x}}+\vec{\boldsymbol{n}}_1]{\vec{\boldsymbol{y}}_1=R_u^{-\frac{1}{2}}\vec{\boldsymbol{y}}}$$

$$\xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Equivalent Discrete Time Channel}} \xrightarrow{\vec{\boldsymbol{y}}_1=R_u^{\frac{1}{2}}\vec{\boldsymbol{x}}+\vec{\boldsymbol{n}}_1}$$

- *Demodulator*:

$$\vec{\boldsymbol{y}} \;=\; [\langle \boldsymbol{y}(t), u_l(t)\rangle]_l$$

  - *Equivalent Discrete Time AWGN Channel:*

$$\vec{\boldsymbol{y}} \;=\; R_u\vec{\boldsymbol{x}} + \vec{\boldsymbol{n}},$$

where $R_u \triangleq [r_u(k,l)]_{l,k}$, $r_u(k,l) = \langle u_k(t), u_l(t)\rangle$, $\vec{\boldsymbol{n}} \sim \mathcal{CN}(\vec{0}, N_0 I)$.

* Inter-symbol Interference:

$$\begin{aligned}\boldsymbol{y}_l &= \langle \boldsymbol{y}(t), u_l(t)\rangle \\ &= \boldsymbol{x}_l r_u(l,l) + \underbrace{\sum_{l_1:l_1\neq l} \boldsymbol{x}_{l_1} r_u(l_1,l)}_{\text{Inter-symbol Interference (ISI)}} + \boldsymbol{n}_l,\end{aligned}$$

– *Demodulator:*

$$\vec{\boldsymbol{y}}_1 = R_u^{-\frac{1}{2}} \left[\langle \boldsymbol{y}(t), u_l(t)\rangle\right]_l$$

* *Equivalent Discrete Time AWGN Channel:*

$$\vec{\boldsymbol{y}}_1 = R_u^{\frac{1}{2}} \vec{\boldsymbol{x}} + \vec{\boldsymbol{n}}_1,$$

where $\vec{\boldsymbol{n}}_1 \sim \mathcal{CN}(\vec{0}, N_0 I)$ is white.

- If $\vec{\boldsymbol{x}}$ is *real* valued,
  – *Demodulator:*

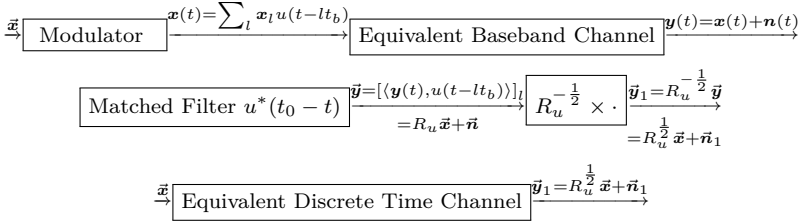
$$\vec{\boldsymbol{y}} = \Re\left[R_u\right]^{-\frac{1}{2}} \left[\Re\left[\langle \boldsymbol{y}(t), u_l(t)\rangle\right]\right]_l$$

– *Equivalent Discrete Time AWGN Channel:*

$$\vec{\boldsymbol{y}} \;=\; \Re\left[R_u\right]^{\frac{1}{2}}\,\vec{\boldsymbol{x}} + \vec{\boldsymbol{n}}_{\mathrm{I}},$$

where $\vec{\boldsymbol{n}}_{\mathrm{I}} \sim \mathcal{N}(\vec{0}, \frac{N_0}{2} I)$.

2.3.7.0.3. *Linear Stream Modulation.*



- *Demodulator:*

$$\begin{aligned}\vec{\boldsymbol{y}} &= [\langle \boldsymbol{y}(t), u(t-lt_b)\rangle]_l \\ &= \left[\boldsymbol{y}(t) * u^*(t_0-t)\big|_{t=t_0+lt_b}\right]_l\end{aligned}$$

- *Equivalent Discrete Time AWGN Channel:* It is the same as that of linear modulation above, where

$$\begin{aligned}r_u(k,l) &= \langle u(t-kt_b), u(t-lt_b)\rangle \\ &= r_u\left((l-k)t_b\right),\end{aligned}$$

$$r_u(\tau) = \int u(t+\tau)u^*(t)dt,$$

$$R_u = \left[r_u\left((l-k)t_b\right)\right]_{l,k},$$

$$\vec{\boldsymbol{n}}_1 \sim \mathcal{CN}(\vec{0}, N_0 I).$$

2.3.7.0.4. *OFDM Modulation.*

- *Demodulator:*

$$\begin{aligned}\vec{\boldsymbol{y}} &= \left[\left\langle \boldsymbol{y}(t), e^{j2\pi(lf_\Delta+f_0)t}u(t)\right\rangle\right]_l \\ &= \left[\int \boldsymbol{y}(t)e^{-j2\pi(lf_\Delta+f_0)t}u^*(t)dt\right]_l .\end{aligned}$$

If

$$u(t) = \begin{cases} c & t_1 \le t \le t_2 \\ 0 & t < t_0 \text{ or } t > t_3 \end{cases},$$

where $t_0 < t_1 < t_2 = t_1 + \tau_h + \frac{1}{f_\Delta} < t_3$, a near sufficient and simple statistic is

$$\vec{\boldsymbol{y}} = \left[\int_{t_1+\tau_h}^{t_1+\tau_h+\frac{1}{f_\Delta}} \boldsymbol{y}(t)e^{-j2\pi(lf_\Delta+f_0)t}cdt\right]_l .$$

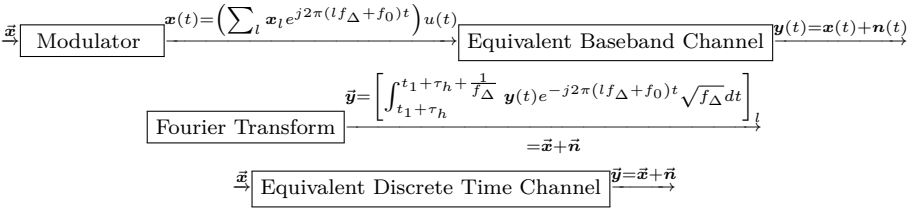
- *Equivalent Discrete Time AWGN Channel:* It is the same as that of linear modulation above, where

$$r_u(k,l) \;=\; \int |u(t)|^2\, e^{-j2\pi(l-k)f_\Delta t} dt.$$

  - If

$$u(t) \;=\; \begin{cases} \sqrt{f_\Delta} & t_1+\tau_h \le t \le t_1+\tau_h+\frac{1}{f_\Delta} \\ 0 & \text{else} \end{cases},$$

    then the pulses are orthonormal and thus $r_u(k,l) = \delta[k-l]$ and $\vec{\boldsymbol{n}} \sim \mathcal{CN}(\vec{0}, N_0 I)$.



2.3.7.0.5. *OCDM+LSM Modulation.*

- *Demodulator:*

$$\vec{\boldsymbol{y}} = \left[\left[\underbrace{\langle \boldsymbol{y}(t), u_k(t - lt_b)\rangle}_{\boldsymbol{y}_{k,l}}\right]_k\right]_l .$$

If the pulses are

$$u_k(t - lt_b) = \sum_{m=1}^{m_s} c_{k,m} u(t - lm_s t_c - mt_c),$$

where m_s is the spreading code length, and the symbol period $t_b = m_s t_c$. Then,

$$\begin{aligned}
\boldsymbol{y}_{k,l} &= \left\langle \boldsymbol{y}(t), \sum_{m=1}^{m_s} c_{k,m} u(t - l m_s t_c - m t_c) \right\rangle \\
&= \sum_{m=1}^{m_s} c_{k,m}^* \left\langle \boldsymbol{y}(t), u(t - l m_s t_c - m t_c) \right\rangle \\
&= \left\langle \left[\left\langle \boldsymbol{y}(t), u(t - l m_s t_c - m t_c) \right\rangle \right]_{m=1:m_s}, \vec{c}_k \right\rangle .
\end{aligned}$$

- *Equivalent Discrete Time AWGN Channel:* It is the same as that of linear modulation above.

$$\begin{aligned}
\boldsymbol{y}_{k,l} &= \langle \boldsymbol{y}(t), u_{k,l}(t) \rangle \\
&= \sum_{k_1, l_1} \boldsymbol{x}_{k_1,l_1} \underbrace{\langle u_{k_1}(t - l_1 t_b), u_k(t - l t_b) \rangle}_{r_u(k_1,k,l_1,l)} + \underbrace{\langle \boldsymbol{n}(t), u_{k,l}(t) \rangle}_{\boldsymbol{n}_{k,l}},
\end{aligned}$$

where

$$\begin{aligned} & \mathrm{E}\left[\boldsymbol{n}_{k,l}\boldsymbol{n}^*_{k_1,l_1}\right] \\ = \quad & N_0 \left\langle u_{k_1,l_1}(t), u_{k,l}(t)\right\rangle . \end{aligned}$$

– If the pulses are orthonormal, i.e.,

$$r_u(k_1,k,l_1,l) \quad = \quad \delta[k_1-k]\delta[l_1-l],$$

then

$$\boldsymbol{y}_{k,l} \quad = \quad \boldsymbol{x}_{k,l} + \boldsymbol{n}_{k,l},$$

where $\boldsymbol{n}_{k,l} \sim \mathcal{CN}\left(0, N_0\right)$, $\forall k, l$, are i.i.d..

$$\xrightarrow{\vec{\boldsymbol{x}}=\left[\left[\boldsymbol{x}_{k,l}\right]_k\right]_l} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)=\sum_k \sum_l \boldsymbol{x}_{k,l} \sum_{m=1}^{m_s} c_{k,m} u(t-lm_s t_c - mt_c)}$$

$$\boxed{\text{Equivalent Baseband Channel}} \xrightarrow{\boldsymbol{y}(t)=\boldsymbol{x}(t)+\boldsymbol{n}(t)}$$

$$\boxed{\text{Matched Filter } u^*(t_0-t)} \xrightarrow[=\vec{\boldsymbol{x}}+\vec{\boldsymbol{n}}]{\vec{\boldsymbol{y}}=\left[\left[\left\langle [\langle \boldsymbol{y}(t), u(t-lm_s t_c - mt_c)\rangle]_{m=1:m_s}, \vec{c}_k \right\rangle\right]_k\right]_l}$$

$$\xrightarrow{\vec{x}} \boxed{\text{Equivalent Discrete Time Channel}} \xrightarrow{\vec{y}=\vec{x}+\vec{n}}$$

Appendices

2.3.A. Homework Problems

PROBLEM 2.3.A.1.

(*Correlation of Waveforms*) Recall the definition of inner product

$$\langle u_1(t), u_2(t) \rangle \triangleq \int u_1(t) u_2^*(t) dt$$

and

$$\text{rect}_{t_0}(t) \triangleq \begin{cases} 1 & -\frac{t_0}{2} \leq t \leq \frac{t_0}{2} \\ 0 & \text{else} \end{cases}.$$

Calculate $\langle u_1(t), u_2(t) \rangle$ for the following cases.

1. $u_1(t) = \text{rect}_{t_0}(t)$, $u_2(t) = u_1(t)$, $t_0 = 2.2$.
2. $u_1(t) = \text{rect}_{t_0}(t)$, $u_2(t) = \text{rect}_{t_0}(t - t_b)$, $t_0 = 2.2$, $t_b = 2$.
3. $u_1(t) = \text{rect}_{t_c}(t) + \text{rect}_{t_c}(t - t_c)$, $u_2(t) = \text{rect}_{t_c}(t) - \text{rect}_{t_c}(t - t_c)$, $t_c = 1$.

4. $u_1(t) = \text{rect}_{t_c}(t) + \text{rect}_{t_c}(t - t_c)$, $u_2'(t) = \text{rect}_{t_c}(t) - \text{rect}_{t_c}(t - t_c)$, $u_2(t) = u_2'(t - \tau)$, $t_c = 1$, $\tau = 0.2$.
5. $u_1(t) = \text{rect}_{\frac{1}{f_\Delta}}(t)$, $u_2(t) = e^{j2\pi l f_\Delta t}\text{rect}_{\frac{1}{f_\Delta}}(t)$, $f_\Delta = 10^5$, $l = 3$.
6. $u_1(t) = \text{rect}_{\frac{1}{f_\Delta}}(t)$, $u_2(t) = e^{j2\pi l f_\Delta (t-\tau)}\text{rect}_{\frac{1}{f_\Delta}}(t)$, $f_\Delta = 10^5$, $l = 3$, $\tau = 0.2\frac{1}{f_\Delta}$.
7. $u_1(t) = \text{rect}_{\frac{1}{f_\Delta}}(t)$, $u_2(t) = e^{j2\pi l (f_\Delta + f_e) t}\text{rect}_{\frac{1}{f_\Delta}}(t)$, $f_\Delta = 10^5$, $f_e = 0.1 f_\Delta$, $l = 3$.

PROBLEM 2.3.A.2.

(*Correlation of Waveforms*) Recall the definition of inner product

$$\langle u_1(t), u_2(t) \rangle \triangleq \int u_1(t) u_2^*(t) dt.$$

Define

$$u(t) \triangleq \sqrt{b}\text{sinc}(tb)$$

Calculate $\langle u_1(t), u_2(t)\rangle$ for the following cases. Hint: Since $b\text{sinc}(tb)$ is the impulse response of an ideal low pass filter with frequency response $\text{rect}_b(f)$, we have $x(t) * b\text{sinc}(tb) = \int x(t-\lambda) b\text{sinc}(\lambda b) d\lambda = x(t)$ for any $x(t)$ whose band is within $\left[-\frac{b}{2}, \frac{b}{2}\right]$.

1. $u_1(t) = u(t)$, $u_2(t) = u(t)$, $b = 5 \times 10^6$.
2. $u_1(t) = u(t)$, $u_2(t) = u(t - \frac{1}{b})$, $b = 5 \times 10^6$.
3. $u_1(t) = u(t)$, $u_2(t) = u(t - \frac{1}{b} - \tau)$, $b = 5 \times 10^6$, $\tau = 0.1\frac{1}{b}$.
4. $u_1(t) = u(t)$, $u_2(t) = u(t - \frac{1}{b} - \tau)$, $b = 5 \times 10^6$, $\tau = -0.1\frac{1}{b}$.

PROBLEM 2.3.A.3.

(*Choice of Whitening Matrix*) For the model of the matched filter output of linear modulation,

$$\vec{\boldsymbol{y}} = R_u \vec{\boldsymbol{x}} + \vec{\boldsymbol{n}},$$

where $\vec{\boldsymbol{n}} = [\boldsymbol{n}_l]_l \sim \mathcal{CN}(\vec{0}, N_0 R_u)$, we know we can use whitening matrix $R_u^{-\frac{1}{2}}$ to whiten the noise in $\vec{\boldsymbol{y}}$ such that

$$\begin{aligned}\vec{\boldsymbol{y}}_1 = R_u^{-\frac{1}{2}}\vec{\boldsymbol{y}} &= R_u^{-\frac{1}{2}} R_u \vec{\boldsymbol{x}} + \underbrace{R_u^{-\frac{1}{2}}\vec{\boldsymbol{n}}}_{\vec{\boldsymbol{n}}_1} \\ &= R_u^{\frac{1}{2}}\vec{\boldsymbol{x}} + \vec{\boldsymbol{n}}_1,\end{aligned}$$

where $\vec{\boldsymbol{n}}_1 \sim \mathcal{CN}(\vec{0}, N_0 I)$, the eigen-decomposition $R_u = V\Lambda V^\dagger$, and $R_u^{-\frac{1}{2}} = V\Lambda^{-\frac{1}{2}}V^\dagger$.

1. Can $A = \Lambda^{-\frac{1}{2}}V^\dagger$ be used as a whitening matrix?
2. If so, which whitening matrix will produce a simpler model when the pulses are near orthogonal? (a)$\vec{\boldsymbol{y}}_1 = R_u^{-\frac{1}{2}}\vec{\boldsymbol{y}}$, or (b) $\vec{\boldsymbol{y}}_2 = A\vec{\boldsymbol{y}}$?

PROBLEM 2.3.A.4.

(*Demodulation of LSM*) The problem uses a simple numeric example to help you go through the procedure of demodulation for linear stream modulation. It is helpful to draw pictures of the waveforms when you solve the problem.

- The transmit signal is

$$\boldsymbol{x}(t) \quad = \quad \sum_{l=0}^{l_s-1} \boldsymbol{x}_l u(t - lt_b),$$

  - where the pulse shape $u(t) = \mathrm{rect}_{t_p}(t - \frac{t_p}{2})$.
- The received signal is

$$\boldsymbol{y}(t) \quad = \quad \boldsymbol{x}(t) + \boldsymbol{n}(t),$$

  - where $\boldsymbol{n}(t)$ is a white circularly symmetric complex Gaussian noise process with power spectral density $\tilde{r}_{\boldsymbol{n}}(f) = N_0$.
- Assume

- $t_p = 2.1$, $t_b = 2$, $l_s = 4$,
- a particular received signal $y(t) = t - \left\lfloor \frac{l_s}{2} \right\rfloor t_b$.

1. Calculate the sufficient statistic

$$\begin{aligned} \vec{y} &= [\langle y(t), u(t - lt_b)\rangle]_l \\ &= \left[y(t) * u^*(t_0 - t)|_{t=t_0+lt_b} \right]_l . \end{aligned}$$

2. Calculate R_u for the equivalent channel model

$$\vec{\boldsymbol{y}} = R_u \vec{\boldsymbol{x}} + \vec{\boldsymbol{n}},$$

   where $\vec{\boldsymbol{n}} = [\boldsymbol{n}_l]_l \sim \mathcal{CN}(\vec{0}, N_0 R_u)$.
3. Calculate the ISI in y_2, the second element of $\vec{y}$, knowing $\vec{x} = [1]_l$ is an all one vector.
4. Calculate the whitening matrix $R_u^{-\frac{1}{2}}$, $R_u^{\frac{1}{2}}$, and $\vec{y}_1 = R_u^{-\frac{1}{2}} \vec{y}$ for the equivalent channel model

$$\vec{\boldsymbol{y}}_1 = R_u^{-\frac{1}{2}} \vec{\boldsymbol{y}} = R_u^{\frac{1}{2}} \vec{\boldsymbol{x}} + \vec{\boldsymbol{n}}_1,$$

where $\vec{\boldsymbol{n}}_1 \sim \mathcal{CN}(\vec{0}, N_0 I)$.

CHAPTER 2.4

Modulator Optimization

2.4.1. Introduction

$$\xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)} \boxed{\text{Equivalent Baseband Channel}} \xrightarrow{\boldsymbol{y}(t)} \boxed{\text{Demodulator}} \xrightarrow{\vec{\boldsymbol{y}}}$$

- How to optimize the modulator to facilitate low complexity receiver and high spectral efficiency?

2.4.1.0.1. *Subjects.*

- Orthogonal pulse design
- Spectrum efficient pulse design
- Reducing ISI

2.4.1.0.2. *Learning Objectives*.

- Can design a pulse to meet various criteria.

2.4.1.0.3. *Organization.*

- Pulse design
  - orthogonality
  - power spectral density
  - ISI

2.4.2. Design Space

$$\xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)} \boxed{\text{Equivalent Baseband Channel}} \xrightarrow{\boldsymbol{y}(t)} \boxed{\text{Demodulator}} \xrightarrow{\vec{\boldsymbol{y}}}$$

- What are the parameters/functions that can be designed in a linear modulator?
  - We consider linear modulation because it does not lose generality.

2.4.2.0.1. *Design space.*

PROBLEM 2.4.2.1. (*Design Space*)

1. What can be optimized/designed in the modulator of a linear modulation?
2. What could be the design objectives?

Solution:

1. *Pulse Design:* For linear modulation,

$$\boldsymbol{x}(t) \quad = \quad \sum_k \boldsymbol{x}_k u_k(t),$$

   all we can optimize/design is the *pulses* $u_k(t)$ because $\boldsymbol{x}_k$'s are designed by the error control coding layer.

2. There are multiple *objectives*/requirements on the pulses:
   - *Tight Spectrum:* We need to minimize the power leakage to the adjacent bands, i.e., fit the power spectral density (PSD) under a mask;
   - *Finite Time Duration:* We do not want the time duration of the pulses to be too long in order to reduce signal processing complexity;
   - *Orthogonality:* We want the pulses to be near orthogonal so that the correlation matrix R_u is near a scaled

identity in order to reduce the decoding complexity of the error control coding layer, considering the equivalent channel model

$$\vec{\boldsymbol{y}} \;=\; R_u^{\frac{1}{2}}\vec{\boldsymbol{x}} + \vec{\boldsymbol{n}},$$

where $R_u \triangleq [r_u(k,l)]_{l,k}$, $r_u(k,l) = \langle u_k(t), u_l(t)\rangle$, $\vec{\boldsymbol{n}} \sim \mathcal{CN}(\vec{0}, N_0 I)$.

- *Reducing ISI:* In suboptimal decoding, if we treat the inter-symbol interference (ISI) as noise, it is even more important to make R_u near a scaled identity in order to make ISI close to zero.
- *Real Symbols:* If $\vec{\boldsymbol{x}}$ is real valued, we want $\Re\left[R_u\right]$ to be near a scaled identity in order to reduce the decoding complexity of the error control coding

layer, considering the equivalent channel model

$$\vec{\boldsymbol{y}} \;=\; \Re\,[R_u]^{\frac{1}{2}}\,\vec{\boldsymbol{x}} + \vec{\boldsymbol{n}}_{\mathrm{I}},$$

where $\vec{\boldsymbol{n}}_{\mathrm{I}} \sim \mathcal{N}(\vec{0}, \frac{N_0}{2} I)$.

$\square$

2.4.3. Pulse Design for Linear Stream Modulation

$$\xrightarrow{\vec{x}} \boxed{\text{Modulator}} \xrightarrow{x(t)} \boxed{\text{Equivalent Baseband Channel}} \xrightarrow{y(t)} \boxed{\text{Demodulator}} \xrightarrow{\vec{y}}$$

- How to design the pulse to fit the spectrum mask and reduce ISI?
- We investigate various aspects of pulse design for linear stream modulation, including
  - spectrum mask, orthogonality, finite time duration, and ISI.

2.4.3.0.1. *Spectrum Mask.*

PROBLEM 2.4.3.1. (**Spectrum Mask**)

Transmit signal should be limited to specified frequency band and should produce as little interference to adjacent bands as possible. Therefore the power spectral density should be under a spectrum mask specified by regulatory authority. For linear stream modulation,

$$\boldsymbol{x}(t) \quad = \quad \sum_{l} \boldsymbol{x}_l u(t - lt_b),$$

how to satisfy the requirement

$$\mathrm{PSD}_{\mathrm{LSM}}(f) \quad \leq \quad \mathrm{Mask}(f)?$$

Solution:

- The PSD of LSM is

$$\mathrm{PSD}_{\mathrm{LSM}}(f) = \frac{e_s}{t_b}\left|\tilde{u}(f)\right|^2 \leq \mathrm{Mask}(f). \tag{2.4.3.1}$$

  Therefore,
  - we should design the pulse shape $u(t)$ such that

$$\frac{\left|\tilde{u}(f)\right|^2}{c_1} \leq \mathrm{Mask}(f),$$

    where the normalizing factor c_1 is chosen such that

$$\max_f \frac{\left|\tilde{u}(f)\right|^2}{c_1} = \max_f \mathrm{Mask}(f),$$

  - Then, the signal power $\frac{e_s}{t_b}$ can be chosen to satisfy equation (2.4.3.1).

□

2.4.3.0.2. *Orthogonal pulse design.*

PROBLEM 2.4.3.2. (**Orthogonal Pulse Design**)
For linear stream modulation,

$$\boldsymbol{x}(t) \quad = \quad \sum_{l} \boldsymbol{x}_l u(t - lt_b)$$

how to design the pulse to achieve zero ISI?

Solution:

- If $\vec{\boldsymbol{x}}$ is *complex or real* valued, to achieve zero ISI, we want $R_u = I$, i.e.,

$$
\begin{aligned}
r_u((k-l)t_b) &= \delta[k-l] \\
&\Longrightarrow \\
r_u(t)\sum_k \delta(t-kt_b) &= \delta(t) \\
&\overset{\mathcal{F}}{\Longrightarrow} \\
|\tilde{u}(f)|^2 * \sum_k \frac{1}{t_b}\delta(f-k\frac{1}{t_b}) &= 1 \\
&\Longrightarrow \\
\frac{1}{t_b}\sum_k |\tilde{u}(f-k\frac{1}{t_b})|^2 &= 1, \qquad (2.4.3.2)
\end{aligned}
$$

i.e., the summation of shifted copies of $|\tilde{u}(f)|^2$ is a constant.

- The above has assumed the pulse energy is 1.
- For example, $\tilde{u}(f) = \sqrt{t_b}\text{rect}_{\frac{1}{t_b}}(f)$ satisfies the requirement.
- For example, if *odd* functions plus a constant are used at the edge, the summation will be constant. Give some examples of your own.
- *Nonzero Crossing of the Pulse:* Note that orthogonal pulses does not necessarily imply zero crossing

$$u(t_0 - lt_b) \quad = \quad c\delta[l]$$

for some t_0. That is, there may be ISI in samples $\boldsymbol{y}(t_0 + lt_b)$, $\forall l$. However, what we want is the sufficient statistic, the matched filter outputs, that has zero ISI.

- If $\vec{\boldsymbol{x}}$ is *real* valued, to achieve zero ISI, we want $\Re\left[R_u\right] = I$, i.e.,

$$
\begin{aligned}
\Re\left[r_u((k-l)t_b)\right] &= \delta[k-l] \\
&\Longrightarrow \\
\Re\left[r_u(t)\sum_k \delta(t-kt_b)\right] &= \delta(t) \\
&\overset{\mathcal{F}}{\Longrightarrow} \\
\frac{|\tilde{u}(f)|^2 + |\tilde{u}(-f)|^2}{2} * \sum_k \frac{1}{t_b}\delta(f - k\frac{1}{t_b}) &= 1 \\
&\Longrightarrow \\
\frac{1}{t_b}\sum_k \frac{|\tilde{u}(f-k\frac{1}{t_b})|^2 + |\tilde{u}(-(f-k\frac{1}{t_b}))|^2}{2} &= 1,
\end{aligned}
$$

i.e., the summation of shifted copies of $|\tilde{u}(f)|^2$ plus $|\tilde{u}(-f)|^2$ is a constant.

- For example, $\tilde{u}(f) = \sqrt{2t_b}\mathrm{rect}_{\frac{1}{2t_b}}(f - \frac{1}{4t_b})$ satisfies the requirement.
- For example, if *odd* functions plus a constant are used at the edge, the summation will be constant. Give some examples of your own.
- *Half Bandwidth:* Note that the smallest bandwidth could be $\frac{1}{2t_b}$, which is half of the bandwidth $\frac{1}{t_b}$ of the complex valued $\vec{\boldsymbol{x}}$ case, where two streams of data are transmitted in the real and imaginary parts of $\vec{\boldsymbol{x}}$. Thus, in terms of bandwidth efficiency, the two cases are the same.

□

PROBLEM 2.4.3.3. (**Square Root Raised Cosine Pulse**) This is an example of orthogonal pulse.

- The *Fourier transform* of the *raised cosine pulse*,

$$
\begin{aligned}
|\tilde{u}(f)|^2 &= \begin{cases} 1 & |f| \leq \frac{1-\alpha}{2} \\ \cos^2\left(\frac{\pi}{2\alpha}\left(|f| - \frac{1-\alpha}{2}\right)\right) & \frac{1-\alpha}{2} < |f| \leq \frac{1+\alpha}{2} \\ 0 & \text{else} \end{cases} \\
&\overset{\text{or}}{=} \begin{cases} 1 & |f| \leq \frac{1-\alpha}{2} \\ \frac{1+\cos\left(\frac{\pi}{\alpha}\left(|f| - \frac{1-\alpha}{2}\right)\right)}{2} & \frac{1-\alpha}{2} < |f| \leq \frac{1+\alpha}{2} \\ 0 & \text{else} \end{cases},
\end{aligned}
$$

satisfies the constraint of 2.4.3.2, $\frac{1}{t_b}\sum_k |\tilde{u}(f - k\frac{1}{t_b})|^2 = 1$, $t_b = 1$, i.e., the sum of shifts of multiple of 1 is a constant, due to the identity $\sin^2(x) + \cos^2(x) = 1$.

- Correspondingly, the *Fourier transform* of the *root raised cosine pulse* is

$$\tilde{u}(f) \quad = \quad \begin{cases} 1 & |f| \leq \frac{1-\alpha}{2} \\ \cos\left(\frac{\pi}{2\alpha}\left(|f| - \frac{1-\alpha}{2}\right)\right) & \frac{1-\alpha}{2} < |f| \leq \frac{1+\alpha}{2} \\ 0 & \text{else} \end{cases} .$$

- Therefore, $u(t)$ is a pulse with orthogonal shifts of multiple of 1.

1. Derive the *root raised cosine pulse* $u(t) = \mathcal{F}^{-1}\left\{\tilde{u}(f)\right\}$.
2. What is the energy of the pulse $e_u = \int |u(t)|^2 \, dt$?
3. Derive the *raised cosine pulse* $r_u(t) = \mathcal{F}^{-1}\left\{|\tilde{u}(f)|^2\right\}$, which is the auto-correlation function of the root raised cosine pulse $u(t)$.
4. Based on $u(t)$, create a unit energy pulse $u(t, t_b)$ with symbol rate $\frac{1}{t_b}$. Find $\tilde{u}(f, t_b)$, $|\tilde{u}(f, t_b)|^2$, and $r_{u(\cdot, t_b)}(t)$.

Solution:

1. The details of the calculation is in the appendix of this chapter. Calculate the inverse Fourier transform

$$
\begin{aligned}
u(t) &= \mathcal{F}^{-1}\left\{\tilde{u}(f)\right\} \\
&= \frac{(1-\alpha)\mathrm{sinc}\left(t(1-\alpha)\right) + \frac{4\alpha}{\pi}\cos\left(\pi t(1+\alpha)\right)}{\left(1-(4\alpha t)^2\right)} \\
&\overset{\text{or}}{=} \frac{\frac{1}{t}\sin\left(\pi t(1-\alpha)\right) + 4\alpha\cos\left(\pi t(1+\alpha)\right)}{\pi\left(1-(4\alpha t)^2\right)},
\end{aligned}
$$

where

$$
u\left(\pm\frac{1}{4\alpha}\right) = \frac{\alpha}{\sqrt{2}}\left(\left(1-\frac{2}{\pi}\right)\cos\left(\frac{\pi}{4\alpha}\right) + \left(1+\frac{2}{\pi}\right)\sin\left(\frac{\pi}{4\alpha}\right)\right),
$$

$$
u(0) = (1-\alpha) + \frac{4\alpha}{\pi}
$$

2.

$$
\begin{aligned}
e_u &= \int |u(t)|^2 \, dt \\
&= \int |\tilde{u}(f)|^2 \, df \\
&= 2\left(\int_0^{\frac{1-\alpha}{2}} 1 \cdot df + \int_{\frac{1-\alpha}{2}}^{\frac{1+\alpha}{2}} \frac{1+\cos\left(\frac{\pi}{\alpha}\left(|f| - \frac{1-\alpha}{2}\right)\right)}{2} df \right) \\
&= 2\left(\frac{1-\alpha}{2} + \int_0^{\alpha} \frac{1+\cos\left(\frac{\pi}{\alpha}(f)\right)}{2} df \right) \\
&= 2\left(\frac{1-\alpha}{2} + \frac{\alpha}{2} \right) \\
&= 1.
\end{aligned}
$$

3. Calculate the inverse Fourier transform

$$\begin{aligned} r_u(t) &= \mathcal{F}^{-1}\left\{|\tilde{u}(f)|^2\right\} \\ &= \frac{\cos(\pi t \alpha)}{\left(1-(2\alpha t)^2\right)} \mathrm{sinc}\,(t)\,, \end{aligned}$$

where

$$r_u\left(\pm\frac{1}{2\alpha}\right) = \frac{\pi}{4}\mathrm{sinc}\left(\frac{1}{2\alpha}\right),$$

$$r_u(0) = 1.$$

4. Unit energy pulse satisfies $\frac{1}{t_b}\sum_k |\tilde{u}(f-k\frac{1}{t_b})|^2 = 1$. Thus we choose

$$|\tilde{u}(f,t_b)|^2 = t_b\,|\tilde{u}(ft_b)|^2\,.$$

Consequently

$$
\begin{aligned}
r_{u(\cdot,t_b)}(t) &= r_u\left(\frac{t}{t_b}\right), \\
\tilde{u}(f,t_b) &= \sqrt{t_b}\tilde{u}(ft_b), \\
u(f,t_b) &= \frac{1}{\sqrt{t_b}}u\left(\frac{t}{t_b}\right),
\end{aligned}
$$

according to the scaling property of Fourier transform.

$\square$

2.4.3.0.3. *Windowed pulse.*

PROBLEM 2.4.3.4. (**Windowed Pulse**)
The power spectral density of linear stream modulation,

$$\boldsymbol{x}(t) = \sum_l \boldsymbol{x}_l u(t - lt_b),$$

is

$$\mathrm{PSD}_{\mathrm{LSM}}(f) = \frac{e_s}{t_b} \left|\tilde{u}(f)\right|^2.$$

To design finite support pulse, we often start with an infinite support pulse $u_1(t)$and multiply it by a finite support window function $w(t)$ to obtain

$$u(t) = u_0(t)w(t).$$

1. How does the window function $w(t)$ affect the power spectral density?

2. How should we design $w(t)$ to minimize the bandwidth and ISI?

Solution:

1. The PSD is

$$\begin{aligned}\mathrm{PSD}_{\mathrm{LSM}}(f) &= \frac{e_s}{t_b}\left|\tilde{u}(f)\right|^2 \\ &= \frac{e_s}{t_b}\left|\tilde{u}_0(f) * \tilde{w}(f)\right|^2 .\end{aligned}$$

   The bandwidth of $\tilde{u}(f)$ is approximately the sum of the bandwidth of $\tilde{u}_0(f)$ and $\tilde{w}(f)$.

2. We want to minimize the bandwidth and the tails of $\tilde{w}(f)$. That implies that $w(t)$ should have a a smooth transition to zero. If $u_0(t)$ is a pulse of zero ISI, we don't want to distort it much to minimize ISI.

□

2.4.3.0.4. *Worst case ISI.*

PROBLEM 2.4.3.5. (**Worst Case ISI**)

- For linear stream modulation,

$$\boldsymbol{x}(t) = \sum_l \boldsymbol{x}_l u(t - lt_b),$$

consider the *equivalent channel*

$$\begin{aligned} \vec{\boldsymbol{y}} &= [\langle \boldsymbol{y}(t), u_l(t) \rangle]_l \\ &= R_u \vec{\boldsymbol{x}} + \vec{\boldsymbol{n}}, \end{aligned}$$

where $R_u \triangleq [r_u(k,l)]_{l,k}$, $r_u(k,l) = \langle u(t - kt_b), u(t - lt_b) \rangle = r_u((l-k)t_b)$, $r_u(\tau) \triangleq \langle u(t+\tau), u(t) \rangle$, $\vec{\boldsymbol{n}} \sim \mathcal{CN}(\vec{0}, N_0 R_u)$. Assume the energy of the pulse is $e_u = r_u(0)$.

- The *ISI* is defined in

$$\begin{aligned}\boldsymbol{y}_l &= \langle \boldsymbol{y}(t), u_l(t)\rangle \\ &= \boldsymbol{x}_l \underbrace{r_u(l,l)}_{e_u} + \underbrace{\sum_{l_1:l_1\neq l} \boldsymbol{x}_{l_1} r_u(l_1,l)}_{\text{Inter-symbol Interference (ISI)}} + \boldsymbol{n}_l.\end{aligned}$$

- Consider BPSK constellation of which $\frac{\boldsymbol{x}_l}{a} \in \{1,-1\}$, $a > 0$. For real valued constellation, the sufficient statistic is $[\Re[\boldsymbol{y}_l]]_l$.

1. What is the worst degradation due to ISI for detecting $\boldsymbol{x}_l = a$ using a threshold detector

$$\hat{\boldsymbol{x}}_l = \begin{cases} a & \Re[\boldsymbol{y}_l] \geq 0 \\ -a & \Re[\boldsymbol{y}_l] < 0 \end{cases}$$

What are the values of $\boldsymbol{x}_{l_1}$, $l_1 \neq l$, that achieves the worst ISI when $\boldsymbol{x}_l = a$?

Solution:

1. If $\boldsymbol{x}_l = a$, the effective signal power for the real valued constellation is

$$
\begin{aligned}
& \left| \boldsymbol{x}_l \underbrace{r_u(l,l)}_{e_u} + \underbrace{\sum_{l_1 : l_1 \neq l} \boldsymbol{x}_{l_1} \Re\left[r_u(l_1,l)\right]}_{\text{Inter-symbol Interference (ISI)}} \right|^2 \\
= & \left| \boldsymbol{x}_l e_u \left(1 + \frac{\sum_{l_1 : l_1 \neq l} \boldsymbol{x}_{l_1} \Re\left[r_u(l_1,l)\right]}{\boldsymbol{x}_l e_u} \right) \right|^2 \\
= & \left| \boldsymbol{x}_l e_u \left(1 + \sum_{l_1 : l_1 \neq l} \frac{\boldsymbol{x}_{l_1}}{a} \frac{\Re\left[r_u(l_1,l)\right]}{e_u} \right) \right|^2 \\
\geq & \left| \boldsymbol{x}_l e_u \left(1 - \sum_{l_1 : l_1 \neq l} \left| \frac{\Re\left[r_u(l_1,l)\right]}{e_u} \right| \right) \right|^2 .
\end{aligned}
$$

The smaller it is, the worse the error probability is.

- *Degradation:* The degradation of the effective signal power due to the ISI is defined to be

$$-20\log_{10}\left(1+\sum_{l_1:l_1\neq l}\frac{\boldsymbol{x}_{l_1}}{a}\frac{\Re\left[r_u(l_1,l)\right]}{e_u}\right)\ \text{dB},$$

  which is the negative power gain due to the ISI.
- The worst degradation of signal power due to the ISI is

$$-20\log_{10}\left(1-\sum_{l_1:l_1\neq l}\left|\frac{\Re\left[r_u(l_1,l)\right]}{e_u}\right|\right)\ \text{dB},$$

  which depends on the summation of the absolute values of the normalized correlation.

- The symbol combination that results in such worst case is

$$\boldsymbol{x}_{l_1} \quad = \quad -\text{sign}\left(r_u(l_1, l)\right) a, \;\; l_1 \neq l.$$

$\square$

2.4.3.0.5. *Pulse design procedure.*

PROBLEM 2.4.3.6. (**Pulse Design Procedure**) Consider the above aspects of spectrum mask, orthogonality, finite time duration, and ISI, how should we design a pulse to

- maximize symbol rate,
- subject to the spectrum mask constraint, time duration constraint, and worst ISI constraint?

Solution:

- A possible procedure is to
  1. design an orthogonal pulse $u_0(t)$ according to the frequency domain insight equation (2.4.3.2),
  2. design a window function $w(t)$ so that $u(t) = u_0(t)w(t)$ satisfies the time duration constraint,
  3. adjust the parameters of $u_0(t)$ and $w(t)$ to maximize symbol rate, while meeting the spectrum mask constraint and worst ISI constraint.
- Alternatively, one may discretize $u(t)$ to a finite dimensional vector and use a constrained optimization algorithms to search for a solution.
- Use your imagination to find your method.

$\square$

2.4.4. Pulse Design for OCDM

- The pulse design for OCDM is the same as the linear stream modulation because an OCDM pulse is a linear stream modulation of the spreading code.

2.4.5. Pulse Design for OFDM

- Recall that the OFDM signal is

$$\tilde{\boldsymbol{x}}(f) = \sum_l \boldsymbol{x}_l \tilde{u}(f - l f_\Delta - f_0)$$

$$\boldsymbol{x}(t) = \left(\sum_l \boldsymbol{x}_l e^{j2\pi(l f_\Delta + f_0)t} \right) u(t),$$

- How to design the time domain window $u(t)$ for OFDM?

2.4.5.0.1. *Pulse shape design for OFDM.*

PROBLEM 2.4.5.1. (**Pulse shape design for OFDM**) Given OFDM signal in frequency and time domain as

$$\tilde{\boldsymbol{x}}(f) = \sum_l \boldsymbol{x}_l \tilde{u}(f - l f_\Delta - f_0)$$

$$\boldsymbol{x}(t) = \left(\sum_l \boldsymbol{x}_l e^{j2\pi(l f_\Delta + f_0)t} \right) u(t),$$

what are the tradeoffs in the design of $u(t)$ to facilitate low complexity demodulation, small bandwidth, and high data rate?

Solution:

- To achieve **low complexity demodulation and small bandwidth**, observe that if

$$u(t) = \begin{cases} c & t_1 \le t \le t_2 \\ \text{smooth transition to } 0 & t_0 < t < t_1 \text{ and } t_2 < t < t_3 \\ 0 & t < t_0 \text{ or } t > t_3 \end{cases} \tag{2.4.5.1}$$

  where $t_0 < t_1 < t_2 = t_1 + \tau_h + \frac{1}{f_\Delta} < t_3$,
  - then the low complexity suboptimal demodulation can be implemented by (Fast) Fourier Transform due to $u(t)$ being a constant in $[t_1, t_2]$;
  - the bandwidth of the power spectral density,

$$\mathrm{PSD}_{\mathrm{OFDM}}(f) = \frac{e_s \sum_l |\tilde{u}(f - l f_\Delta - f_0)|^2}{\text{duration of } u(t)},$$

will be small due to fast die-off of $\tilde{u}(f)$ as a result of the smooth transition to zero.

- Thus, (2.4.5.1) will be our design template.
- To achieve **high data rate**,
  - we should minimize the time duration $t_3 - t_0$.
  - However, $t_1 - t_0$ and $t_3 - t_2$ need to be large enough to allow smooth transition, and thus, small bandwidth.
  - Note that $t_2 - t_1$ is τ_h larger than $\frac{1}{f_\Delta}$. After we study demodulation in fading channels, we will see that τ_h is there to cover the delay spread so that the demodulator can have low complexity.
  - $\frac{1}{f_\Delta}$ is chosen to make the channel nearly time invariant within one OFDM symbol period.

$\square$

2.4.5.0.2. *Design example.*

EXAMPLE 2.4.1. An example of $u(t)$ to meet the above criteria is

$$u(t) = \begin{cases} c\frac{\cos\left(\pi\frac{t-t_1}{t_1-t_0}\right)+1}{2} & t_0 < t < t_1 \\ c & t_1 \leq t \leq t_2 \\ c\frac{\cos\left(\pi\frac{t-t_2}{t_3-t_2}\right)+1}{2} & t_2 < t < t_3 \\ 0 & t < t_0 \text{ or } t > t_3 \end{cases},$$

with raised cosine for the smooth transition. $t_1 - t_0$ and $t_3 - t_2$ need to be fine tuned to minimize the bandwidth for given data rate.

2.4.6. Summary

2.4.6.0.1. *Pulse design.*

- For linear modulation, the only thing we can optimize are the pulses $u_l(t)$.

2.4.6.0.2. *Design criteria of $u(t)$ for linear stream modulation and OCDM.*

1. $\tilde{u}(f)$ should be close to square to fit under the FCC mask and to contain as much energy as possible.
2. $\tilde{u}(f)$ should have an odd function edge to achieve a constant after shifting for orthogonality, i.e.,

$$\frac{1}{t_b}\sum_k |\tilde{u}(f - k\frac{1}{t_b})|^2 \quad = \quad 1.$$

3. $\tilde{u}(f)$ should have smooth edges to have fast die-off in time domain.
4. When a time domain window is used to make $u(t)$ have finite support, the window
   4.1. should approach zero smoothly at the edge and
   4.2. should be close to flat in the middle in order not to distort the pulse much.

2.4.6.0.3. *Design criteria of $u(t)$ for OFDM.*

- $$u(t) = \begin{cases} c & t_1 \le t \le t_2 \\ \text{smooth transition to } 0 & t_0 < t < t_1 \text{ and } t_2 < t < t_3, \\ 0 & t < t_0 \text{ or } t > t_3 \end{cases}$$

  - where $t_0 < t_1 < t_2 = t_1 + \tau_h + \frac{1}{f_\Delta} < t_3$;
  - the constant c and τ_h are to reduce demodulation complexity in AWGN and fading channels;
  - $\frac{1}{f_\Delta}$ is chosen to make the channel nearly time invariant within one OFDM symbol period;
  - the smooth transition is to reduce bandwidth;
    * $t_1 - t_0$ and $t_3 - t_2$ need to be large enough to allow the smooth transition;

- $t_3 - t_0$ should be minimized to increase OFDM symbol rate.

Appendices

2.4.A. Homework Problems

PROBLEM 2.4.A.1.

(*Pulse Design for Linear Steam Modulation with Complex Valued Symbols*; [Modified from Project 16.9.1 of [1]]) This project examines the design of a realistic pulse shape to be used with linear modulation that will meet a spectral emissions mask in Fig. 2.4.A.1. The spectral emissions mask is given in a Matlab file at the end of this problem. This spectral emissions mask is the one used for GSM handsets. So this problem is one of great practical interest.

- Consider a linear stream modulation having the form

$$\boldsymbol{x}(t) \quad = \quad \sum_{l} \boldsymbol{x}_l u(t - lt_b).$$

- The design is constrained in that the pulse, $u(t)$, must not extend for longer than $t_u = 40$ µs in time.

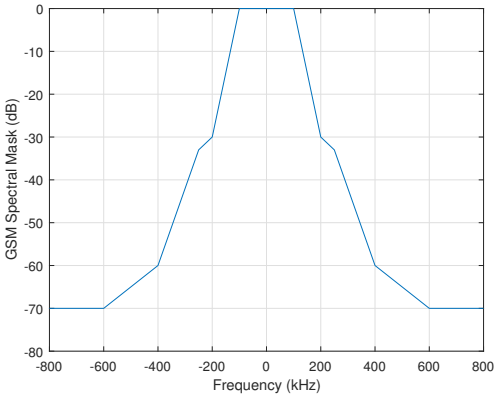


FIGURE 2.4.A.1. GSM Spectral Mask

- For the remainder of the project it is assumed that the pulse shape designed above is used with BPSK modulation.

- Any finite length pulse shape which achieves this spectral mask will have to produce some inter-symbol interference (ISI), which can be made very small with a well designed pulse shape.

1. Assume $x_k = 1$,
   1.1. derive the formula of the worst case degradation of the effective signal power for an arbitrary pulse shape $u(t)$ whose autocorrelation function is $r_u(\tau) = \langle u(t+\tau), u(t) \rangle$.
   1.2. Identify the formula of x_l, $l \neq k$, which achieves this worse case degradation for an arbitrary pulse shape with BPSK modulation.
2. Design a pulse shape and corresponding symbol rate such that
   - the worse case (over all ISI patterns) degradation to the effective SNR is less than 0.25 dB,

- and the spectral mask given in Fig. 2.4.A.1 is met by the transmission. To compare with the spectrum mask, you need to choose e_u such that $\max_f 10 \log_{10} \left(|\tilde{u}(f)|^2 \right) = 0$ dB.
- Grades will be assigned proportional to the achieved symbol rate (the higher the better!). Anyone who beats the posted solution will get a bonus of 10 points for the project!
- Your grade will be penalized if a common solution to yours is found among your classmates solutions in direct proportion to the number of identical solutions. This is to encourage each of you to work independently. Any computer code (e.g., Matlab) should be turned in with the project write-up.

2.1. List your design steps and design parameter values.

2.2. Plot $u(t)$ and $r_u(\tau)$ in one figure, $\tilde{u}(f)$ and$|\tilde{u}(f)|^2$ in one figure, and $10\log_{10}\left(|\tilde{u}(f)|^2\right)$ and Mask(f) in one figure, showing the spectral mask is met.

2.3. For $\vec{x} = [1{-}111{-}11{-}1{-}11]^T$, plot the transmitted complex envelope $x(t)$ for the pulse shape that was designed. Plot the real component of the matched filter output, on which mark the symbol time sampling points.

Matlab file plotting the spectral mask:

```
close all
clear all
% specifying mask
freqm=[-800 -600 -400 -250 -200 -100 0.0000 100 200 250 ...
    400 600 800];
mask=[-70 -70 -60 -33 -30 0 0.0000 0 -30 -33 -60 -70 -70];
```

```
plot(freqm, mask)
xlabel('Frequency (kHz)')
ylabel('GSM Spectral Mask (dB)')
ylim([-80 0])
grid on
```

2.4.B. Calculation of the Root Raised Cosine Pulse

PROBLEM 2.4.B.1. (**Square Root Raised Cosine Pulse**) This is an example of orthogonal pulse.

- The *Fourier transform* of the *raised cosine pulse*,

$$
\begin{aligned}
|\tilde{u}(f)|^2 &= \begin{cases} 1 & |f| \leq \frac{1-\alpha}{2} \\ \cos^2\left(\frac{\pi}{2\alpha}\left(|f| - \frac{1-\alpha}{2}\right)\right) & \frac{1-\alpha}{2} < |f| \leq \frac{1+\alpha}{2} \\ 0 & \text{else} \end{cases} \\
&\overset{\text{or}}{=} \begin{cases} 1 & |f| \leq \frac{1-\alpha}{2} \\ \frac{1+\cos\left(\frac{\pi}{\alpha}\left(|f| - \frac{1-\alpha}{2}\right)\right)}{2} & \frac{1-\alpha}{2} < |f| \leq \frac{1+\alpha}{2} \\ 0 & \text{else} \end{cases},
\end{aligned}
$$

satisfies the constraint of 2.4.3.2, $\frac{1}{t_b}\sum_k |\tilde{u}(f - k\frac{1}{t_b})|^2 = 1$, $t_b = 1$, i.e., the sum of shifts of multiple of 1 is a constant, due to the identity $\sin^2(x) + \cos^2(x) = 1$.

- Correspondingly, the *Fourier transform* of the *root raised cosine pulse* is

$$\tilde{u}(f) = \begin{cases} 1 & |f| \leq \frac{1-\alpha}{2} \\ \cos\left(\frac{\pi}{2\alpha}\left(|f| - \frac{1-\alpha}{2}\right)\right) & \frac{1-\alpha}{2} < |f| \leq \frac{1+\alpha}{2} \\ 0 & \text{else} \end{cases}.$$

- Therefore, $u(t)$ is a pulse with orthogonal shifts of multiple of 1.

1. Derive the *root raised cosine pulse* $u(t) = \mathcal{F}^{-1}\{\tilde{u}(f)\}$.
2. What is the energy of the pulse $e_u = \int |u(t)|^2 \, dt$?

3. Derive the *raised cosine pulse* $r_u(t) = \mathcal{F}^{-1}\left\{|\tilde{u}(f)|^2\right\}$, which is the auto-correlation function of the root raised cosine pulse $u(t)$.
4. Based on $u(t)$, create a unit energy pulse $u(t, t_b)$ with symbol rate $\frac{1}{t_b}$. Find $\tilde{u}(f, t_b)$, $|\tilde{u}(f, t_b)|^2$, and $r_{u(\cdot,t_b)}(t)$.

Solution:

1. Calculate the inverse Fourier transform

$$
\begin{aligned}
u(t) &= \mathcal{F}^{-1}\left\{\tilde{u}(f)\right\} \\
&= \mathcal{F}^{-1}\left\{\mathrm{rect}_{1-\alpha}(f) + \tilde{g}_1(f) + \tilde{g}_1^*(-f)\right\} \\
&= (1-\alpha)\,\mathrm{sinc}\left(t\left(1-\alpha\right)\right) + g_1(t) + g_1^*(t) \\
&= (1-\alpha)\,\mathrm{sinc}\left(t\left(1-\alpha\right)\right) + \Re\left[2g_1(t)\right] \\
&= (1-\alpha)\,\mathrm{sinc}\left(t\left(1-\alpha\right)\right) + \\
&\quad \frac{-\frac{1}{2\alpha}\cos\left(\pi t(1+\alpha)\right) - 2\pi t^2(1-\alpha)\mathrm{sinc}\left(t(1-\alpha)\right)}{2\pi\left(t^2 - \left(\frac{1}{4\alpha}\right)^2\right)} \\
&= \frac{-\left(\frac{1}{4\alpha}\right)^2(1-\alpha)\mathrm{sinc}\left(t(1-\alpha)\right) - \frac{1}{4\pi\alpha}\cos\left(\pi t(1+\alpha)\right)}{\left(t^2 - \left(\frac{1}{4\alpha}\right)^2\right)} \\
&= \frac{(1-\alpha)\mathrm{sinc}\left(t(1-\alpha)\right) + \frac{4\alpha}{\pi}\cos\left(\pi t(1+\alpha)\right)}{\left(1-(4\alpha t)^2\right)} \\
&\overset{\text{or}}{=} \frac{\frac{1}{t}\sin\left(\pi t(1-\alpha)\right) + 4\alpha\cos\left(\pi t(1+\alpha)\right)}{\left(\qquad\qquad\right)}.
\end{aligned}
$$

- When the denominator is zero at $|t| = \frac{1}{4\alpha}$, the value of $u(t)$ is

$$
\begin{aligned}
u(t) &= \frac{(1-\alpha)\mathrm{sinc}\left(t(1-\alpha)\right) + \frac{4\alpha}{\pi}\cos\left(\pi t(1+\alpha)\right)}{\left(1-(4\alpha t)^2\right)} \\
&= \frac{\frac{1}{\pi t}\sin\left(\pi t(1-\alpha)\right) + \frac{4\alpha}{\pi}\cos\left(\pi t(1+\alpha)\right)}{\left(1-(4\alpha t)^2\right)} \\
&\xrightarrow{|t|\to\frac{1}{4\alpha}} \frac{\frac{\pi(1-\alpha)}{\pi t}\cos\left(\pi t(1-\alpha)\right) - \frac{1}{\pi t^2}\sin\left(\pi t(1-\alpha)\right) - 4\alpha(1+\alpha)\sin\left(\pi t(\right.}{-32\alpha^2 t} \\
&= \frac{\frac{\pi(1-\alpha)}{\pi\frac{1}{4\alpha}}\cos\left(\pi\frac{1}{4\alpha}(1-\alpha)\right) - 4\alpha(1+\alpha)\sin\left(\pi\frac{1}{4\alpha}(1+\alpha)\right) - \frac{(4\alpha)^2}{\pi}\mathrm{si}}{-8\alpha} \\
&= \frac{(1-\alpha)\cos\left(\frac{\pi}{4\alpha}(1-\alpha)\right) - (1+\alpha)\sin\left(\frac{\pi}{4\alpha}(1+\alpha)\right) - \frac{4\alpha}{\pi}\sin\left(\frac{\pi}{4\alpha}(\right.}{-2} \\
&= \frac{(1-\alpha)\cos\left(\frac{\pi}{4\alpha}(1-\alpha)\right) - (1+\alpha)\sin\left(\frac{\pi}{4\alpha}(1-\alpha)+\frac{\pi}{2}\right) - \frac{4\alpha}{\pi}\sin}{-2}
\end{aligned}
$$

using the L'Hôpital's rule.

- If $t = 0$, we have

$$
\begin{aligned}
u(0) &= \frac{(1-\alpha)\mathrm{sinc}\left(0(1-\alpha)\right) + \frac{4\alpha}{\pi}\cos\left(\pi 0(1+\alpha)\right)}{\left(1-(4\alpha 0)^2\right)} \\
&\overset{\text{or}}{=} (1-\alpha) + \frac{4\alpha}{\pi}
\end{aligned}
$$

- In the above derivation,

$$
\begin{aligned}
\tilde{g}_1(f) &= \begin{cases} \cos\left(\frac{\pi}{2\alpha}\left(f - \frac{1-\alpha}{2}\right)\right) & \frac{1-\alpha}{2} < f \leq \frac{1+\alpha}{2} \\ 0 & \text{else} \end{cases}, \\
&= \tilde{g}_2(f - \frac{1}{2})
\end{aligned}
$$

$$
\begin{aligned}
\tilde{g}_2(f) &= \cos\left(\frac{\pi}{2\alpha}\left(f+\frac{\alpha}{2}\right)\right)\mathrm{rect}_\alpha(f) \\
&= \frac{1}{2}\left(e^{j\left(\frac{\pi}{2\alpha}\left(f+\frac{\alpha}{2}\right)\right)} + e^{-j\left(\frac{\pi}{2\alpha}\left(f+\frac{\alpha}{2}\right)\right)}\right)\mathrm{rect}_\alpha(f),
\end{aligned}
$$

$$
\begin{aligned}
2g_2(t) &= \mathcal{F}^{-1}\left\{2\tilde{g}_2(f)\right\} \\
&= e^{-j\frac{\pi}{4}}\alpha\mathrm{sinc}\left(\left(t-\frac{1}{4\alpha}\right)\alpha\right)+ \\
&\quad e^{j\frac{\pi}{4}}\alpha\mathrm{sinc}\left(\left(t+\frac{1}{4\alpha}\right)\alpha\right) \\
&= e^{-j\frac{\pi}{4}}\alpha\frac{e^{j\pi\left(t-\frac{1}{4\alpha}\right)\alpha}-e^{-j\pi\left(t-\frac{1}{4\alpha}\right)\alpha}}{2j\pi\left(t-\frac{1}{4\alpha}\right)\alpha}+ \\
&\quad e^{j\frac{\pi}{4}}\alpha\frac{e^{j\pi\left(t+\frac{1}{4\alpha}\right)\alpha}-e^{-j\pi\left(t+\frac{1}{4\alpha}\right)\alpha}}{2j\pi\left(t+\frac{1}{4\alpha}\right)\alpha} \\
&= \alpha\frac{e^{j\pi\left(t\alpha-\frac{1}{2}\right)}-e^{-j\pi t\alpha}}{2j\pi\left(t-\frac{1}{4\alpha}\right)\alpha}+ \\
&\quad \alpha\frac{e^{j\pi\left(t\alpha+\frac{1}{2}\right)}-e^{-j\pi t\alpha}}{2j\pi\left(t+\frac{1}{4\alpha}\right)\alpha} \\
&= \alpha\frac{-je^{j\pi t\alpha}-e^{-j\pi t\alpha}}{\ldots}+
\end{aligned}
$$

$$
\begin{aligned}
2g_1(t) &= 2g_2(t)e^{j2\pi\frac{1}{2}t} \\
&= \alpha\frac{-je^{j\pi t(1+\alpha)} - e^{j\pi t(1-\alpha)}}{2j\pi\left(t-\frac{1}{4\alpha}\right)\alpha} + \\
&\quad \alpha\frac{je^{j\pi t(1+\alpha)} - e^{j\pi t(1-\alpha)}}{2j\pi\left(t+\frac{1}{4\alpha}\right)\alpha} \\
&= \alpha\frac{-\left(t+\frac{1}{4\alpha}\right)je^{j\pi t(1+\alpha)} - \left(t+\frac{1}{4\alpha}\right)e^{j\pi t(1-\alpha)}}{2j\pi\left(t^2-\left(\frac{1}{4\alpha}\right)^2\right)\alpha} + \\
&\quad \alpha\frac{\left(t-\frac{1}{4\alpha}\right)je^{j\pi t(1+\alpha)} - \left(t-\frac{1}{4\alpha}\right)e^{j\pi t(1-\alpha)}}{2j\pi\left(t^2-\left(\frac{1}{4\alpha}\right)^2\right)\alpha} \\
&= \frac{-\frac{1}{2\alpha}je^{j\pi t(1+\alpha)} - 2te^{j\pi t(1-\alpha)}}{2j\pi\left(t^2-\left(\frac{1}{4\alpha}\right)^2\right)},
\end{aligned}
$$

$$
\begin{aligned}
\Re\left[2g_1(t)\right] &= \frac{-\frac{1}{2\alpha}\cos\left(\pi t(1+\alpha)\right) - 2t\sin\left(\pi t(1-\alpha)\right)}{2\pi\left(t^2 - \left(\frac{1}{4\alpha}\right)^2\right)} \\
&= \frac{-\frac{1}{2\alpha}\cos\left(\pi t(1+\alpha)\right) - 2\pi t^2(1-\alpha)\mathrm{sinc}\left(t(1-\alpha)\right)}{2\pi\left(t^2 - \left(\frac{1}{4\alpha}\right)^2\right)}.
\end{aligned}
$$

2.

$$
\begin{aligned}
e_u &= \int |u(t)|^2 \, dt \\
&= \int |\tilde{u}(f)|^2 \, df \\
&= 2\left(\int_0^{\frac{1-\alpha}{2}} 1 \cdot df + \int_{\frac{1-\alpha}{2}}^{\frac{1+\alpha}{2}} \frac{1+\cos\left(\frac{\pi}{\alpha}\left(|f| - \frac{1-\alpha}{2}\right)\right)}{2} df \right) \\
&= 2\left(\frac{1-\alpha}{2} + \int_0^{\alpha} \frac{1+\cos\left(\frac{\pi}{\alpha}(f)\right)}{2} df \right) \\
&= 2\left(\frac{1-\alpha}{2} + \frac{\alpha}{2} \right) \\
&= 1.
\end{aligned}
$$

3. Calculate the inverse Fourier transform

$$
\begin{aligned}
r_u(t) &= \mathcal{F}^{-1}\left\{\left|\tilde{u}(f)\right|^2\right\} \\
&= \mathcal{F}^{-1}\left\{\mathrm{rect}_{1-\alpha}(f) + \tilde{g}_3(f) + \tilde{g}_3^*(-f)\right\} \\
&= (1-\alpha)\,\mathrm{sinc}\left(t\,(1-\alpha)\right) + g_3(t) + g_3^*(t) \\
&= (1-\alpha)\,\mathrm{sinc}\left(t\,(1-\alpha)\right) + \Re\left[2g_3(t)\right] \\
&= (1-\alpha)\,\mathrm{sinc}\left(t\,(1-\alpha)\right) + \\
&\quad \frac{-\frac{1}{2\alpha^2}\sin\left(\pi t(1+\alpha)\right) - \left(4t^2 - \frac{1}{2\alpha^2}\right)\sin\left(\pi t(1-\alpha)\right)}{4\pi t\left(t^2 - \frac{1}{4\alpha^2}\right)} \\
&= \frac{4\left(t^2 - \frac{1}{4\alpha^2}\right)\sin\left(\pi t\,(1-\alpha)\right) - \frac{1}{2\alpha^2}\sin\left(\pi t(1+\alpha)\right) - \left(4t^2 - \frac{1}{2\alpha^2}\right)\sin}{4\pi t\left(t^2 - \frac{1}{4\alpha^2}\right)} \\
&= \frac{-\frac{1}{2\alpha^2}\sin\left(\pi t(1+\alpha)\right) - \frac{1}{2\alpha^2}\sin\left(\pi t(1-\alpha)\right)}{4\pi t\left(t^2 - \frac{1}{4\alpha^2}\right)} \\
&= \frac{\sin\left(\pi t(1+\alpha)\right) + \sin\left(\pi t(1-\alpha)\right)}{2\pi t\left(1 - (2\alpha t)^2\right)}
\end{aligned}
$$

- When the denominator is zero at $|t| = \frac{1}{2\alpha}$, the value of $u(t)$ is

$$\begin{aligned} r_u(t) &= \frac{\cos(\pi t \alpha)}{\left(1-(2\alpha t)^2\right)} \operatorname{sinc}(t) \\ &\xrightarrow{|t| \to \frac{1}{2\alpha}} \left. \frac{-\pi\alpha \sin(\pi t \alpha)}{(-8\alpha^2 t)} \right|_{t=\frac{1}{2\alpha}} \operatorname{sinc}\left(\frac{1}{2\alpha}\right) \\ &= \frac{\pi}{4} \operatorname{sinc}\left(\frac{1}{2\alpha}\right), \end{aligned}$$

  using the L'Hôpital's rule.
- If $t = 0$, we have

$$r_u(0) = \frac{\cos(\pi 0 \alpha)}{\left(1-(2\alpha 0)^2\right)} \operatorname{sinc}(0) = 1.$$

- In the above derivation,

$$\begin{aligned}
\tilde{g}_3(f) &= \begin{cases} \frac{1+\cos\left(\frac{\pi}{\alpha}\left(f-\frac{1-\alpha}{2}\right)\right)}{2} & \frac{1-\alpha}{2} < f \leq \frac{1+\alpha}{2} \\ 0 & \text{else} \end{cases}, \\
&= \tilde{g}_4(f - \frac{1}{2})
\end{aligned}$$

$$\begin{aligned}
\tilde{g}_4(f) &= \frac{1+\cos\left(\frac{\pi}{\alpha}\left(f+\frac{\alpha}{2}\right)\right)}{2}\text{rect}_\alpha(f) \\
&= \frac{1}{4}\left(2 + e^{j\left(\frac{\pi}{\alpha}\left(f+\frac{\alpha}{2}\right)\right)} + e^{-j\left(\frac{\pi}{\alpha}\left(f+\frac{\alpha}{2}\right)\right)}\right)\text{rect}_\alpha(f),
\end{aligned}$$

$$
\begin{aligned}
2g_4(t) &= \mathcal{F}^{-1}\left\{2\tilde{g}_4(f)\right\} \\
&= \alpha\mathrm{sinc}\left(t\alpha\right) + \frac{1}{2}e^{-j\frac{\pi}{2}}\alpha\mathrm{sinc}\left(\left(t-\frac{1}{2\alpha}\right)\alpha\right) + \\
&\quad \frac{1}{2}e^{j\frac{\pi}{2}}\alpha\mathrm{sinc}\left(\left(t+\frac{1}{2\alpha}\right)\alpha\right) \\
&= \alpha\frac{e^{j\pi t\alpha}-e^{-j\pi t\alpha}}{2j\pi t\alpha} + \\
&\quad -\alpha\frac{e^{j\pi\left(t-\frac{1}{2\alpha}\right)\alpha}-e^{-j\pi\left(t-\frac{1}{2\alpha}\right)\alpha}}{4\pi\left(t-\frac{1}{2\alpha}\right)\alpha} + \\
&\quad \alpha\frac{e^{j\pi\left(t+\frac{1}{2\alpha}\right)\alpha}-e^{-j\pi\left(t+\frac{1}{2\alpha}\right)\alpha}}{4\pi\left(t+\frac{1}{2\alpha}\right)\alpha} \\
&= \frac{e^{j\pi t\alpha}-e^{-j\pi t\alpha}}{2j\pi t} - \frac{-je^{j\pi t\alpha}-je^{-j\pi t\alpha}}{4\pi\left(t-\frac{1}{2\alpha}\right)} + \frac{je^{j\pi t\alpha}+je^{-j\pi t\alpha}}{4\pi\left(t+\frac{1}{2\alpha}\right)} \\
&= \frac{\left(2\left(t^2-\frac{1}{4\alpha^2}\right)-t\left(t+\frac{1}{2\alpha}\right)-t\left(t-\frac{1}{2\alpha}\right)\right)e^{j\pi t\alpha}}{4j\pi t\left(t^2-\frac{1}{}\right.} -
\end{aligned}
$$

$$
\begin{aligned}
2g_3(t) &= 2g_4(t)e^{j2\pi\frac{1}{2}t} \\
&= \frac{-\frac{1}{2\alpha^2}e^{j\pi t(1+\alpha)} - \left(4t^2 - \frac{1}{2\alpha^2}\right)e^{j\pi t(1-\alpha)}}{4j\pi t\left(t^2 - \frac{1}{4\alpha^2}\right)}
\end{aligned}
$$

$$
\Re\left[2g_3(t)\right] = \frac{-\frac{1}{2\alpha^2}\sin\left(\pi t(1+\alpha)\right) - \left(4t^2 - \frac{1}{2\alpha^2}\right)\sin\left(\pi t(1-\alpha)\right)}{4\pi t\left(t^2 - \frac{1}{4\alpha^2}\right)}.
$$

4. Unit energy pulse satisfies $\frac{1}{t_b}\sum_k \left|\tilde{u}(f - k\frac{1}{t_b})\right|^2 = 1$. Thus we choose

$$
\left|\tilde{u}(f, t_b)\right|^2 = t_b\left|\tilde{u}(ft_b)\right|^2.
$$

Consequently

$$
r_{u(\cdot,t_b)}(t) = r_u\left(\frac{t}{t_b}\right),
$$

$$\begin{aligned}\tilde{u}(f, t_b) &= \sqrt{t_b}\tilde{u}(f t_b), \\ u(f, t_b) &= \frac{1}{\sqrt{t_b}} u\left(\frac{t}{t_b}\right),\end{aligned}$$

according to the scaling property of Fourier transform.

$\square$

CHAPTER 2.5

Demodulator for Fading Channels and MIMO Fading Channels

2.5.1. Introduction

$$\xrightarrow{\vec{x}} \boxed{\text{Modulator}} \xrightarrow{x(t)} \boxed{\text{Equivalent Baseband Channel}} \xrightarrow{y(t)} \boxed{\text{Demodulator}} \xrightarrow{\vec{y}}$$

2.5.1.0.1. *Topics and organization.*

- We study the demodulator design for slowly time varying flat fading channels,
- time invariant frequency selective fading channels,
- and MIMO fading channels.

2.5.1.0.2. *Learning objectives.*

- Design demodulator for (MIMO) frequency flat and selective fading channels.
- Understand the equivalent discrete time channel model of MIMO fading channels.

2.5.2. Demodulator for Linear Modulation in Flat Fading Channels

$$\xrightarrow{\vec{\boldsymbol{i}}} \boxed{\text{Encoder}} \xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)} \boxed{\text{Fading Channel}} \xrightarrow{\boldsymbol{y}(t)=\boldsymbol{h}(t)\boldsymbol{x}(t)+\boldsymbol{n}(t)} \boxed{\text{Dem}}$$

- How to design demodulators for linear modulation in time invariant and slowly time varying flat fading channels?
- What are the resulting equivalent discrete time channel model for linear modulation in time invariant and slowly time varying flat fading channels?

2.5.2.0.1. *Demodulator for Linear Modulation in Flat Fading Channels.*

PROBLEM 2.5.2.1. (**Demodulator for Linear Modulation in Flat Fading Channels**) Consider the time invariant flat fading channel

$$\xrightarrow{\vec{\boldsymbol{i}}} \boxed{\text{Encoder}} \xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)} \boxed{\text{Fading Channel}} \xrightarrow{\boldsymbol{y}(t)=\boldsymbol{h}\boldsymbol{x}(t)+\boldsymbol{n}(t)} \boxed{\text{Demod}}$$

- where $\vec{\boldsymbol{i}} \in \{0,1\}^n$ is the message to be sent or the information vector;
- $\vec{\boldsymbol{x}} = [\boldsymbol{x}_l]_l = \vec{x}_{\vec{\boldsymbol{i}}} = \left[x_{\vec{\boldsymbol{i}},l}\right]_l$ is the code word;
- $\boldsymbol{x}(t) = \sum_l \boldsymbol{x}_l u_l(t)$ is the baseband signal with linear modulation and is band limited within $\left(-\frac{b}{2}, \frac{b}{2}\right)$;
- $\boldsymbol{h}$ is the time invariant fading coefficient;
- $\boldsymbol{y}(t)$ is the output of the equivalent baseband AWGN channel;

- $\boldsymbol{n}(t)$ is the circularly symmetric complex Gaussian noise process with power spectral density $\tilde{r}_{\boldsymbol{n}}(f) = N_0 \mathrm{rect}_b(f)$.

1. Derive the optimal demodulator and the corresponding equivalent channel model such that
   1.1. $\vec{\boldsymbol{y}}$ is a sufficient statistic for detecting $\vec{\boldsymbol{i}}$ when $\boldsymbol{h} = h$ is assumed to be known;
   1.2. or $\vec{\boldsymbol{y}}$ is a sufficient statistic for detecting both $\vec{\boldsymbol{i}}$ and $\boldsymbol{h}$ jointly.
2. Repeat the above for linear stream modulation, $\boldsymbol{x}(t) = \sum_l \boldsymbol{x}_l u(t - lt_b)$, when the channel changes slowly relative to the symbol period t_b with received signal

$$\boldsymbol{y}(t) \quad = \quad \boldsymbol{h}(t)\boldsymbol{x}(t) + \boldsymbol{n}(t).$$

Solution:

1. - *Converting to AWGN Channel:* Observe that

$$\begin{aligned}\boldsymbol{y}(t) &= \boldsymbol{h}\boldsymbol{x}(t) + \boldsymbol{n}(t) \\ &= \sum_l \underbrace{\boldsymbol{h}\boldsymbol{x}_l}_{\boldsymbol{x}'_l} u_l(t) + \boldsymbol{n}(t) \\ &= \sum_l \boldsymbol{x}'_l u_l(t) + \boldsymbol{n}(t),\end{aligned}$$

   which corresponds to an AWGN channel with redefined symbols $[\boldsymbol{x}'_l]_l$.

   - For either joint detection of $\vec{\boldsymbol{i}}$ and $\boldsymbol{h}$ or detection of $\vec{\boldsymbol{i}}$ conditioned on $\boldsymbol{h} = h$, the likelihood function will be conditioned on both $\vec{\boldsymbol{i}}$ and $\boldsymbol{h}$. Applying the results of AWGN channels, we obtain

– the sufficient statistic, which is the *demodulator* output, as

$$\vec{\boldsymbol{y}}_1 = R_u^{-\frac{1}{2}} \left[\langle \boldsymbol{y}(t), u_l(t)\rangle\right]_l,$$

where $R_u \triangleq [r_u(k,l)]_{l,k}$, $r_u(k,l) = \langle u_k(t), u_l(t)\rangle$.

– *the equivalent discrete time fading channel* is

$$\vec{\boldsymbol{y}}_1 = R_u^{\frac{1}{2}} \boldsymbol{h}\vec{\boldsymbol{x}} + \vec{\boldsymbol{n}}_1,$$

where $\vec{\boldsymbol{n}}_1 \sim \mathcal{CN}(\vec{0}, N_0 I)$ is white.

2. • *Converting to AWGN Channel:* Use an approximation

$$\begin{aligned} \boldsymbol{y}(t) &= \boldsymbol{h}(t)\boldsymbol{x}(t) + \boldsymbol{n}(t) \\ &\approx \sum_l \underbrace{\boldsymbol{h}(lt_b)\boldsymbol{x}_l}_{\boldsymbol{x}'_l} u(t - lt_b) + \boldsymbol{n}(t) \qquad (2.5.2.1) \\ &= \sum_l \boldsymbol{x}'_l u(t - lt_b) + \boldsymbol{n}(t), \end{aligned}$$

which corresponds to an AWGN channel with redefined symbols $[\boldsymbol{x}'_l]_l$.

- For either joint detection of $\vec{\boldsymbol{i}}$ and $[\boldsymbol{h}(lt_b)]_l$ or detection of $\vec{\boldsymbol{i}}$ conditioned on $[\boldsymbol{h}(lt_b)]_l = [h_l]_l$, the likelihood function will be conditioned on both $\vec{\boldsymbol{i}}$ and $[\boldsymbol{h}(lt_b)]_l$. Applying the results of AWGN channels, we obtain
  - the sufficient statistic, which is the *demodulator* output, as

    $$\vec{\boldsymbol{y}}_1 \quad = \quad R_u^{-\frac{1}{2}} \left[\langle \boldsymbol{y}(t), u(t - lt_b)\rangle\right]_l ,$$

    where $R_u \triangleq [r_u(k,l)]_{l,k}$, $r_u(k,l) = \langle u(t - kt_b), u(t - lt_b)\rangle$.
  - *the equivalent discrete time fading channel* is

    $$\vec{\boldsymbol{y}}_1 \quad = \quad R_u^{\frac{1}{2}} \left[\boldsymbol{h}(lt_b)\boldsymbol{x}_l\right]_l + \vec{\boldsymbol{n}}_1,$$

    where $\vec{\boldsymbol{n}}_1 \sim \mathcal{CN}(\vec{0}, N_0 I)$ is white.

□

2.5.3. Demodulator for OFDM in Time Invariant Frequency Selective Fading Channels

$$\xrightarrow{\vec{\boldsymbol{i}}} \boxed{\text{Encoder}} \xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)} \boxed{\text{Fading Channel}} \xrightarrow{\boldsymbol{y}(t)=\boldsymbol{h}(t)*\boldsymbol{x}(t)+\boldsymbol{n}(t)} \boxed{\text{Demodulator}} \xrightarrow{\vec{\boldsymbol{y}}}$$

- How to design a low complexity demodulator for OFDM in time invariant frequency selective fading channels?
- What are the resulting equivalent discrete time channel model for OFDM in time invariant frequency selective fading channels?

2.5.3.0.1. *Demodulator for OFDM in Time Invariant Frequency Selective Fading Channels*.

PROBLEM 2.5.3.1. (**Low Complexity Demodulator for OFDM in Time Invariant Frequency Selective Fading Channels**) Consider the time invariant frequency selective fading channel

$$\xrightarrow{\vec{\boldsymbol{i}}} \boxed{\text{Encoder}} \xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)} \boxed{\text{Fading Channel}} \xrightarrow{\boldsymbol{y}(t)=\boldsymbol{h}(t)*\boldsymbol{x}(t)+\boldsymbol{n}(t)} \boxed{\text{Den}}$$

- where $\vec{\boldsymbol{i}} \in \{0,1\}^n$ is the message to be sent or the information vector;
- $\vec{\boldsymbol{x}} = [\boldsymbol{x}_l]_l = \vec{x}_{\vec{\boldsymbol{i}}} = \left[x_{\vec{\boldsymbol{i}},l}\right]_l$ is the code word;
-

$$\boldsymbol{x}(t) = \left(\sum_l \boldsymbol{x}_l e^{j2\pi(lf_\Delta+f_0)t}\right) u(t), \tag{2.5.3.1}$$

$$(\tilde{\boldsymbol{x}}(f) \quad = \quad \sum_l \boldsymbol{x}_l \tilde{u}(f - l f_\Delta - f_0)),$$

is the baseband signal with OFDM modulation and is band limited within $\left(-\frac{b}{2}, \frac{b}{2}\right)$;

- $$u(t) \quad = \quad \begin{cases} c_u & t_1 \leq t \leq t_2 \\ 0 & t < t_0 \text{ or } t > t_3 \end{cases},$$

  $t_0 < t_1 < t_2 = t_1 + \tau_h + \frac{1}{f_\Delta} < t_3$;
- $\boldsymbol{h}(t)$ is the time invariant impulse response of the channel with support $[0, \tau_h]$;
- $\boldsymbol{y}(t)$ is the output of the equivalent baseband AWGN channel;
- $\boldsymbol{n}(t)$ is the circularly symmetric complex Gaussian noise process with power spectral density $\tilde{r}_{\boldsymbol{n}}(f) = N_0 \mathrm{rect}_b(f)$.

Derive the low complexity demodulator and the corresponding zero ISI equivalent channel model such that $\vec{\boldsymbol{y}}$ is a near sufficient statistic for detecting $\vec{\boldsymbol{i}}$ when the samples of $\tilde{\boldsymbol{h}}(f)$ are assumed to be known.

Solution:

2.5.3.0.2. *Converting the fading to the AWGN channel.*

- Let's examine the equivalent transmit signal to see whether we can convert it to an AWGN channel problem. Observe that

$$
\begin{aligned}
& \boldsymbol{x}(t) * \boldsymbol{h}(t) \\
= & \int_0^{T_h} \sum_l \boldsymbol{x}_l u(t-\lambda) e^{j2\pi(lf_\Delta + f_0)(t-\lambda)} \boldsymbol{h}(\lambda) d\lambda \\
= & \begin{cases} \sum_l \boldsymbol{x}_l e^{j2\pi(lf_\Delta + f_0)t} c_u \int_0^{T_h} e^{-j2\pi(lf_\Delta + f_0)\lambda} \boldsymbol{h}(\lambda) d\lambda & t_1 + \tau_h \leq t \leq t_2 \\ * & \text{else} \end{cases} \\
= & \begin{cases} \sum_l \underbrace{c_u \tilde{\boldsymbol{h}}(lf_\Delta + f_0) \boldsymbol{x}_l}_{\boldsymbol{x}'_l} e^{j2\pi(lf_\Delta + f_0)t} & t_1 + \tau_h \leq t \leq t_2 \\ * & \text{else} \end{cases}
\end{aligned}
$$

$$
\begin{aligned}
&= \sum_l \boldsymbol{x}'_l e^{j2\pi(lf_\Delta+f_0)t}\mathrm{rect}_{\frac{1}{f_\Delta}}\left(t-\left(t_1+\tau_h+\frac{1}{2f_\Delta}\right)\right) \qquad (2.5.3.2)\\
&\quad + \begin{cases} 0 & t_1+\tau_h \le t \le t_2 \\ * & \text{else} \end{cases}\\
&= \mathcal{F}^{-1}\left\{\sum_l \boldsymbol{x}'_l \frac{1}{f_\Delta}\mathrm{sinc}\left((f-lf_\Delta-f_0)\frac{1}{f_\Delta}\right)e^{-j2\pi(f-lf_\Delta-f_0)\left(t_1+\tau_h+\frac{1}{2f_\Delta}\right)}\right\}\\
&\quad + \begin{cases} 0 & t_1+\tau_h \le t \le t_2 \\ * & \text{else} \end{cases}.
\end{aligned}
$$

- After discarding signals outside $[t_1+\tau_h, t_2]$, the signal is a linear modulation in time domain with orthogonal pulses,

$$
e^{j2\pi(lf_\Delta+f_0)t}\mathrm{rect}_{\frac{1}{f_\Delta}}\left(t-\left(t_1+\tau_h+\frac{1}{2f_\Delta}\right)\right), \forall l,
$$

and a LSM in frequency domain with equivalent symbol $\boldsymbol{x}_l' = \boldsymbol{h}(lf_\Delta + f_0)\boldsymbol{x}_l$ and orthogonal sinc pulses.

- Thus, the channel becomes an AWGN channel.

2.5.3.0.3. *Demodulator and equivalent discrete time channel.*

- The optimal demodulator of the signal after the discarding is the same as the optimal demodulator for linear modulation.
  - The demodulator is sub-optimal overall because we have discarded signal outside $[t_1 + \tau_h, t_2]$.
  - For each time index, the demodulator and the equivalent discrete time channel model is obtained by matched filtering, which is a Fourier transform and frequency sampling, as

$$\begin{aligned}
\boldsymbol{y}_{1,l} &= \sqrt{f_\Delta} \int_{t_1+\tau_h}^{t_2} \boldsymbol{y}(t) e^{-j2\pi(lf_\Delta + f_0)t} dt \\
&= \sqrt{f_\Delta} \int_{t_1+\tau_h}^{t_2} \left(\boldsymbol{x}(t) * \boldsymbol{h}(t) + \boldsymbol{n}(t)\right) e^{-j2\pi(lf_\Delta + f_0)t} dt \\
&= \tilde{\boldsymbol{h}}(lf_\Delta + f_0)\sqrt{e_u}\boldsymbol{x}_l + \boldsymbol{n}_{1,l}, \qquad (2.5.3.3)
\end{aligned}$$

where $\boldsymbol{n}_l \sim \mathcal{CN}(0, N_0)$ are i.i.d. and $e_u = c_u^2(t_2 - t_1 - \tau_h) = c_u^2 \frac{1}{f_\Delta}$.

- Therefore, the equivalent discrete time channel model is

$$\vec{\boldsymbol{y}}_1 = \left[\tilde{\boldsymbol{h}}(l f_\Delta + f_0)\sqrt{e_u}\boldsymbol{x}_l\right]_l + \vec{\boldsymbol{n}}_1,$$

  - where $\vec{\boldsymbol{n}}_1 \sim \mathcal{CN}(\vec{0}, N_0 I)$.

□

2.5.3.0.4. *A frequency domain sampling view of OFDM.*

- Time varying flat fading channels is the dual of time invariant frequency selective fading channels as one can be obtained by exchanging the time and frequency indexes of another.
- Recall that in time varying flat fading channels, the equivalent discrete time channel is approximately affected by time samples of the fading process, as seen in (2.5.2.1).
  - Is it possible to make it exact?
- In the following problem, we achieve the exact sampling for time invariant frequency selective fading channels using OFDM.
  - Exchanging the time and frequency indexes, we can obtain the exact result for flat fading channels.

PROBLEM 2.5.3.2. (OFDM viewed as a sampling using Dirac functions in frequency domain)

- In time invariant frequency selective fading channels, we hope for the signal to exactly sample the fading process as

$$\begin{aligned} & \tilde{\boldsymbol{h}}(f)\tilde{\boldsymbol{x}}_{f_\Delta}(f) \\ = \; & \tilde{\boldsymbol{h}}(f)\sum_l \boldsymbol{x}_l \delta\left(f-(lf_\Delta+f_0)\right) \\ = \; & \sum_l \tilde{\boldsymbol{h}}\left(lf_\Delta+f_0\right)\boldsymbol{x}_l \delta\left(f-(lf_\Delta+f_0)\right). \end{aligned}$$

- However, $\boldsymbol{x}_{f_\Delta}(t) = \sum_l \boldsymbol{x}_l e^{j2\pi(lf_\Delta+f_0)t}$ will have infinite time duration.
- Is it possible to achieve the above with finite duration transmit signal?

Solution:

- Yes.
- Observe that the received signal without noise,

$$\begin{aligned} & \boldsymbol{h}(t) * \boldsymbol{x}_{f_\Delta}(t) \\ = \; & \boldsymbol{h}(t) * \sum_l \boldsymbol{x}_l e^{j2\pi(lf_\Delta + f_0)t} \end{aligned}$$

  is periodic after compensating off f_0.
- Therefore, as long as we receive one period of it, it is equivalent to receive all of it.
- This can be achieved by sending τ_h more than one period $\frac{1}{f_\Delta}$ of $\boldsymbol{x}_{f_\Delta}(t)$, i.e., by transmitting signal

$$\boldsymbol{x}_{f_\Delta}(t)\mathrm{rect}_{\tau_h + \frac{1}{f_\Delta}}\left(t - \left(\frac{t_1 + t_2}{2}\right)\right),$$

  where $t_2 = t_1 + \tau_h + \frac{1}{f_\Delta}$.

- At the receiver, carving out one period of the signal as

$$(\boldsymbol{h}(t) * \boldsymbol{x}_{f_\Delta}(t)) \operatorname{rect}_{\frac{1}{f_\Delta}} \left(t - \left(t_1 + \tau_h + \frac{1}{2 f_\Delta} \right) \right).$$

- The above is performed in (2.5.A.1) and (2.5.3.2).
  - Thus, it is not surprising to obtain the equivalent discrete time channel model in (2.5.3.3).

□

2.5.4. Demodulator for MIMO Fading Channels

- When a transmitter and a receiver have multiple antennas, it is a multiple-input multiple-output (MIMO) system.
  - In contrast, single antenna communication systems are single-input single-output (SISO) systems.
- How is demodulators for MIMO fading channels different from those of SISO fading channels?
- What are their equivalent discrete time channel models?

2.5.4.0.1. *What are MIMO systems?*

- In a traditional multi-antenna beamforming transmitter, the antennas transmit one up-converted baseband signal with different phase shifts.
  - The receiver linearly combine the signals from multiple receive antennas before inputting it into the I/Q down converter to produce a single baseband signal.
  - Therefore, effectively, they are SISO systems in terms of baseband channel model.
- In MIMO systems, the antennas transmit signals up converted from different baseband signals.
  - Each receive antenna's signal is fed to a separate I/Q down converter, resulting in a different baseband received signal.

2.5.4.0.2. *Demodulator for MIMO fading channel with arbitrary modulation.*

- We derive the optimal demodulator for the case of *arbitrary* modulation first.

PROBLEM 2.5.4.1. (**Demodulator for MIMO fading channel with arbitrary modulation**) Consider the baseband channel model for MIMO fading channels given in Part 1, Section 1.4.11.0.2:

$$\left[\boldsymbol{x}_{k_{\mathrm{t}}}(t)\right]_{k_{\mathrm{t}}} = \left[x_{\vec{\boldsymbol{i}},k_t}(t)\right]_{k_{\mathrm{t}}} \xrightarrow{\quad} \boxed{\text{Channel}} \xrightarrow{\left[\boldsymbol{y}_{k_{\mathrm{r}}}(t)\right]_{k_{\mathrm{r}}} = \left[\sum_{k_{\mathrm{t}}} \int \boldsymbol{h}_{k_{\mathrm{r}},k_{\mathrm{t}}}(t,t-\lambda)\boldsymbol{x}_{k_{\mathrm{t}}}(\lambda)d\lambda + \boldsymbol{n}_{k_{\mathrm{r}}}(t)\right]_{k_{\mathrm{r}}}}$$

- where $\vec{\boldsymbol{i}} \in \{0,1\}^n$ is the message to be sent or the information vector;
- $\boldsymbol{x}_{k_{\mathrm{t}}}(t) = x_{\vec{\boldsymbol{i}},k_t}(t)$ is the transmit signal of the k_t-th antenna and is band limited within $\left(-\frac{b}{2}, \frac{b}{2}\right)$;

- $\boldsymbol{h}_{k_\mathrm{r},k_\mathrm{t}}(t,\tau)$ is the time varying channel impulse response from the k_t-th transmit antenna to the k_r-th receive antenna;
- $\boldsymbol{y}_{k_r}(t)$ is the received signal of the k_r-th receive antenna;
- $\boldsymbol{n}_{k_r}(t)$ is the circularly symmetric complex Gaussian noise process of the k_r-th receive antenna with power spectral density $\tilde{r}_{\boldsymbol{n}}(f) = N_0 \mathrm{rect}_b(f)$ and is independent across k_r.

1. Derive the optimal demodulator whose output is a sufficient statistic for detecting $\vec{\boldsymbol{i}}$ conditioned on $\boldsymbol{h}_{k_r,k_t}(t,\tau) = h_{k_r,k_t}(t,\tau)$.

Solution:

1. - *Converting to AWGN Channel:* Observe that, conditioned on $\boldsymbol{h}_{k_r,k_t}(t,\tau) = h_{k_r,k_t}(t,\tau)$, the signal without noise is

$$x'_{\vec{\boldsymbol{i}},k_r}(t) \quad = \quad \sum_{k_\mathrm{t}} \int h_{k_\mathrm{r},k_\mathrm{t}}(t, t-\lambda) x_{\vec{\boldsymbol{i}},k_\mathrm{t}}(\lambda) d\lambda.$$

   - Thus, we have an AWGN channel

$$\Big[\boldsymbol{y}_{k_\mathrm{r}}(t)\Big]_{k_\mathrm{r}} \quad = \quad \Big[x'_{\vec{\boldsymbol{i}},k_r}(t) + \boldsymbol{n}_{k_\mathrm{r}}(t)\Big]_{k_\mathrm{r}}$$

   - Additionally, since $\boldsymbol{n}_{k_r}(t)$ is independent across k_r, the likelihood function of $\Big[\boldsymbol{y}_{k_\mathrm{r}}(t)\Big]_{k_\mathrm{r}}$ conditioned on $\vec{\boldsymbol{i}} = \vec{i}$ and $\boldsymbol{h}_{k_r,k_t}(t,\tau) = h_{k_r,k_t}(t,\tau)$ is a product of the likelihood functions (2.3.4.5) corresponding to the receive antennas.

- Therefore, by Theorem 2.3.1 and 2.3.2 and the sufficient statistic (2.3.7.1) of the AWGN channel, the demodulator,

$$\text{correlation: } \vec{\boldsymbol{y}} \quad = \quad \left[\left[\left\langle \boldsymbol{y}(t), x'_{\vec{i},k_r}(t) \right\rangle\right]_{k_r}\right]_{\vec{i}\in\{0,1\}^n},$$

  provides the sufficient statistic to detect $\vec{\boldsymbol{i}}$ conditioned on $\boldsymbol{h}_{k_r,k_t}(t,\tau) = h_{k_r,k_t}(t,\tau)$.

$\square$

2.5.4.0.3. *Demodulator for MIMO flat fading channel with linear and linear stream modulation.*

- We derive the optimal demodulator for the case of linear modulation.
  - It is conditioned on knowing the fading channel coefficients. In practice, the channel coefficients can be obtained by estimation via sending pilot symbols, known to the receiver.

PROBLEM 2.5.4.2. (**Demodulator for MIMO flat fading channel with linear and linear stream modulation**) Consider the baseband channel model for MIMO flat fading channels given in Part 1, Section 1.4.11.0.2:

$$\xrightarrow{\left[\boldsymbol{x}_{k_\mathrm{t}}(t)\right]_{k_\mathrm{t}}=\left[\sum_l \boldsymbol{x}_{k_t,l}u_l(t)\right]_{k_\mathrm{t}}} \boxed{\text{Channel}} \xrightarrow{\left[\boldsymbol{y}_{k_\mathrm{r}}(t)\right]_{k_\mathrm{r}}=\left[\sum_{k_\mathrm{t}} \boldsymbol{h}_{k_\mathrm{r},k_\mathrm{t}}(t)\boldsymbol{x}_{k_\mathrm{t}}(t)+\boldsymbol{n}_{k_\mathrm{r}}(t)\right]_{k_\mathrm{r}}}$$

- where $\vec{\boldsymbol{i}} \in \{0,1\}^n$ is the message to be sent or the information vector;
- $\boldsymbol{x}_{k_\mathrm{t}}(t) = \sum_l \boldsymbol{x}_{k_t,l} u_l(t)$ is the transmit signal of the k_t-th antenna and is band limited within $\left(-\frac{b}{2}, \frac{b}{2}\right)$;
- $u_l(t)$'s are orthonormal;
- $\boldsymbol{h}_{k_\mathrm{r},k_\mathrm{t}}(t)$ is the time varying channel coefficient of the k_t-th transmit antenna and the k_r-th receive antenna;
- $\boldsymbol{y}_{k_t}(t)$ is the received signal of the k_r-th receive antenna;
- $\boldsymbol{n}_{k_r}(t)$ is the circularly symmetric complex Gaussian noise process of the k_r-th receive antenna with power spectral density $\tilde{r}_{\boldsymbol{n}}(f) = N_0 \mathrm{rect}_b(f)$ and is independent across k_r.

1. Assume $\boldsymbol{h}_{k_r,k_t}(t) = h_{k_r,k_t}$.
   1.1. Derive the optimal demodulator whose output is a sufficient statistic for detecting $\vec{\boldsymbol{i}}$.

1.2. Derive the equivalent discrete time channel model for orthonormal pulses.

2. Repeat the above for linear stream modulation, $\boldsymbol{x}_{k_t}(t) = \sum_l \boldsymbol{x}_{k_t,l} u(t - lt_b)$, when the channel changes slowly relative to the symbol period t_b and thus $\boldsymbol{h}_{k_r,k_t}(t) \approx h_{k_r,k_t}(lt_b)$, $\forall t \in [lt_b - \frac{t_b}{2}, lt_b + \frac{t_b}{2}]$.

Solution:

1. 1.1.
   - *Converting to AWGN Channel:* Observe that, conditioned on $\boldsymbol{h}_{k_r,k_t}(t) = h_{k_r,k_t}$, the signal without noise is

$$\sum_{l} \underbrace{\sum_{k_\mathrm{t}} h_{k_\mathrm{r},k_\mathrm{t}} \boldsymbol{x}_{k_t,l}}_{x'_{\vec{i},l}} u_l(t)$$

     - Thus, we have an AWGN channel

$$\left[\boldsymbol{y}_{k_\mathrm{r}}(t)\right]_{k_\mathrm{r}} = \left[\sum_{l} x'_{\vec{i},l} u_l(t) + \boldsymbol{n}_{k_\mathrm{r}}(t)\right]_{k_\mathrm{r}}$$

       with LSM.
   - Based on the results of Problem 2.5.4.1 of arbitrary modulation and Problem 2.5.2.1 of SISO flat

fading channel, the demodulator is

$$\vec{\boldsymbol{y}} = \left[R_u^{-\frac{1}{2}} \left[\underbrace{\left\langle \boldsymbol{y}_{k_r}(t), u_l(t) \right\rangle}_{\boldsymbol{y}_{k_r,l}} \right]_{k_r} \right]_l ,$$

where $R_u = I$ due to the pulses being orthonormal.

1.2. Based on similar reasons, the equivalent discrete time channel model is

$$\left[\underbrace{\left[\boldsymbol{y}_{k_r,l} \right]_{k_r}}_{\vec{\boldsymbol{y}}_l} \right]_l = \left[\underbrace{\left[\boldsymbol{h}_{k_r,k_t} \right]_{k_r,k_t}}_{\boldsymbol{H}} \right]_{l,l} \left[\underbrace{\left[\boldsymbol{x}_{k_t,l} \right]_{k_t}}_{\vec{\boldsymbol{x}}_l} \right]_l + \left[\underbrace{\left[\boldsymbol{n}_{k_r,l} \right]_{k_r}}_{\vec{\boldsymbol{n}}_l} \right]_l ,$$

i.e.,

$$\vec{\boldsymbol{y}}_l = \boldsymbol{H}\vec{\boldsymbol{x}}_l + \vec{\boldsymbol{n}}_l, \forall l,$$

where $\boldsymbol{n}_{k_r,l} \sim \mathcal{CN}(\vec{0}, N_0)$ is i.i.d. across k_r and l.

2. Similarly, we can obtain the following for slowly time varying flat fading channel with LSM.

   2.1. The demodulator is

$$\vec{\boldsymbol{y}} = \left[R_u^{-\frac{1}{2}} \left[\underbrace{\left\langle \boldsymbol{y}_{k_r}(t), u(t - lt_b) \right\rangle}_{\boldsymbol{y}_{k_r,l}} \right]_{k_r} \right]_l .$$

   2.2. Based on similar reasons, the equivalent discrete time channel model is

$$\left[\underbrace{\left[\boldsymbol{y}_{k_r,l} \right]_{k_r}}_{\vec{\boldsymbol{y}}_l} \right]_l = \left[\underbrace{\left[\boldsymbol{h}_{k_r,k_t}(lt_b) \right]_{k_r,k_t}}_{\boldsymbol{H}_l} \right]_{l,l} \left[\underbrace{\left[\boldsymbol{x}_{k_t,l} \right]_{k_t}}_{\vec{\boldsymbol{x}}_l} \right]_l + \left[\underbrace{\left[\boldsymbol{n}_{k_r,l} \right]_{k_r}}_{\vec{\boldsymbol{n}}_l} \right]_l ,$$

i.e.,

$$\vec{\boldsymbol{y}}_l = \boldsymbol{H}_l \vec{\boldsymbol{x}}_l + \vec{\boldsymbol{n}}_l, \forall l,$$

where $\boldsymbol{n}_{k_r,l} \sim \mathcal{CN}(\vec{0}, N_0)$ is i.i.d. across k_r and l.

$\square$

2.5.4.0.4. *Demodulator for MIMO time invariant frequency selective fading channel with OFDM modulation.*

- We derive the optimal demodulator and equivalent discrete time channel model for OFDM modulation.

PROBLEM 2.5.4.3. (**Demodulator for MIMO time invariant frequency selective fading channel with OFDM modulation**) Consider the baseband channel model for MIMO time invariant frequency selective fading channel with OFDM modulation given in Part 1, Section 1.4.11.0.2:

$$\left[\boldsymbol{x}_{k_{\mathrm{t}}}(t)\right]_{k_{\mathrm{t}}}=\left[\sum_{l}\boldsymbol{x}_{k_{t},l}e^{j2\pi(lf_{\Delta}+f_{0})t}u(t)\right]_{k_{\mathrm{t}}} \xrightarrow{\quad} \boxed{\text{Channel}} \xrightarrow{\left[\boldsymbol{y}_{k_{\mathrm{r}}}(t)\right]_{k_{\mathrm{r}}}=\left[\sum_{k_{\mathrm{t}}}\boldsymbol{h}_{k_{\mathrm{r}},k_{\mathrm{t}}}(t)*\boldsymbol{x}_{k_{\mathrm{t}}}(t)+\boldsymbol{n}_{k_{\mathrm{r}}}(t\right.}$$

- where $\vec{\boldsymbol{i}} \in \{0,1\}^{n}$ is the message to be sent or the information vector;

- $\boldsymbol{x}_{k_\mathrm{t}}(t) = \sum_l \boldsymbol{x}_{k_t,l} e^{j2\pi(lf_\Delta + f_0)t} u(t)$ is the transmit signal of the k_t-th antenna and is band limited within $\left(-\frac{b}{2}, \frac{b}{2}\right)$;
-
$$u(t) \quad = \quad \begin{cases} c_u & t_1 \leq t \leq t_2 \\ 0 & t < t_0 \text{ or } t > t_3 \end{cases},$$

  $t_0 < t_1 < t_2 = t_1 + \tau_h + \frac{1}{f_\Delta} < t_3$, $c_u = \sqrt{f_\Delta}$;
- $\boldsymbol{h}_{k_\mathrm{r},k_\mathrm{t}}(t)$ is the time invariant channel impulse response of the k_t-th transmit antenna and the k_r-th receive antenna;
- Assume $\tilde{\boldsymbol{h}}_{k_\mathrm{r},k_\mathrm{t}}(lf_\Delta + f_0) = \tilde{h}_{k_\mathrm{r},k_\mathrm{t}}(lf_\Delta + f_0)$ is known to the receiver;
- $\boldsymbol{y}_{k_r}(t)$ is the received signal of the k_r-th receive antenna;
- $\boldsymbol{n}_{k_r}(t)$ is the circularly symmetric complex Gaussian noise process of the k_r-th receive antenna with power spectral density $\tilde{r}_{\boldsymbol{n}}(f) = N_0 \mathrm{rect}_b(f)$ and is independent across k_r.

1. Derive a low complexity demodulator whose output is a near sufficient statistic for detecting $\vec{\boldsymbol{i}}$
2. Derive the equivalent discrete time channel model.

Solution:

1. Based on the results of Problem 2.5.3.1 of OFDM modulation, the signal in $[t_1, t_2]$ is a linear modulation with orthonormal pulses. The result in Problem 2.5.4.2 can be applied to obtain the low complexity demodulator as

$$\vec{\boldsymbol{y}} = \left[\left[\underbrace{\int_{t_1+\tau_h}^{t_1+\tau_h+\frac{1}{f_\Delta}} \boldsymbol{y}_{k_r}(t) e^{-j2\pi(lf_\Delta+f_0)t}\sqrt{f_\Delta}dt}_{\boldsymbol{y}_{k_r,l}}\right]_{k_r}\right]_l .$$

2. Based on similar reasons, the equivalent discrete time channel model is

$$\left[\underbrace{\left[\boldsymbol{y}_{k_r,l}\right]_{k_r}}_{\vec{\boldsymbol{y}}_l}\right]_l = \left[\underbrace{\left[\tilde{\boldsymbol{h}}_{k_r,k_t}(lf_\Delta + f_0)\right]_{k_r,k_t}}_{\boldsymbol{H}_l}\right]_{l,l} \left[\underbrace{\left[\boldsymbol{x}_{k_t,l}\right]_{k_t}}_{\vec{\boldsymbol{x}}_l}\right]_l + \left[\underbrace{\left[\boldsymbol{n}_{k_r,l}\right]_{k_r}}_{\vec{\boldsymbol{n}}_l}\right]_l ,$$

i.e.,

$$\vec{\boldsymbol{y}}_l \quad = \quad \boldsymbol{H}_l \vec{\boldsymbol{x}}_l + \vec{\boldsymbol{n}}_l, \forall l,$$

where $\boldsymbol{n}_{k_r,l} \sim \mathcal{CN}(\vec{0}, N_0)$ is i.i.d. across k_r and l, the pulse energy is $e_u = c_u^2 \frac{1}{f_\Delta} = 1$.

$\square$

2.5.5. Summary

2.5.5.0.1. *Demodulators and equivalent discrete time channel models.*

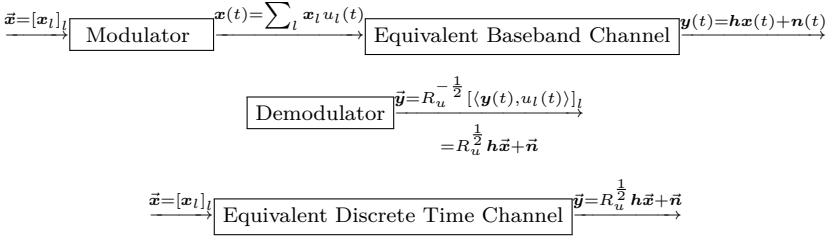
$$\xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Equivalent Discrete Time Channel}} \xrightarrow{\vec{\boldsymbol{y}}=\boldsymbol{H}\vec{\boldsymbol{x}}+\vec{\boldsymbol{n}}}$$

- The equivalent discrete time channel model that replaces the demodulator, the baseband channel, and the demodulator has a **general** form

$$\vec{\boldsymbol{y}} \quad = \quad \boldsymbol{H}\vec{\boldsymbol{x}} + \vec{\boldsymbol{n}},$$

  where the element in $\vec{\boldsymbol{n}}$ are i.i.d. with distribution $\mathcal{CN}(0, N_0)$.
- The demodulator that produces $\vec{\boldsymbol{y}}$ and the $\boldsymbol{H}$ matrix are given for each of the following cases, where $\boldsymbol{n}(t)$ is the circularly symmetric complex Gaussian noise process with power spectral density $\tilde{r}_{\boldsymbol{n}}(f) = N_0 \mathrm{rect}_b(f)$.

2.5.5.0.2. *Linear modulation and time invariant flat fading channel.*



- For

$$\boldsymbol{y}(t) \quad = \quad \boldsymbol{h}\sum_l \boldsymbol{x}_l u_l(t) + \boldsymbol{n}(t),$$

the demodulator is

$$\vec{\boldsymbol{y}} \quad = \quad R_u^{-\frac{1}{2}}\left[\langle \boldsymbol{y}(t), u_l(t)\rangle\right]_l,$$

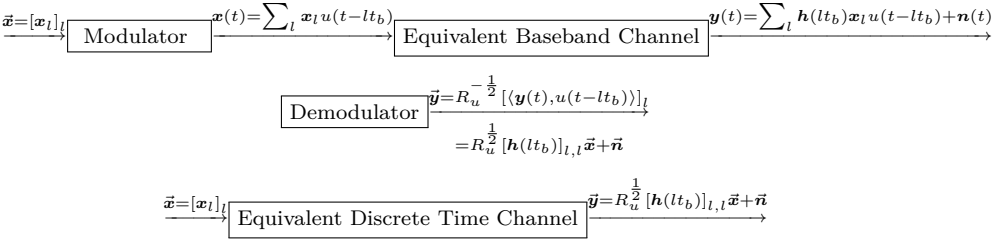
–

$$\boldsymbol{H} = R_u^{\frac{1}{2}} \boldsymbol{h};$$

–

$$\vec{\boldsymbol{x}} = [\boldsymbol{x}_l]_l \, .$$

2.5.5.0.3. *LSM and the slowly time varying flat fading channel.*



- For

$$\boldsymbol{y}(t) = \sum_l \boldsymbol{h}(lt_b)\boldsymbol{x}_l u(t-lt_b) + \boldsymbol{n}(t),$$

the demodulator is

$$\vec{\boldsymbol{y}} = R_u^{-\frac{1}{2}}\left[\langle \boldsymbol{y}(t), u(t-lt_b)\rangle\right]_l .$$

- In the equivalent channel,

$$\boldsymbol{H} \quad = \quad R_u^{\frac{1}{2}} \left[\boldsymbol{h}(lt_b)\right]_{l,l}$$

where $\left[\boldsymbol{h}(lt_b)\right]_{l,l}$ is a diagonal matrix and

$$\vec{\boldsymbol{x}} \quad = \quad \left[\boldsymbol{x}_l\right]_l .$$

2.5.5.0.4. *OFDM modulation and time invariant frequency selective fading channel.*

$$\xrightarrow{\vec{\boldsymbol{x}}=[\boldsymbol{x}_l]_l} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)=\left(\sum_l \boldsymbol{x}_l e^{j2\pi(lf_\Delta+f_0)t}\right)u(t)} \boxed{\text{Equivalent Baseband Channel}} \xrightarrow{\boldsymbol{y}(t)=\boldsymbol{h}(t)*\boldsymbol{x}(t)+\boldsymbol{n}(t)}$$

$$\boxed{\text{Fourier Transform}} \xrightarrow[=\left[\tilde{\boldsymbol{h}}(lf_\Delta+f_0)\right]_{l,l}\vec{\boldsymbol{x}}+\vec{\boldsymbol{n}}]{\vec{\boldsymbol{y}}=\left[\int_{t_1+\tau_h}^{t_1+\tau_h+\frac{1}{f_\Delta}} \boldsymbol{y}(t)e^{-j2\pi(lf_\Delta+f_0)t}\sqrt{f_\Delta}dt\right]_l}$$

$$\xrightarrow{\vec{\boldsymbol{x}}=[\boldsymbol{x}_l]_l} \boxed{\text{Equivalent Discrete Time Channel}} \xrightarrow{\vec{\boldsymbol{y}}=\left[\tilde{\boldsymbol{h}}(lf_\Delta+f_0)\right]_{l,l}\vec{\boldsymbol{x}}+\vec{\boldsymbol{n}}}$$

- For

$$\boldsymbol{y}(t) = \boldsymbol{h}(t) * \left(\sum_l \boldsymbol{x}_l e^{j2\pi(lf_\Delta+f_0)t}\right) u(t) + \boldsymbol{n}(t),$$

$$u(t) = \begin{cases} \sqrt{f_\Delta} & t_1 \le t \le t_2 \\ 0 & t < t_0 \text{ or } t > t_3 \end{cases},$$

$t_0 < t_1 < t_2 = t_1 + \tau_h + \frac{1}{f_\Delta} < t_3$,

– the low complexity demodulator is

$$\begin{aligned}
\vec{\boldsymbol{y}} &= \left[\left\langle \boldsymbol{y}(t), e^{j2\pi(lf_\Delta+f_0)t}\sqrt{f_\Delta}\mathrm{rect}_{\frac{1}{f_\Delta}}\left(t-\left(t_1+\tau_h+\frac{1}{2f_\Delta}\right)\right)\right\rangle\right]_l \\
&= \left[\mathcal{F}\left\{\boldsymbol{y}(t)\sqrt{f_\Delta}\mathrm{rect}_{\frac{1}{f_\Delta}}\left(t-\left(t_1+\tau_h+\frac{1}{2f_\Delta}\right)\right)\right\}(lf_\Delta+f_0)\right]_l \\
&= \left[\int_{t_1+\tau_h}^{t_1+\tau_h+\frac{1}{f_\Delta}} \boldsymbol{y}(t)e^{-j2\pi(lf_\Delta+f_0)t}\sqrt{f_\Delta}dt\right]_l .
\end{aligned}$$

- In the equivalent channel,

$$\boldsymbol{H} = \left[\tilde{\boldsymbol{h}}(lf_\Delta+f_0)\right]_{l,l}$$

is a diagonal matrix and

$$\vec{\boldsymbol{x}} = [\boldsymbol{x}_l]_l .$$

2.5.5.0.5. *Linear modulation with orthonormal pulses and MIMO constant fading channel.*

$$\xrightarrow{\vec{\boldsymbol{x}}=\left[\left[\boldsymbol{x}_{k_t,l}\right]_{k_t}\right]_l} \boxed{\text{Modulator}} \xrightarrow{\left[\boldsymbol{x}_{k_t}(t)\right]_{k_t}=\left[\sum_l \boldsymbol{x}_{k_t,l}u_l(t)\right]_{k_t}} \boxed{\text{Equivalent Baseband Channel}}$$

$$\xrightarrow{\left[\boldsymbol{y}_{k_r}(t)\right]_{k_r}=\left[\sum_{k_t}\boldsymbol{h}_{k_r,k_t}\boldsymbol{x}_{k_t}(t)+\boldsymbol{n}_{k_r}(t)\right]_{k_r}} \boxed{\text{Demodulator}} \xrightarrow[=\left[\left[\boldsymbol{h}_{k_r,k_t}\right]_{k_r,k_t}\right]_{l,l}\vec{\boldsymbol{x}}+\vec{\boldsymbol{n}}]{\vec{\boldsymbol{y}}=\left[\left[\langle \boldsymbol{y}_{k_r}(t),u_l(t)\rangle\right]_{k_r}\right]_l}$$

$$\xrightarrow{\vec{\boldsymbol{x}}=\left[\left[\boldsymbol{x}_{k_t,l}\right]_{k_t}\right]_l} \boxed{\text{Equivalent Discrete Time Channel}} \xrightarrow{\vec{\boldsymbol{y}}=\left[\left[\boldsymbol{h}_{k_r,k_t}\right]_{k_r,k_t}\right]_{l,l}\vec{\boldsymbol{x}}+\vec{\boldsymbol{n}}}$$

- For

$$\left[\boldsymbol{y}_{k_r}(t)\right]_{k_r} = \left[\sum_{k_t}\boldsymbol{h}_{k_r,k_t}\boldsymbol{x}_{k_t,l}u_l(t)+\boldsymbol{n}_{k_r}(t)\right]_{k_r},$$

where $\boldsymbol{n}_{k_\mathrm{r}}(t)$ are independent random processes across k_r, the demodulator is

$$\vec{\boldsymbol{y}} = \left[\left[\left\langle \boldsymbol{y}_{k_r}(t), u_l(t) \right\rangle\right]_{k_r}\right]_l .$$

- In the equivalent channel,

$$\boldsymbol{H} = \left[\left[\boldsymbol{h}_{k_r,k_t}\right]_{k_r,k_t}\right]_{l,l}$$

and

$$\vec{\boldsymbol{x}} = \left[\left[\boldsymbol{x}_{k_t,l}\right]_{k_t}\right]_l .$$

2.5.5.0.6. *Linear stream modulation with orthonormal pulses and MIMO slowly time varying fading channel.*

$$\xrightarrow{\vec{\boldsymbol{x}}=\left[\left[\boldsymbol{x}_{k_t,l}\right]_{k_t}\right]_l} \boxed{\text{Modulator}} \xrightarrow{\left[\boldsymbol{x}_{k_t}(t)\right]_{k_t}=\left[\sum_l \boldsymbol{x}_{k_t,l}u(t-lt_b)\right]_{k_t}} \boxed{\text{Equivalent Baseband Channel}}$$

$$\xrightarrow{\left[\boldsymbol{y}_{k_r}(t)\right]_{k_r}=\left[\sum_{k_t}\boldsymbol{h}_{k_r,k_t}(t)\boldsymbol{x}_{k_t}(t)+\boldsymbol{n}_{k_r}(t)\right]_{k_r}=\left[\sum_{k_t}\boldsymbol{h}_{k_r,k_t}(lt_b)\boldsymbol{x}_{k_t,l}u(t-lt_b)+\boldsymbol{n}_{k_r}(t)\right]_{k_r}} \boxed{\text{Demodulator}} \xrightarrow{\vec{\boldsymbol{y}}=\left[\left[\langle \boldsymbol{y}_{k_r}(t),,u(t-lt_b)\rangle\right]_{k_r}\right]_l=\left[\left[\boldsymbol{h}_{k_r,k_t}(lt_b)\right]_{k_r,k_t}\right]_{l,l}\vec{\boldsymbol{x}}+\vec{\boldsymbol{n}}}$$

$$\xrightarrow{\vec{\boldsymbol{x}}=\left[\left[\boldsymbol{x}_{k_t,l}\right]_{k_t}\right]_l} \boxed{\text{Equivalent Discrete Time Channel}} \xrightarrow{\vec{\boldsymbol{y}}=\left[\left[\boldsymbol{h}_{k_r,k_t}(lt_b)\right]_{k_r,k_t}\right]_{l,l}\vec{\boldsymbol{x}}+\vec{\boldsymbol{n}}}$$

- For

$$\left[\boldsymbol{y}_{k_r}(t)\right]_{k_r} = \left[\sum_{k_t}\boldsymbol{h}_{k_r,k_t}(lt_b)\boldsymbol{x}_{k_t,l}u(t-lt_b)+\boldsymbol{n}_{k_r}(t)\right]_{k_r},$$

the demodulator is

$$\vec{\boldsymbol{y}} \quad = \quad \left[\left[\left\langle \boldsymbol{y}_{k_r}(t), u(t - lt_b) \right\rangle\right]_{k_r}\right]_l .$$

is the demodulator;

- In the equivalent channel,

$$\boldsymbol{H} \quad = \quad \left[\left[\boldsymbol{h}_{k_r,k_t}(lt_b)\right]_{k_r,k_t}\right]_{l,l}$$

is a block diagonal matrix; and

$$\vec{\boldsymbol{x}} \quad = \quad \left[\left[\boldsymbol{x}_{k_t,l}\right]_{k_t}\right]_l .$$

2.5.5.0.7. *OFDM modulation and MIMO time invariant frequency selective fading channel.*

$$\xrightarrow{\vec{\boldsymbol{x}}=\left[\left[\boldsymbol{x}_{k_t,l}\right]_{k_t}\right]_l} \boxed{\text{Modulator}} \xrightarrow{\left[\boldsymbol{x}_{k_t}(t)\right]_{k_t}=\left[\left(\sum_l \boldsymbol{x}_{k_t,l}e^{j2\pi(lf_\Delta+f_0)t}\right)u(t)\right]_{k_t}}$$

$$\boxed{\text{Equivalent Baseband Channel}} \xrightarrow{\left[\boldsymbol{y}_{k_r}(t)\right]_{k_r}=\left[\sum_{k_t}\boldsymbol{h}_{k_r,k_t}(t)*\boldsymbol{x}_{k_t}(t)+\boldsymbol{n}_{k_r}(t)\right]_{k_r}}$$

$$=\left[\sum_{k_t}\boldsymbol{h}_{k_r,k_t}(t)*\left(\sum_l \boldsymbol{x}_{k_t,l}e^{j2\pi(lf_\Delta+f_0)t}u(t)\right)+\boldsymbol{n}_{k_r}(t)\right]_{k_r}$$

$$\boxed{\text{Demodulator}} \xrightarrow{\vec{\boldsymbol{y}}=\left[\left[\int_{t_1+\tau_h}^{t_1+\tau_h+\frac{1}{f_\Delta}}\boldsymbol{y}_{k_r}(t)e^{-j2\pi(lf_\Delta+f_0)t}\sqrt{f_\Delta}\,dt\right]_{k_r}\right]_l}$$

$$=\left[\left[\tilde{\boldsymbol{h}}_{k_r,k_t}(lf_\Delta+f_0)\right]_{k_r,k_t}\right]_{l,l}\vec{\boldsymbol{x}}+\vec{\boldsymbol{n}}$$

$$\xrightarrow{\vec{\boldsymbol{x}}=\left[\left[\boldsymbol{x}_{k_t,l}\right]_{k_t}\right]_l} \boxed{\text{Equivalent Discrete Time Channel}} \xrightarrow{\vec{\boldsymbol{y}}=\left[\left[\tilde{\boldsymbol{h}}_{k_r,k_t}(lf_\Delta+f_0)\right]_{k_r,k_t}\right]_{l,l}\vec{\boldsymbol{x}}+\vec{\boldsymbol{n}}}$$

- For

$$\left[\boldsymbol{y}_{k_\mathrm{r}}(t)\right]_{k_\mathrm{r}} = \left[\sum_{k_\mathrm{t}} \boldsymbol{h}_{k_\mathrm{r},k_\mathrm{t}}(t) * \left(\sum_{l} \boldsymbol{x}_{k_t,l} e^{j2\pi(lf_\Delta+f_0)t} u(t)\right) + \boldsymbol{n}_{k_\mathrm{r}}(t)\right]_{k_\mathrm{r}},$$

$$u(t) = \begin{cases} \sqrt{f_\Delta} & t_1 \le t \le t_2 \\ 0 & t < t_0 \text{ or } t > t_3 \end{cases},$$

$t_0 < t_1 < t_2 = t_1 + \tau_h + \frac{1}{f_\Delta} < t_3,$

  – the low complexity demodulator is

$$\begin{aligned} \vec{\boldsymbol{y}} &= \left[\left[\left\langle \boldsymbol{y}_{k_r}(t), e^{j2\pi(lf_\Delta+f_0)t}\sqrt{f_\Delta}\mathrm{rect}_{\frac{1}{f_\Delta}}\left(t - \left(t_1 + \tau_h + \frac{1}{2f_\Delta}\right)\right)\right\rangle\right]_{k_r}\right]_l \\ &= \left[\left[\int_{t_1+\tau_h}^{t_1+\tau_h+\frac{1}{f_\Delta}} \boldsymbol{y}_{k_r}(t) e^{-j2\pi(lf_\Delta+f_0)t}\sqrt{f_\Delta}dt\right]_{k_r}\right]_l. \end{aligned}$$

- In the equivalent channel,

$$\boldsymbol{H} = \left[\left[\tilde{\boldsymbol{h}}_{k_r,k_t}(l f_\Delta + f_0)\right]_{k_r,k_t}\right]_{l,l},$$

is a block diagonal matrix, and

$$\vec{\boldsymbol{x}} = \left[\left[\boldsymbol{x}_{k_t,l}\right]_{k_t}\right]_l.$$

Appendices

2.5.A. Homework Problems

PROBLEM 2.5.A.1.

(**Optimal Demodulator for OFDM in Time Invariant Frequency Selective Fading Channels**) Consider the time invariant frequency selective fading channel

$$\xrightarrow{\vec{\boldsymbol{i}}} \boxed{\text{Encoder}} \xrightarrow{\vec{\boldsymbol{x}}} \boxed{\text{Modulator}} \xrightarrow{\boldsymbol{x}(t)} \boxed{\text{Fading Channel}} \xrightarrow{\boldsymbol{y}(t)=\boldsymbol{h}(t)*\boldsymbol{x}(t)+\boldsymbol{n}(t)} \boxed{\text{Demodulator}} \xrightarrow{\vec{\boldsymbol{y}}}$$

- where $\vec{\boldsymbol{i}} \in \{0,1\}^n$ is the message to be sent or the information vector;
  - $\vec{\boldsymbol{x}} = [\boldsymbol{x}_l]_l = \vec{x}_{\vec{\boldsymbol{i}}} = \left[x_{\vec{\boldsymbol{i}},l}\right]_l$ is the code word;
  -

$$\boldsymbol{x}(t) = \left(\sum_l \boldsymbol{x}_l e^{j2\pi(lf_\Delta+f_0)t}\right) u(t), \tag{2.5.A.1}$$

$$(\tilde{\boldsymbol{x}}(f) = \sum_l \boldsymbol{x}_l \tilde{u}(f - lf_\Delta - f_0)),$$

is the baseband signal with OFDM modulation and is band limited within $\left(-\frac{b}{2}, \frac{b}{2}\right)$;

–

$$u(t) = \begin{cases} c_u & t_1 \leq t \leq t_2 \\ 0 & t < t_0 \text{ or } t > t_3 \end{cases},$$

$t_0 < t_1 < t_2 = t_1 + \tau_h + \frac{1}{f_\Delta} < t_3$;

– $\boldsymbol{h}(t) = h(t)$ is the time invariant impulse response of the channel with support $[0, \tau_h]$ and is known to the receiver;

– $\boldsymbol{y}(t)$ is the output of the equivalent baseband AWGN channel;

– $\boldsymbol{n}(t)$ is the circularly symmetric complex Gaussian noise process with power spectral density $\tilde{r}_{\boldsymbol{n}}(f) = N_0 \mathrm{rect}_b(f)$.

1. Derive the optimal demodulator.

2. Find the equivalent channel model such that $\vec{\boldsymbol{y}}$ is a sufficient statistic for detecting $\vec{\boldsymbol{i}}$. How is it different from the channel model from the low complexity receiver?

PROBLEM 2.5.A.2.

(**2x1 constant MIMO flat fading channel**) Consider a 2 transmit antenna and 1 receive antenna, constant flat fading channel and linear modulation with orthonormal pulses $u_l(t)$'s. The receiver knows that $h_{1,1} = 10$, $h_{1,2} = -1$.

1. Find the optimal demodulator.
2. Find the equivalent discrete time channel model.

Part 3

Error Control Encoder and Decoder (Codec) Layer: From Information Bits to Discrete Time Signals and Back